ISRO INTERVIEW PREPARATION

Space Technology for Agriculture

A complete, point-wise study guide with diagrams, equations, research basis and model answers

SIF & FLEX • NISAR • Evapotranspiration (SEBAL / METRIC / ALEXI) • Eddy covariance

Crop & soil parameter retrieval • Meteorology & El Niño • Statistics & ML • Digital twins

Crop-residue burning • Conservation & regenerative agriculture • Carbon credits & MRV

Precision agriculture & VRT • FASAL, YES-TECH & PMFBY • IARI Agricultural Physics • IAIRO

Prepared: October 2026

Contents

1. How to use this guide 5

1.1 Self-introduction: “a short introduction, then your own research” 5
1.2 Agricultural Physics: the “1/2 line” rule 6
1.3 Engineering work at IARI 8
1.4 IAIRO — India AI Research Organisation 8

2. Remote-sensing foundations (quick refresher) 10

2.1 Electromagnetic radiation and the key laws 10
2.2 Four resolutions and the trade-off 11
2.3 Indian Earth-observation satellites relevant to agriculture 11
2.4 Vegetation and other indices (formula sheet) 12
2.5 Accuracy assessment basics 12

3. Sun-Induced Fluorescence (SIF) and the FLEX mission 14

3.1 Physics: the fate of absorbed light 14
3.2 Active vs passive fluorescence 15
3.3 How SIF is retrieved: the in-filling principle 16
3.4 SIF satellite missions 17
3.5 FLEX: mission details 17
3.6 Ground measurement of SIF 18
3.7 Agricultural applications of SIF 19
3.8 Limitations and research challenges 20
3.9 Interview questions on SIF and FLEX 20

4. NISAR: crop mapping and soil-moisture retrieval 22

4.1 SAR fundamentals you must be able to explain 22
4.2 NISAR mission specifications 23
4.3 Crop mapping with NISAR 23
4.4 Soil-moisture retrieval with NISAR 25
4.5 Other agricultural uses of NISAR 26
4.6 Interview questions on NISAR 27

5. Retrieving crop and soil parameters 28

5.1 Three families of retrieval methods 28
5.2 Crop parameters 28
5.3 Radiative transfer models (RTMs) 30
5.4 Soil parameters 31
5.5 Digital soil mapping (DSM): Dr. Bappa Das's line of work 32
5.6 Where parameter retrieval is heading 33
5.7 Interview questions 33

6. Evapotranspiration: SEBAL, METRIC, ALEXI and scaling 34

6.1 Definitions you must keep distinct 34
6.2 Measuring ET on the ground 34
6.3 The surface energy balance (SEB) 34
6.4 SEBAL (Bastiaanssen et al. 1998) 35
6.5 METRIC (Allen et al. 2007) 36
6.6 ALEXI and DisALEXI (Anderson et al. 1997, 2007) 37
6.7 Scaling instantaneous ET to daily ET 38
6.8 Eddy covariance (EC) and NRSC's work 39
6.9 Satellite ET products to know 41
6.10 Interview questions on ET 41

7. Meteorology for agriculture, including El Niño and "Super El Niño"....43
7.1 Core concepts....43
7.2 Indian summer monsoon (ISM)....43
7.3 ENSO, El Niño, La Niña and "Super El Niño"....44
7.4 Indian Ocean Dipole (IOD) and MJO....45
7.5 How El Niño affects Indian agriculture....45
7.6 Forecasts and agro-advisories in India....46
7.7 Interview questions....46
8. Statistics and machine learning for agricultural remote sensing....48
8.1 Statistical foundations....48
8.2 Machine-learning algorithms....48
8.3 The bias–variance trade-off, overfitting and validation....50
8.4 ML vs process-based models....51
8.5 Interview questions....52
9. Digital twins vs crop simulation models....53
9.1 Crop simulation models: a refresher....53
9.2 What makes something a digital twin....54
9.3 Data assimilation strategies....55
9.4 Digital-twin initiatives....56
9.5 Interview questions....57
10. Crop-residue burning: detection, FRP, burnt area and management....58
10.1 Why farmers burn and why it matters....58
10.2 Active-fire detection from thermal data....58
10.3 Fire radiative power (FRP)....59
10.4 The daily fire bulletin (CREAMS-IARI / NRSC)....60
10.5 Burnt-area mapping with Sentinel-2's two SWIR bands....61
10.6 Residue management options....63
10.7 Interview questions....65
11. Conservation agriculture, regenerative agriculture and biogeochemical cycles....66
11.1 CA in India: status and benefits....66
11.2 Identifying CA wheat fields from satellite data (the paper by Dr. Rajkumar)....67
11.3 Carbon cycle in agroecosystems....69
11.4 Other biogeochemical cycles....69
11.5 Interview questions....70
12. Carbon credits and MRV (monitoring, reporting, verification)....71
12.1 Types of carbon credits....71
12.2 How credits are assigned: the project cycle....72
12.3 MRV methodology for soil-carbon projects....73
12.4 Biochar and carbon credits....74
12.5 Who buys carbon credits?....74
12.6 The situation in India....75
12.7 IARI and Division of Agricultural Physics involvement in MRV....75
12.8 Interview questions....76
13. Precision agriculture and variable-rate technology....77
13.1 Principles....77
13.2 Tools and technologies....77
13.3 Variable-rate technology (VRT)....78
13.4 Applications....79
13.5 Applicability and adoption in India....80

13.6 ICAR Network Programme on Precision Agriculture (NePPA) and IARI's role....80
13.7 Space-based precision agriculture and the ISRO–SAC context....80
13.8 Interview questions....81

14. Crop insurance and yield estimation (the likely main focus)....82

14.1 The Indian crop-statistics and insurance ecosystem....82
14.2 FASAL: Forecasting Agricultural output using Space, Agro-meteorology and Land-based observations....83
14.3 Yield models used in FASAL and YES-TECH....84
14.4 PMFBY in brief....85
14.5 Yield-index approach for cereals vs weather-based approach for horticulture....87
14.6 YES-TECH: Yield Estimation System based on Technology....88
14.7 Yield modelling at insurance-unit (GP) level: a design you can present....89
14.8 What we (ISRO and research) can contribute to crop insurance....90
14.9 Key concepts in insurance economics....91
14.10 Interview questions on insurance and yield....91

15. Worked numerical examples....93

15.1 FAO-56 reference ET (ET_0)....93
15.2 SEBAL instantaneous and daily ET for one pixel....93
15.3 Spectral indices....94
15.4 Fire radiative power, biomass and emissions....94
15.5 Semi-physical (LUE) wheat yield....94
15.6 Insurance claim and smart sampling....95
15.7 Carbon credits from a zero-till + no-burn wheat farm....96
15.8 Accuracy statistics from a confusion matrix....96
15.9 GDD and phenology....96
15.10 Water-cloud model inversion....96

16. Research-paper summaries (concepts and evidence)....98

16.1 Evapotranspiration....98
16.2 SIF and FLEX....99
16.3 SAR and soil moisture....99
16.4 Fire and burnt area....100
16.5 Crop yield and parameter retrieval....101
16.6 Soil, carbon and conservation agriculture....102

17. Mock interview: long-form model answers and programme timeline....104

17.1 Insurance and yield (expected main focus)....104
17.2 Residue burning....105
17.3 ET, SIF and NISAR....106
17.4 Carbon, CA and precision agriculture....106
17.5 Opening and closing questions....107

18. Indian programmes, data portals and upcoming missions....108

18.1 Operational and digital-agriculture programmes....108
18.2 Data portals....108
18.3 Recent and upcoming missions relevant to agriculture....109
18.4 Topic-wise question bank (practise aloud)....109

19. Rapid revision: formula sheet, abbreviations and rapid-fire Q&A....111

19.1 One-page formula sheet....111
19.2 Abbreviations....111
19.3 Rapid-fire questions (one-line answers)....113

20. References and further reading....117

1. How to use this guide

This guide covers every topic in your interview notes. Each chapter follows the same pattern, so you can revise any topic in the same way: **(i) one-line definition**, **(ii) underlying concept and physics**, **(iii) equations and diagrams**, **(iv) Indian programmes and IARI/ISRO work**, **(v) research basis (key papers)**, **(vi) limitations and future direction**, and **(vii) likely interview questions with model answers**.

Table 1. Topic map and expected weight in the interview

Topic	What the board is likely to probe	Priority
Crop insurance & yield estimation	FASAL models, YES-TECH, smart sampling of CCEs, GP-level ensemble, threshold yield, your contribution	Very high (main focus)
Crop residue burning	Active-fire detection, FRP, daily bulletin, dNBR/BAI burnt area, residue management (in-situ / ex-situ)	High
Evapotranspiration	SEB, SEBAL vs METRIC vs ALEXI, hot/cold pixel, instantaneous→daily scaling, eddy covariance	High
SIF & FLEX	Physics of fluorescence, FLORIS, tandem with Sentinel-3, FLD retrieval, ground measurement, GPP	High
NISAR	L/S-band, crop mapping, soil-moisture retrieval, water cloud model	High
Crop & soil parameters	LAI, chlorophyll, N, water; salinity indices; digital soil mapping; PROSAIL	Medium–High
Carbon credits & MRV, CA/RA	Sequestration vs reduction, VM0042, biochar, buyers, CCTS India, CA detection paper	Medium–High
Precision agriculture & VRT	Principles, tools, NePPA, applicability in India	Medium
Meteorology, Statistics & ML	ENSO/Super El Niño, IOD, monsoon; regression, validation, RF/XGBoost/LSTM	Medium
Digital twins vs crop models	Definitions, data assimilation, closed loop	Medium
Agricultural Physics, IARI engineering, IAIRO	One-line definitions, objectives, achievements	Opening questions

> **Before the interview:** Verify the latest numbers on the morning of the interview: fire counts, NISAR product releases, FLEX commissioning status and PMFBY weights. These change month to month. Everything here is current to October 2026 as far as public sources show.

1.1 Self-introduction: “a short introduction, then your own research”

Keep the introduction to about 90 seconds. Use this structure and replace the bracketed parts with your own details:

1. **Who you are (1 line):** “I am [name], a [M.Sc./Ph.D.] scholar in Agricultural Physics at ICAR-IARI, New Delhi, specialising in remote sensing and crop modelling.”

2. **Your research problem (2 lines):** "My work estimates [crop yield / ET / residue burning / soil moisture] at [field / GP] scale by combining [Sentinel-1/2, SIF, weather] with [semi-physical / ML / crop models]."
3. **Method (2 lines):** name the data, the model, the validation and the accuracy (for example "RMSE 0.42 t/ha, R^2 0.78 against CCE yields").
4. **Result and relevance (1 line):** connect it to an ISRO or national programme (FASAL, YES-TECH, the CREAMS fire bulletin, NISAR cal/val).
5. **Why ISRO (1 line):** "I want to take research-grade retrievals to operational, national-scale products."

> **Tip:** Every statement you make in the introduction invites a follow-up question. Mention only methods you can derive on a whiteboard, for example the LUE equation, the NBR formula, the energy balance or Random Forest.

1.2 Agricultural Physics: the "1/2 line" rule

One-line definition: Agricultural Physics applies the principles of physics (energy, mass and momentum transfer, radiation, and electromagnetic interaction) to understand, measure and manage the soil–plant–atmosphere continuum (SPAC) for sustainable crop production.

Half-line version: "Physics of the soil–plant–atmosphere system for agriculture."

Objectives of the Division of Agricultural Physics (ICAR-IARI)

- Understand **energy and water exchange** in the SPAC: radiation, ET, soil heat and water flow.
- Develop **remote-sensing and GIS methods** for crop inventory, crop condition, yield forecasting, drought, residue burning and soil health.
- Advance **agricultural meteorology**: weather-based agro-advisories, crop–weather modelling and climate-change impact assessment.
- Study **soil physical health**: structure, compaction, hydraulic properties, conservation tillage and water-saving irrigation.
- Apply **biophysics**: hyperspectral phenotyping, stress detection, and nano-sensors for plants and soil.
- **Human-resource development**: M.Sc. and Ph.D. programmes in Agricultural Physics at the IARI Graduate School.

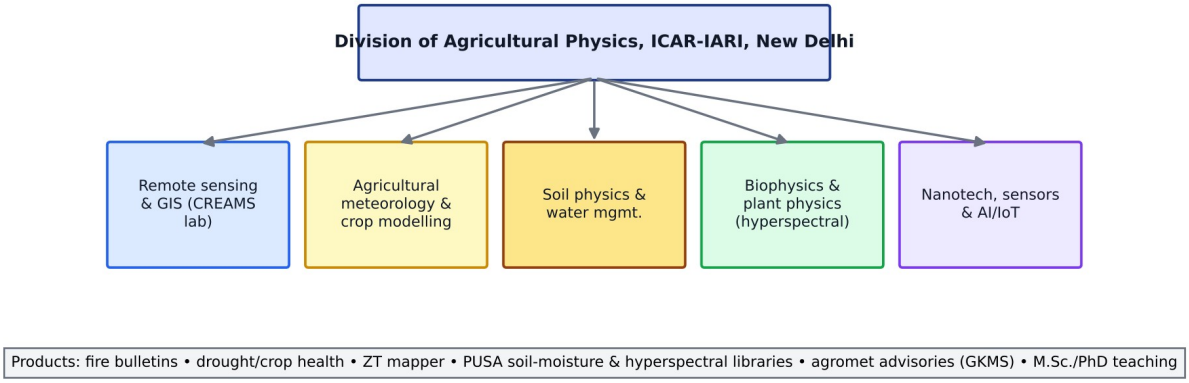


Figure 1. Structure and research areas of the Division of Agricultural Physics, ICAR-IARI (schematic).

Achievements to quote

Area	Achievement or product
Crop-residue burning	The CREAMS laboratory (Consortium for Research on Agroecosystem Monitoring and Modelling from Space) issues daily paddy-residue fire bulletins for Punjab, Haryana, UP, Delhi, Rajasthan and MP from MODIS/VIIRS active fires. It co-developed the ISRO–ICAR standard protocol for monitoring burning events, and maps wheat and rice burnt area. Bulletins go to MoA&FW, CAQM and state departments.
Conservation agriculture	Zero-Tillage Mapper: a time series of ZT-wheat adoption over about 63,000 fields in Fazilka (Punjab), 2018–2023 (Sehgal & Dhakar, CREAMS).
Crop monitoring	Drought and crop-health assessment, satellite-based crop monitoring for national initiatives, and support for crop area and yield estimation.
Hyperspectral & biophysics	Spectral libraries and indices for wheat yellow rust, nitrogen and water stress; hyperspectral soil-property estimation (R. N. Sahoo and colleagues).
Agrometeorology	Gramin Krishi Mausam Sewa (GKMS) agro-advisories for Delhi NCR; crop–weather relations and climate-change impact studies.
Soil physics	Soil compaction, hydraulic properties, irrigation scheduling, and tillage and residue effects on soil moisture and temperature.
Precision agriculture	IARI is the lead institute of ICAR-NePPA (Network Programme on Precision Agriculture, launched 8 Sep 2021, 16 partner institutes).

> **Name to remember:** Dr. Vinay Kumar Sehgal heads the Division of Agricultural Physics and leads the CREAMS lab, which supplies residue-burning reports to the Ministry of Agriculture. Your notes mention a paper by “Dr. Vinay” that you couldn’t read clearly; Dr. Sehgal is the most likely author. Check his recent publications on CA mapping, residue burning or MRV before the interview.

1.3 Engineering work at IARI

IARI's Division of Agricultural Engineering, together with the Post-Harvest and Food Science divisions and the Water Technology Centre, works on machines and devices for small farms. Know three or four examples with a one-line description of each:

Technology	One-line description
Pusa Zero Energy Cool Chamber (ZECC)	A double brick-wall chamber with a wet sand cavity that keeps fruit and vegetables 10–15 °C cooler than ambient using evaporative cooling and no electricity.
Pusa-Farm SunFridge	A solar-powered on-farm cold room for perishables, with thermal storage for night-time cooling.
Conservation-agriculture machinery	Zero-till drills, raised-bed planters and residue-handling attachments adapted for rice–wheat systems (with PAU, CIMMYT and other partners).
Precision application	Sensor-based (NDVI) variable-rate N applicators, drone spraying trials and IoT soil-moisture-based irrigation controllers.
Water technology	Pressurised micro-irrigation, fertigation scheduling and wastewater reuse (Water Technology Centre).
PUSA Decomposer (microbiology, not engineering)	A fungal consortium (several *Trichoderma*/*Aspergillus*-type cellulolytic strains) sprayed on paddy stubble that decomposes it in about 20–25 days. Often asked together with residue management.

> **Caution:** Before quoting a specific machine and its developer, check it against the latest IARI Annual Report. The board will appreciate precise claims more than broad ones.

1.4 IAIRO — India AI Research Organisation

- **What:** a national, execution-first AI research institution, modelled "in the spirit of ISRO, but for AI".
- **Where and when:** at GIFT City, Gandhinagar. Gujarat gave in-principle approval in late December 2025, operations started on 1 January 2026, and the formal launch was on **30 January 2026**.
- **Legal form:** a Section 8 non-profit run as a **public-private partnership**, positioned as a Focused Research Organisation under the **IndiaAI Mission** (MeitY).
- **Funding:** about ₹300 crore over five years, from the Gujarat government, the Centre and industry (the Indian Pharmaceutical Alliance has been reported as a partner).
- **Leadership:** Founding Director Dr. Amit Sheth, author of the launch position paper "Sovereign AI for India's Strategic Autonomy".
- **Mandate:** frontier AI research, sovereign IP, retaining Indian AI talent, translating research into deployments, and evidence-based policy. Compute will combine on-premise GPUs with the IndiaAI Cloud.
- **Agriculture relevance:** foundation models for Earth observation, such as an Indian counterpart to NASA–IBM Prithvi, trained on Resourcesat, EOS-04, NISAR and Sentinel

archives. Possible uses are crop-type mapping, yield nowcasting, pest-outbreak prediction and multilingual advisory (Bhashini).

Q. Why does India need IAIRO when IITs and IndiaAI already exist?

A. Universities optimise for papers and startups optimise for short-term revenue. IAIRO is meant to fill the gap between them with mission-mode, long-horizon, sovereign AI research that is translated into public-good deployments, the way ISRO did for space.

Q. How could IAIRO help ISRO's agriculture applications?

A. It could train geospatial foundation models on Indian satellite archives, build benchmark datasets such as CCE-linked yield labels and crop-type ground truth, and provide GPU compute for national-scale inference, for example GP-level yield under YES-TECH.

2. Remote-sensing foundations (quick refresher)

Every specialised topic later in the guide (SIF, NISAR, ET, burning, yield) rests on a few physical ideas. The board often opens with one of them before going deeper.

2.1 Electromagnetic radiation and the key laws

- **Planck's law:** spectral radiance of a black body, $B(\lambda,T) = 2hc^2/\lambda^5 \cdot 1/(e^{(hc/\lambda kT)} - 1)$.
- **Wien's displacement law:** $\lambda_{max} = 2898/T$ µm. The Sun (≈5800 K) peaks at 0.5 µm, the Earth (≈300 K) at about 9.7 µm and a fire (≈1000 K) at about 2.9 µm, which is why fires are detected at 3.9 µm.
- **Stefan–Boltzmann law:** $M = \varepsilon\sigma T^4$ with $\sigma = 5.67\times10^{-8}$ W m^{-2} K^{-4}. This is the basis of the outgoing longwave term in the energy balance.
- **Kirchhoff's law:** at thermal equilibrium, emissivity $\varepsilon(\lambda)$ = absorptivity $\alpha(\lambda)$. For opaque bodies, reflectance = $1 - \varepsilon$.
- **Atmospheric windows:** VIS–NIR (0.4–1.3 µm), SWIR (1.55–1.75 and 2.05–2.4 µm), MIR (3–5 µm), TIR (8–14 µm) and microwave (>1 cm, nearly all-weather).

Table 2. Spectral regions and agricultural information content

Region	Range	What it tells you about crops and soil
Blue	0.45–0.52 µm	Chlorophyll and carotenoid absorption; aerosol correction
Green	0.52–0.60 µm	Green reflectance peak; chlorophyll content (GCI, GNDVI)
Red	0.63–0.69 µm	Strong chlorophyll absorption (red well)
Red edge	0.69–0.75 µm	Position shifts with chlorophyll and N (REP, NDRE, CI-red-edge); Sentinel-2 B5–B7
NIR	0.75–1.3 µm	Mesophyll scattering, LAI and biomass; high for healthy canopies
SWIR	1.55–2.35 µm	Leaf water (NDWI/LSWI), cellulose/lignin (residue, CAI), burn scars (NBR), soil minerals and salinity
MIR	3–5 µm	Active fires, FRP
TIR	8–14 µm	LST, ET (SEBAL/METRIC/ALEXI), crop water stress (CWSI)
Microwave	1 mm–1 m	SAR: structure, biomass, soil moisture, flooding; radiometers (SMAP, SMOS) for soil moisture

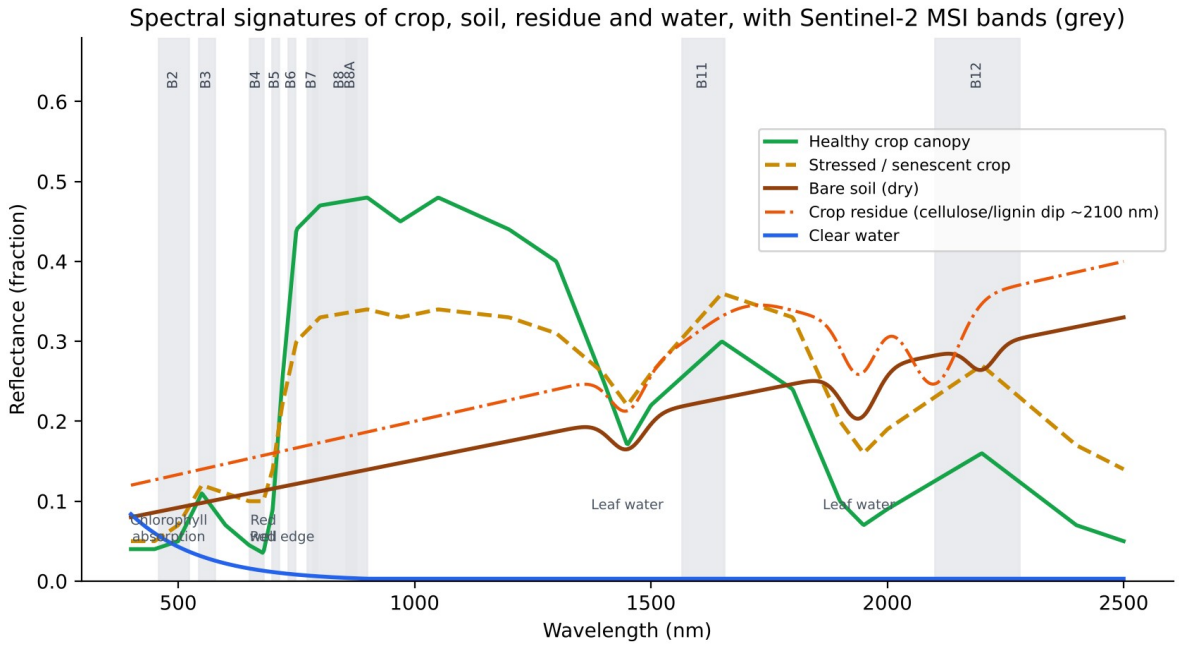


Figure 2. Typical spectral reflectance curves of healthy and stressed crop, bare soil, crop residue and water, with Sentinel-2 MSI band positions.

2.2 Four resolutions and the trade-off

- **Spatial:** pixel size, e.g. LISS-IV 5.8 m, Sentinel-2 10–20 m, Landsat 30 m, MODIS 250–1000 m.
- **Spectral:** number and width of bands, e.g. multispectral (4–13 bands) or hyperspectral (PRISMA, EnMAP, HySIS: >200 bands at about 10 nm).
- **Radiometric:** quantisation, e.g. 10-bit LISS-III, 12-bit Sentinel-2 and 14-bit Landsat-9, which set how finely brightness differences can be distinguished.
- **Temporal:** revisit, e.g. Sentinel-2 5 days (2 satellites), Sentinel-1 6–12 days, NISAR 12 days, MODIS daily and INSAT every 15–30 min.
- **Trade-off:** fine spatial resolution usually means a narrow swath and less frequent revisit. **Data fusion** (STARFM/ESTARFM), **harmonised products** (HLS) and **constellations** address this.

2.3 Indian Earth-observation satellites relevant to agriculture

Table 3. ISRO missions used in agricultural applications

Satellite	Sensor / key specs	Agricultural use
Resourcesat-2/2A	LISS-IV 5.8 m, LISS-III 23.5 m, AWiFS 56 m (5-day)	Crop acreage (FASAL), horticulture (CHAMAN), cropping system analysis
EOS-04 (RISAT-1A)	C-band SAR 5.4 GHz, multi-pol	Kharif rice and jute mapping under clouds, flood mapping
EOS-06 (Oceansat-3)	OCM-3, SSTM	Ocean colour and SST feeding monsoon forecasts
NISAR (with NASA)	L + S-band SAR, 12-day	Crop mapping, soil moisture, biomass,

Satellite	Sensor / key specs	Agricultural use
		subsidence
INSAT-3D/3DR/3DS	Imager + Sounder; 6 channels; 15–30 min	Insolation, LST, rainfall (HEM/IMSRA), fog, ET (MOSDAC)
Cartosat-2/3	Panchromatic < 1 m	Field boundaries, cadastral overlay, infrastructure
EOS-07 / EOS-08	Microwave sounders, GNSS-R (EOS-08)	Atmospheric profiles; GNSS-R for soil moisture and flood detection
HySIS	Hyperspectral VNIR–SWIR	Soil and crop biochemistry research

2.4 Vegetation and other indices (formula sheet)

Table 4. Frequently used spectral indices (Sentinel-2 band numbers in brackets)

Index	Formula	Sensitive to
NDVI	(NIR − R)/(NIR + R) = (B8 − B4)/(B8 + B4)	Greenness and biomass; saturates at LAI > 3
EVI	2.5(NIR − R)/(NIR + 6R − 7.5B + 1)	Biomass at high LAI; corrects for soil and atmosphere
SAVI	(1 + L)(NIR − R)/(NIR + R + L), L = 0.5	Sparse canopies; reduces soil background
NDRE	(NIR − RE)/(NIR + RE) = (B8A − B5)/(B8A + B5)	Chlorophyll and N at high LAI
CIred-edge	NIR/RE − 1 = B7/B5 − 1	Canopy chlorophyll content (Gitelson)
MTCI	(B6 − B5)/(B5 − B4)	Chlorophyll (MERIS/OLCI)
NDWI / LSWI	(NIR − SWIR1)/(NIR + SWIR1) = (B8 − B11)/(B8 + B11)	Leaf and canopy water; flooding at rice transplanting
NDTI	(SWIR1 − SWIR2)/(SWIR1 + SWIR2) = (B11 − B12)/(B11 + B12)	Crop residue cover, tillage
CAI	0.5(R2.0 + R2.2) − R2.1 (hyperspectral)	Cellulose absorption: residue vs soil
NBR	(NIR − SWIR2)/(NIR + SWIR2) = (B8 − B12)/(B8 + B12)	Burn scars; dNBR for severity
BAI	$1/[(0.1 - R)^2 + (0.06 - NIR)^2]$	Charcoal signal after burning
MIRBI	10·SWIR2 − 9.8·SWIR1 + 2	Burnt area in SWIR (Trigg & Flasse)
PRI	(R531 − R570)/(R531 + R570)	Xanthophyll cycle, LUE
SI / NDSI (salinity)	√(B·R); (R − NIR)/(R + NIR)	Surface salt crust

2.5 Accuracy assessment basics

- **Confusion matrix:** rows are reference classes and columns are classified classes.
- **Overall accuracy (OA)** = Σ diagonal / N.
- **Producer's accuracy** = 1 − omission error. **User's accuracy** = 1 − commission error.

- **Kappa** κ = (po − pe)/(1 − pe). Many reviewers now prefer reporting OA together with area-adjusted estimates and their confidence intervals (Olofsson et al. 2014).
- **Continuous variables:** RMSE, MAE, bias, R^2, nRMSE (%) and Willmott's d. Always validate on independent data.

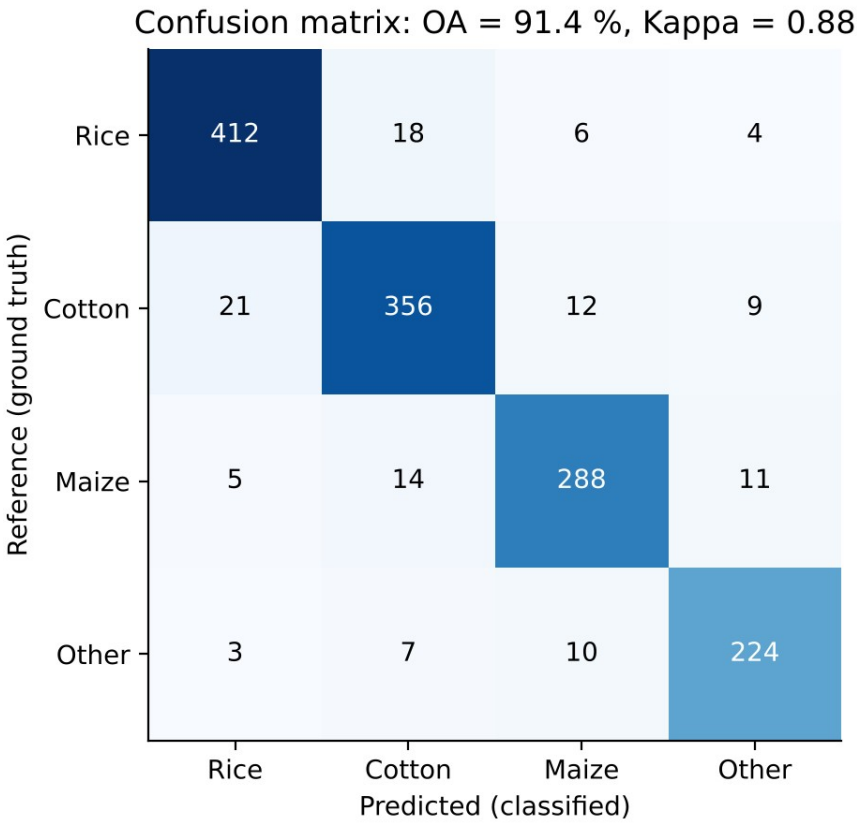


Figure 3. Example confusion matrix for a crop classification, with overall accuracy and kappa.

3. Sun-Induced Fluorescence (SIF) and the FLEX mission

One-line definition: Sun-induced chlorophyll fluorescence is the faint red and far-red light (650–800 nm) re-emitted by chlorophyll-a after absorbing sunlight. It is a direct, physiological signal of photosynthetic activity, unlike reflectance indices such as NDVI, which show only greenness (potential photosynthesis).

3.1 Physics: the fate of absorbed light

A photon absorbed by the light-harvesting antennae of photosystem II (PSII) or PSI has three competing de-excitation pathways:

1. **Photochemistry (ΦP):** electron transport drives ATP/NADPH production and CO_2 fixation. This is the useful path and corresponds to GPP.
2. **Non-photochemical quenching (ΦN, NPQ):** excess energy is dissipated as heat through the xanthophyll cycle (violaxanthin → zeaxanthin). This rises under stress and high light; PRI tracks it.
3. **Fluorescence (ΦF):** about 0.5–3 % of absorbed energy is re-emitted at longer wavelengths. Emission peaks at **~685 nm (red, mostly PSII)** and **~740 nm (far-red, PSI + PSII)**.

$$\Phi P + \Phi N + \Phi F = 1 \quad \text{(energy conservation of absorbed photons)}$$

$$SIF(\lambda) = PAR \cdot fAPAR \cdot \Phi F(\lambda) \cdot fesc(\lambda) \qquad GPP = PAR \cdot fAPAR \cdot \Phi P \;\; (= APAR \cdot LUEp)$$

Combining the two equations gives **GPP = SIF · (ΦP / (ΦF · fesc))**. The relationship between SIF and GPP is close to linear at canopy and ecosystem scale. Much of that linearity comes from the shared APAR term, and the ratio ΦP/ΦF varies under strong stress (Porcar-Castell et al. 2014; Mohammed et al. 2019).

Fate of light absorbed by chlorophyll: why SIF is a direct proxy for photosynthesis

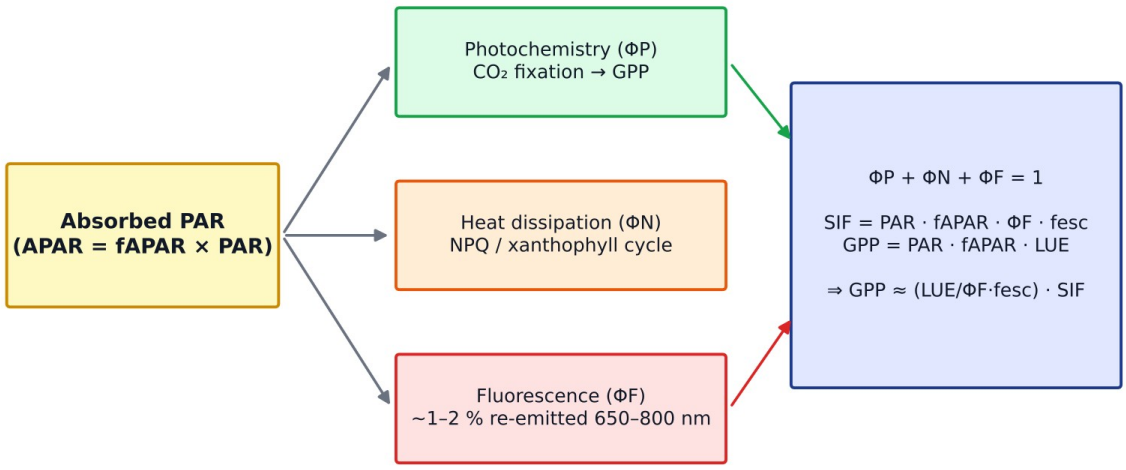


Figure 4. Partitioning of absorbed photosynthetically active radiation into photochemistry, heat and fluorescence. Concept redrawn after Porcar-Castell et al. (2014) and Mohammed et al. (2019).

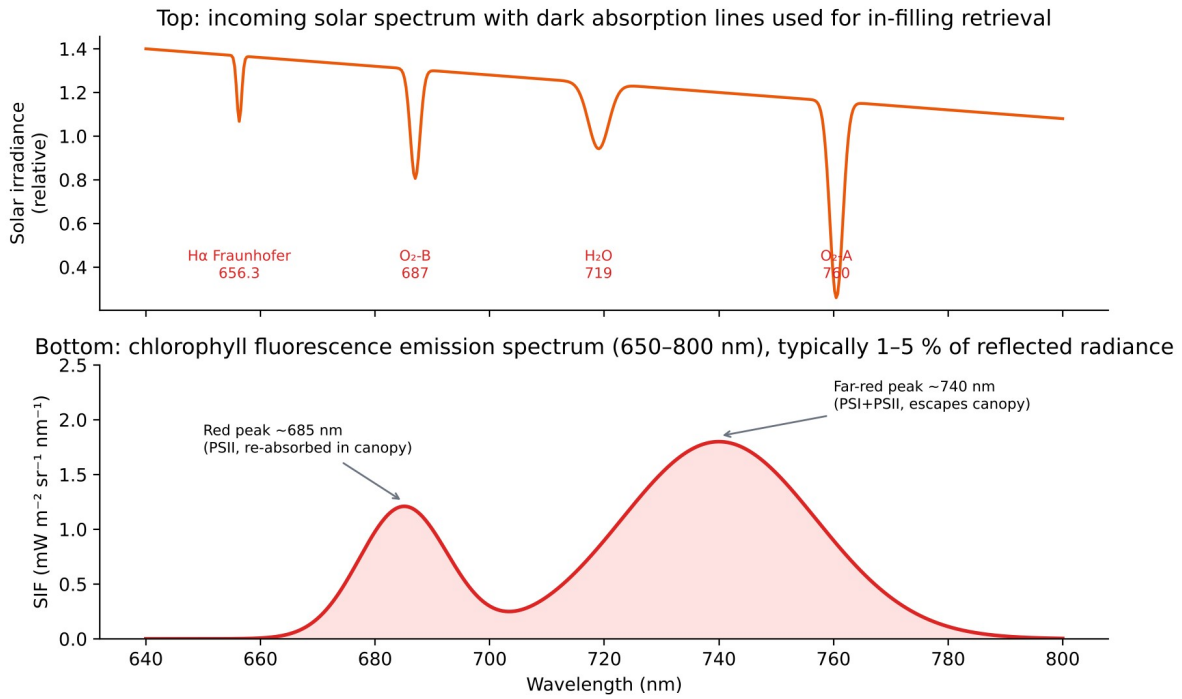


Figure 5. Top: solar irradiance with the Fraunhofer and telluric (O_2-A, O_2-B) absorption bands used to retrieve SIF. Bottom: the double-peaked fluorescence emission spectrum. Schematic redrawn after Meroni et al. (2009).

Why are the two peaks different?

- The **red peak (685 nm)** is strongly **re-absorbed** by chlorophyll inside the leaf and canopy, so its top-of-canopy value is much lower. It is more sensitive to PSII and to stress-driven changes in ΦF.
- The **far-red peak (740 nm)** escapes the canopy (high fesc) and depends strongly on canopy structure (LAI, leaf angle, clumping). It is the peak most satellites (OCO-2, TROPOMI, GOME-2) retrieve.
- The **red/far-red ratio** is used as an indicator of chlorophyll content and stress.

3.2 Active vs passive fluorescence

Aspect	Active (PAM, LIFT, LIF)	Passive SIF (sun-induced)
Light source	Artificial: saturating pulse or laser	Sunlight
Scale	Leaf (cm) to a few metres (LIFT)	Canopy to globe (tower, airborne, satellite)
Parameters	Fv/Fm (max PSII efficiency, ≈0.83 healthy), ΦPSII, qP, NPQ	SIF magnitude (mW m^{-2} sr^{-1} nm^{-1}), full spectrum
Measurement	Needs dark adaptation for Fv/Fm	No contact; continuous under sunlight
Use	Mechanistic plant physiology	Large-scale GPP, stress, yield

3.3 How SIF is retrieved: the in-filling principle

SIF is only about 1–5 % of the reflected radiance in the NIR, so it cannot be separated by reflectance alone. Retrieval works in **dark absorption lines** where incident light is very low. In those lines the relative contribution of emitted fluorescence is large and it "fills in" the line.

- **Telluric O_2 bands:** O_2-A (760 nm, far-red) and O_2-B (687 nm, red). These are deep and wide, so they suit tower, airborne and FLEX retrievals, but they need accurate atmospheric correction above about 100 m height.
- **Solar Fraunhofer lines:** Hα 656 nm, K I 770 nm and several lines at 745–758 nm. They are narrow and insensitive to atmosphere and clouds, and they are used by GOSAT, OCO-2 and TROPOMI.

Retrieval algorithms

Method	Principle	Notes
FLD (Plascyk 1975)	Two bands (inside and outside the line); assumes reflectance and F are constant across them. F = (Eout·Lin − Ein·Lout)/(Eout − Ein)	Simple; biased if reflectance changes
3FLD / iFLD	Uses two shoulders or corrects for the spectral slope of reflectance and F	Lower bias
SFM (spectral fitting; Meroni 2009)	Fits parametric shapes (polynomial reflectance + Gaussian/Lorentzian F) across the whole band	Gives the full spectrum; needs ≤0.3 nm resolution
SVD / PCA data-driven (Guanter 2012; Joiner 2013)	Learns the atmospheric/solar spectral variability from non-fluorescing targets (deserts, clouds); SIF is the residual	Standard for satellites
Physical / RTM inversion	Couples atmospheric RT (MODTRAN/libRadtran) with canopy RT (SCOPE)	Used for FLEX L2

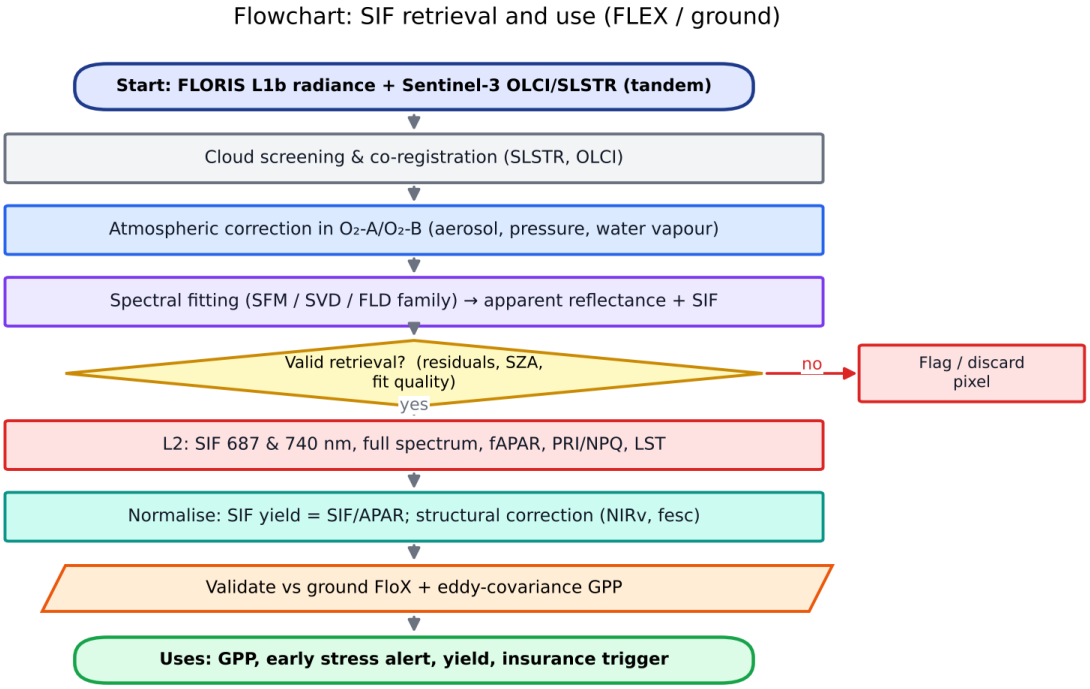


Figure 6. Flowchart of SIF retrieval and its use, from FLEX/Sentinel-3 radiance to agricultural applications.

3.4 SIF satellite missions

Table 5. Spaceborne SIF sources

Mission	Launch	Resolution	Remarks
GOSAT (TANSO-FTS)	2009	~10 km footprints, sparse	First global SIF maps (Frankenberg 2011; Joiner 2011)
GOME-2 (MetOp-A/B/C)	2006+	40×80 km	Long record, used for US Corn Belt GPP/yield (Guan 2016)
OCO-2 / OCO-3	2014 / 2019	1.3×2.25 km, narrow swath	Fine footprint; OCO-3 on ISS samples the diurnal cycle
TROPOMI (Sentinel-5P)	2017	3.5×5.5 km, daily global	Workhorse for regional crop monitoring
TanSat (China)	2016	2×2 km	CO_2 + SIF
FLEX (ESA EE-8)	15 Sep 2026	300 m, 150 km swath	First mission designed for fluorescence; full spectrum and both peaks

3.5 FLEX: mission details

Parameter	Detail
Full name	FLuorescence EXplorer, ESA's 8th Earth Explorer (selected 2015)
Launch	15 September 2026, Vega-C flight VV30, Kourou (French Guiana), launched together with Copernicus Sentinel-3C. Your notes have “Sep 2026”, which is correct.

Parameter	Detail
Orbit	Sun-synchronous, about 815 km, 10:00 local descending node, tandem with Sentinel-3, a few seconds apart on the same ground track
Instrument	FLORIS (FLuORescence Imaging Spectrometer, Leonardo): pushbroom, 500–780 nm
Spectral sampling	High-resolution channel ~0.1 nm sampling / 0.3 nm resolution in O_2-B (677–697 nm) and O_2-A (740–780 nm); low-resolution 0.5–2 nm elsewhere
Spatial	300 m × 300 m pixels; 150 km swath
Revisit	27-day repeat (global land between about 75° N and 56° S); combined with Sentinel-3 for context
Platform	OHB System AG (Germany); planned life 3.5 years plus extension
Role of Sentinel-3	OLCI gives reflectance, aerosol and water vapour; SLSTR gives LST and cloud screening, plus canopy temperature for NPQ and stress interpretation
Products	L1b radiance; L2: total SIF, SIF at 687 and 740 nm, full spectrum, fAPAR, PRI/NPQ proxy, canopy temperature; L3/L4 photosynthesis (GPP) and stress indicators
Airborne demonstrator	HyPlant (Forschungszentrum Jülich), flown in FLEXSense campaigns

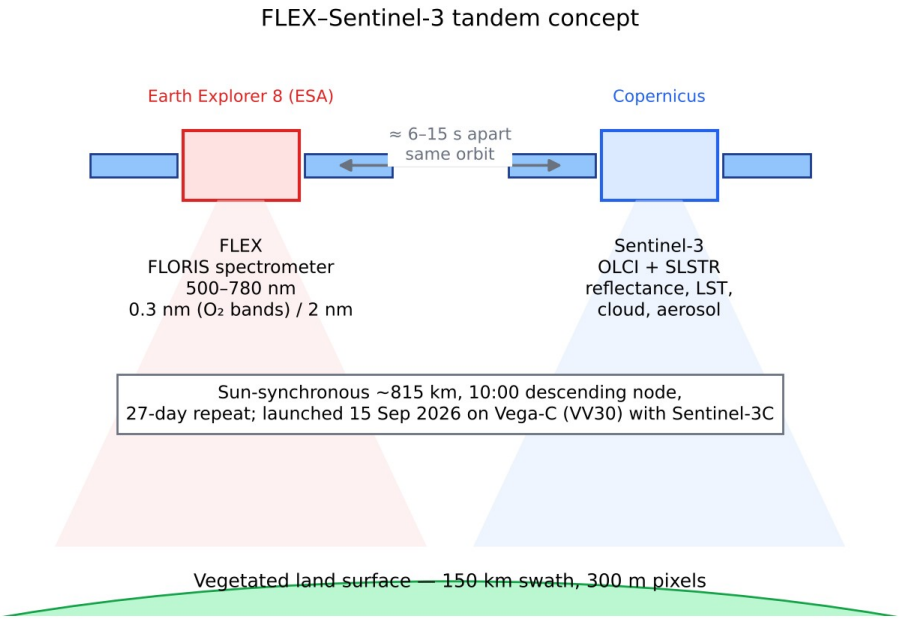


Figure 7. FLEX flying in tandem with Sentinel-3 (schematic based on Drusch et al. 2017 and ESA mission documentation).

> **Why FLEX is different:** Unlike OCO-2 and TROPOMI, FLEX resolves **both** fluorescence peaks and the full emission spectrum at **300 m**, close to field scale for large Indian plots and village scale for smallholders. This allows the photosynthetic *efficiency* signal to be separated from the APAR signal, which earlier missions could not do.

3.6 Ground measurement of SIF

Ground (proximal) SIF systems give continuous canopy measurements for process understanding and for **FLEX calibration and validation**.

- **Instrument:** two fibre-optic channels feed a high-resolution spectrometer, such as an Ocean Insight QE Pro (FWHM ≈ 0.3 nm, SNR > 1000) for fluorescence plus a broader-range unit for reflectance. Examples are **FloX** (JB Hyperspectral), **Piccolo Doppio**, **PhotoSpec** and **FluoSpec2**.
- **Geometry:** an upward cosine diffuser measures irradiance E(λ), and a downward bare fibre (about 25° FOV) or a rotating mirror measures canopy radiance L(λ). Readings alternate every few seconds and are cross-calibrated with a reference panel.
- **Thermal stability:** spectrometers sit in a Peltier-cooled housing because dark current and wavelength shifts corrupt the 0.1 nm-scale retrieval.
- **Height:** sensors are mounted 2–10 m above the canopy. Above about 20 m, O_2-band atmospheric absorption between sensor and canopy must be corrected.
- **Retrieval:** FLD/3FLD/iFLD or SFM at O_2-A and O_2-B. Quality control removes changing-cloud records and low-sun-angle data.
- **Ancillary data:** an eddy-covariance tower (GPP, ET), PAR sensors, leaf-level PAM fluorometry (Fv/Fm, NPQ) and LAI measurements.
- **Networks:** the ESA FLEX cal/val network (European Fluorescence Network), ChinaSpec and the US SIF network. In India, a ground SIF system next to an existing flux tower (NRSC/ISRO, IARI farm) would make an ideal **FLEX cal/val super-site**.

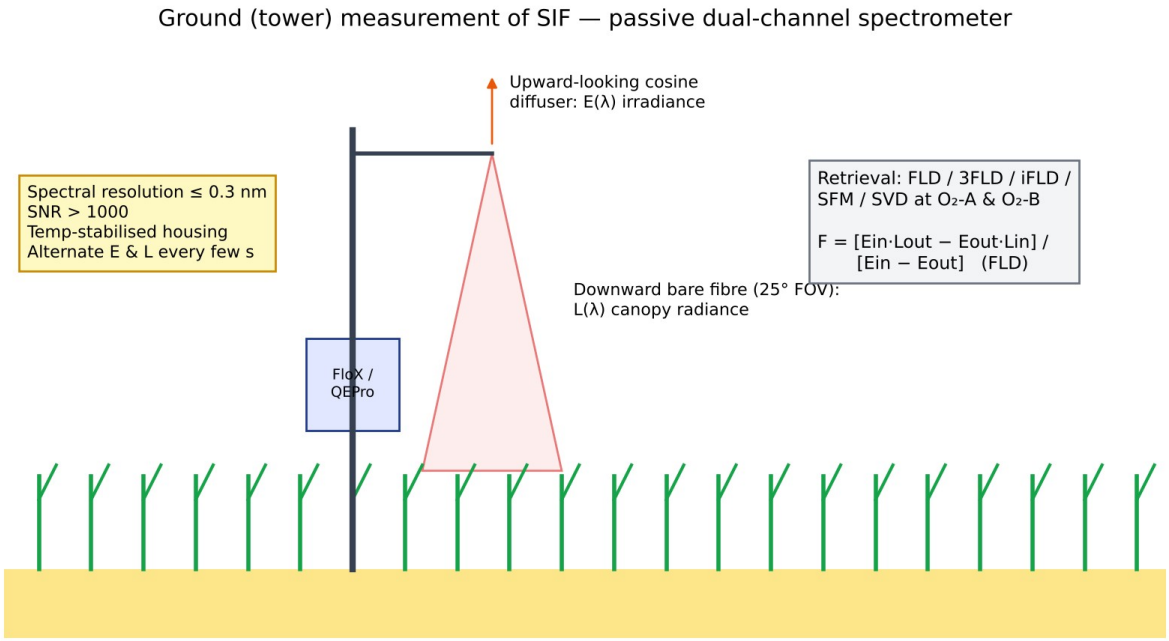


Figure 8. Tower-based passive SIF measurement: irradiance and radiance channels, and the FLD principle.

$$\text{FLD:}\quad F = (E_{out} \cdot L_{in} - E_{in} \cdot L_{out}) / (E_{out} - E_{in}) \qquad (\text{in} = \text{inside line, out} = \text{shoulder})$$

3.7 Agricultural applications of SIF

- **GPP estimation:** a near-linear SIF–GPP relationship holds across crops. C4 crops (maize, sugarcane) have higher GPP per unit SIF than C3 crops (wheat, rice).

- **Yield forecasting:** Guan et al. (2016) showed GOME-2 SIF explained US Corn Belt yields better than NDVI or EVI. TROPOMI SIF is now used for county-level yields, and FLEX's 300 m data could reach GP level.
- **Early stress detection:** SIF drops within hours to days of drought, heat or nitrogen stress, before visible greening loss (a lag of 1–3 weeks in NDVI). Sun et al. (2015) documented this for the 2012 US drought.
- **Heat waves:** helps separate heat stress (an ΦF/NPQ change) from water stress, together with Sentinel-3 LST.
- **Nitrogen status:** the red/far-red ratio and SIF yield (SIF/APAR) respond to N supply.
- **Irrigation efficiency:** SIF/ET gives a photosynthesis-based estimate of water productivity.
- **Inputs to crop models and digital twins:** SIF can be assimilated as a process-level observation of photosynthesis.

3.8 Limitations and research challenges

- The signal is weak (0.5–3 mW m^{-2} sr^{-1} nm^{-1}), so it needs high SNR and fine spectral resolution. Satellite SIF is noisy and must be averaged in time or space.
- **Structural confounding:** fesc depends on LAI, leaf angle and clumping. Methods to normalise it include the NIRv/FCVI approach and SIF/(NIRv·PAR).
- Sun-view geometry, the overpass time (one snapshot a day) and the need to scale to daily values (as with ET).
- Mixed pixels at 300 m in fragmented Indian smallholdings (median holding about 1 ha).
- Atmospheric correction in the O_2 bands (aerosols, surface pressure).

3.9 Interview questions on SIF and FLEX

Q. Why is SIF better than NDVI for detecting stress?

A. NDVI measures chlorophyll quantity and canopy structure, so it changes only once leaves yellow or wilt. SIF responds to the actual photosynthetic flux. When stomata close or heat damages PSII, ΦP and ΦF drop within hours, long before any change in pigments.

Q. Is SIF a measure of photosynthesis or of APAR?

A. Both. At canopy scale SIF ≈ APAR × ΦF × fesc. Much of the SIF–GPP correlation comes from APAR, and the physiological component (ΦF) adds information under stress. FLEX measures both peaks together with PRI, fAPAR and LST, so it can separate the two parts.

Q. Why fly FLEX in tandem with Sentinel-3?

A. SIF retrieval needs simultaneous reflectance (to model the background), aerosol and water vapour (for O_2-band atmospheric correction), cloud screening, and LST to interpret NPQ and stress. Sentinel-3 OLCI and SLSTR provide all of these a few seconds apart on the same ground track.

Q. How would you validate FLEX SIF over India?

A. Set up FloX-type ground systems in 300 m-homogeneous wheat and rice fields at IARI and at an NRSC flux-tower site. Measure continuously, upscale with Sentinel-2 or UAV hyperspectral data to 300 m, and compare at overpass time. Report bias, RMSE and the linearity of SIF with eddy-covariance GPP.

Q. What is Fv/Fm?

A. The maximum quantum efficiency of PSII, measured on dark-adapted leaves with a PAM fluorometer: (Fm − F0)/Fm. It is about 0.83 in healthy C3 leaves and decreases under photoinhibition.

Q. What are the O_2-A and O_2-B bands?

A. Absorption bands of atmospheric oxygen centred at 760 nm (A) and 687 nm (B). They make deep dark wells in the incoming spectrum, where the fluorescence in-filling signal is relatively large.

4. NISAR: crop mapping and soil-moisture retrieval

One-line definition: NISAR (NASA–ISRO Synthetic Aperture Radar) is the first dual-frequency (L- and S-band) imaging radar satellite. It gives all-weather, day-and-night, 12-day repeat observations of the Earth's surface for ecosystems, cryosphere, solid-earth deformation and **agriculture**.

4.1 SAR fundamentals you must be able to explain

- **Active microwave:** the radar transmits pulses and records the backscatter σ^0 (dB). It works through clouds, which is critical for kharif crops during the monsoon.
- **Synthetic aperture:** the motion of the platform synthesises a long antenna, giving azimuth resolution of about L/2 (antenna length), independent of range.
- **Wavelength vs penetration:** X (3 cm) < C (5.6 cm, Sentinel-1/EOS-04) < S (9.4 cm) < L (24 cm) < P (70 cm). Longer wavelengths penetrate deeper into canopy and soil.
- **Polarisation:** HH, VV (co-pol) and HV/VH (cross-pol). Cross-pol is generated by **volume (multiple) scattering**, so it is a good biomass and canopy indicator.
- **Dielectric constant (ε):** dry soil ε ≈ 3–5 and water ε ≈ 80. Backscatter rises with soil moisture, which is the basis of SM retrieval.
- **Geometric effects:** foreshortening, layover and shadow; incidence-angle dependence (normalise to a reference angle, e.g. 40°).
- **Speckle:** coherent interference noise. Reduce it with multilooking, Lee/Refined-Lee/Gamma-MAP filters or temporal filtering.
- **Interferometry (InSAR):** phase difference gives deformation. **Coherence** drops when the field is ploughed, sown or harvested, so it can be used to detect tillage and harvest.
- **Polarimetry:** decompositions such as Freeman–Durden (surface/double/volume), Cloude–Pottier (H/A/α) and Yamaguchi describe scattering mechanisms.

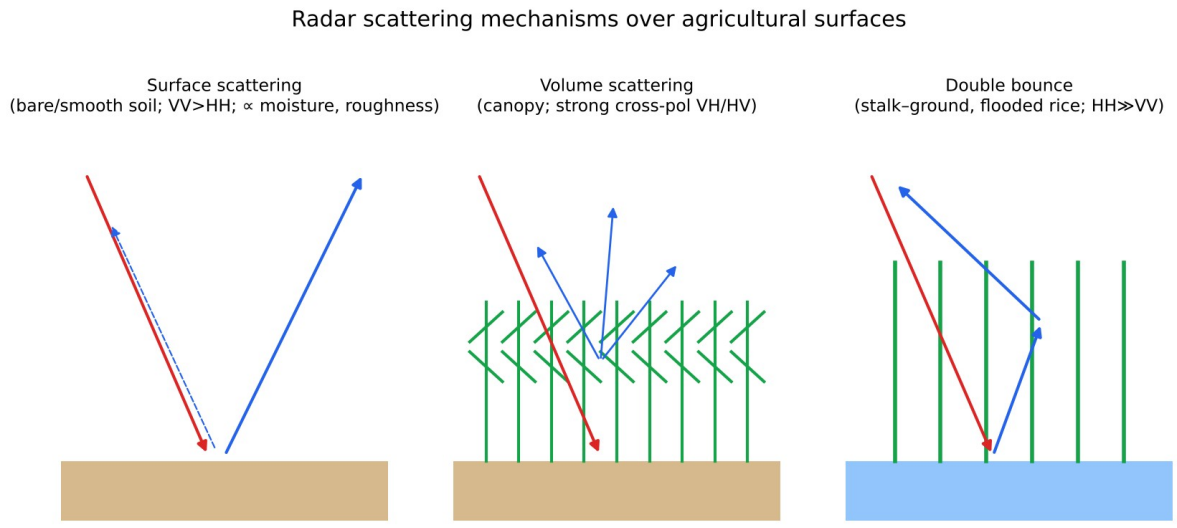


Figure 9. Surface, volume and double-bounce scattering mechanisms in agricultural fields (after Freeman–Durden decomposition concepts).

4.2 NISAR mission specifications

Parameter	Value
Partners	NASA/JPL (L-band SAR, reflector, GPS, data handling) and ISRO (S-band SAR, I-3K spacecraft bus, GSLV launch, operations)
Launch	30 July 2025, GSLV-F16, Satish Dhawan Space Centre, Sriharikota
Orbit	Sun-synchronous, dawn–dusk, about 747 km, 98.4° inclination
Repeat	12 days (about 6 days using ascending and descending passes)
Frequencies	L-band 1.257 GHz (24 cm), global coverage; S-band 3.2 GHz (9.4 cm), focused on India and selected sites
Antenna	12 m deployable mesh reflector; SweepSAR digital beamforming
Swath / resolution	~242 km swath; 3–10 m (mode dependent)
Polarisation	Single, dual and quad (full) polarimetric modes
Data	Free and open; through NASA ASF DAAC and ISRO Bhoonidhi
Mission life	3 years (nominal); fuel for longer

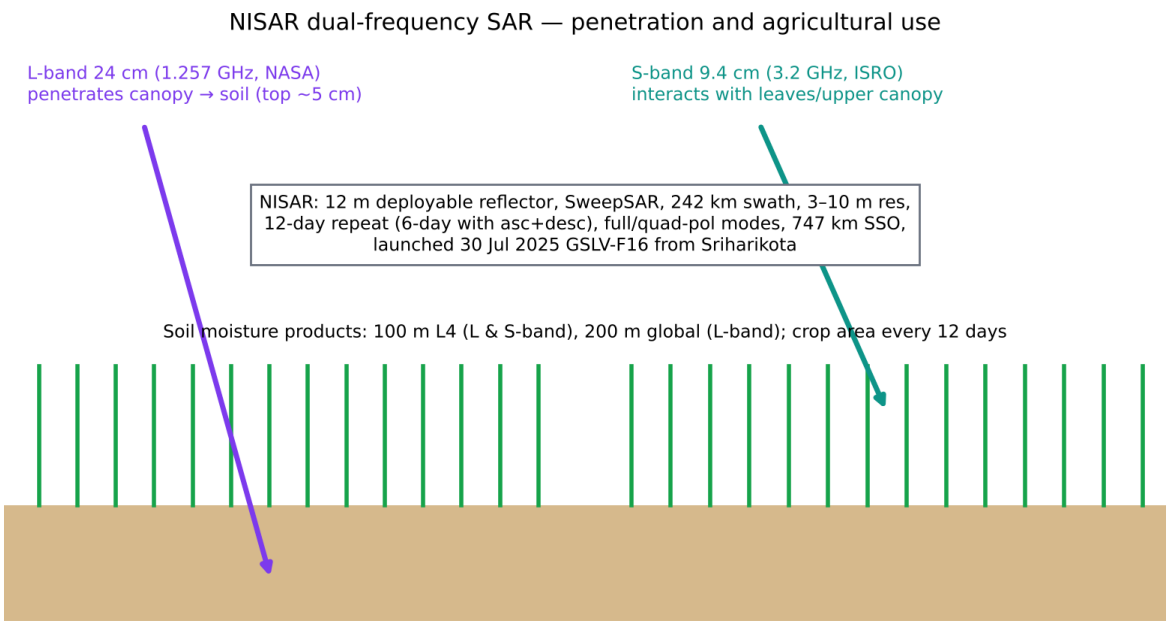


Figure 10. NISAR L-band (deeper penetration) and S-band (canopy) interaction with crops and soil.

4.3 Crop mapping with NISAR

Physical basis

- As a crop grows, **volume scattering** from leaves and stems raises cross-pol (HV/VH) backscatter, and attenuation changes co-pol.
- **Rice:** at puddling and transplanting, the flooded field reflects specularly, giving very low σ^0 (−22 to −26 dB in VH). As tillers grow, stem–water **double-bounce** raises HH strongly. HH/VV > 1 is diagnostic of rice.

- **Wheat and mustard:** vertical stems attenuate VV more than HH. The VH/VV ratio rises with biomass.
- **Sugarcane:** high, stable backscatter all year from a tall, dense canopy (L-band saturates later, at about 100 t/ha biomass).
- **L vs S vs C:** L-band sees stems and soil, and saturates late. S-band (like C) is sensitive to leaves and the upper canopy and helps separate leafy crops. Together they help discriminate crop types and growth stages.

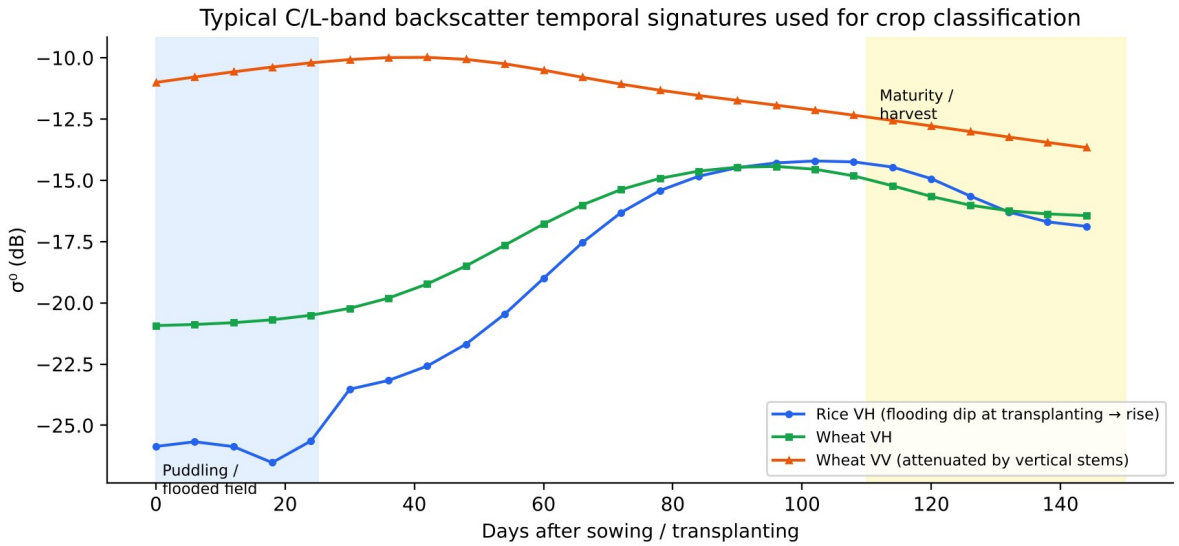


Figure 11. Backscatter time series for rice and wheat, showing the flooding dip at rice transplanting and the rise with canopy growth.

Workflow (as used for EOS-04 and Sentinel-1, adapted to NISAR)

1. Download **GCOV** (geocoded polarimetric covariance) or RSLC products.
2. Radiometric terrain correction (γ^0), speckle filtering and conversion to dB.
3. Build a **multi-temporal stack** (12-day composites) and derive features: HH, HV, VV, ratios, the **Radar Vegetation Index RVI = 8σHV/(σHH + σVV + 2σHV)**, polarimetric entropy and alpha, and coherence.
4. Collect training samples from field surveys, CROPIC/Digital Crop Survey geotagged photos or Bhuvan.
5. Classify with **Random Forest, SVM or a deep-learning model (1-D CNN / LSTM / Transformer on time series)**. Fuse with Sentinel-2 where cloud-free.
6. Validate with independent samples: OA, kappa and area-adjusted accuracy.
7. Estimate area by stratified sampling (FASAL approach) and update district and state acreage.

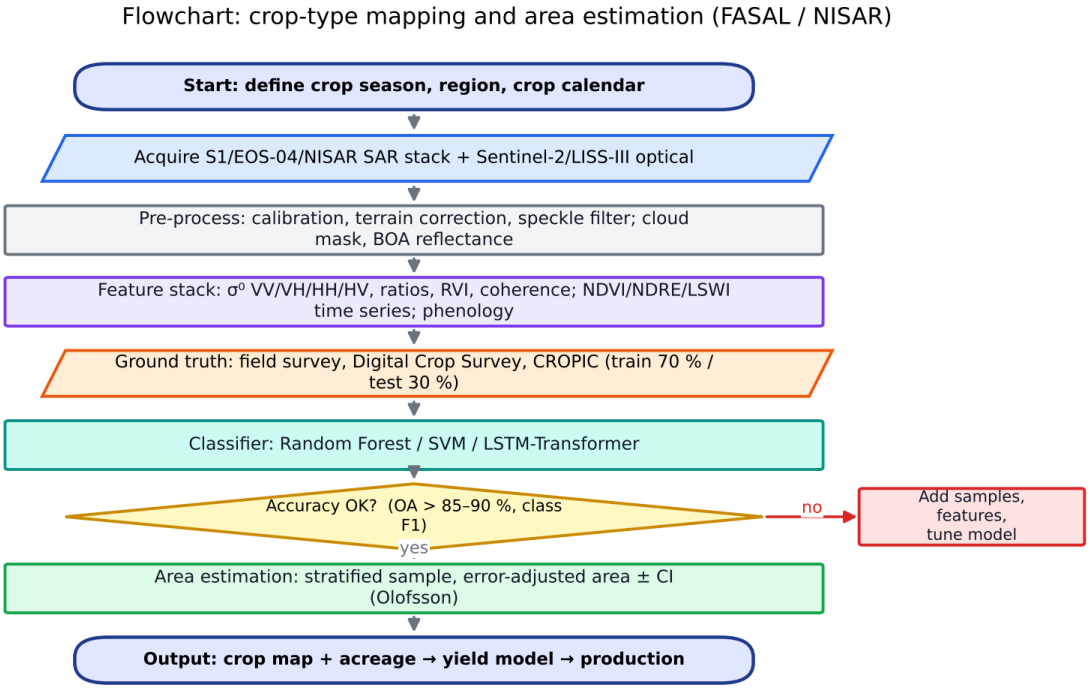


Figure 12. Flowchart of crop-type mapping and area estimation (FASAL / NISAR).

> **Official product:** The **NISAR Crop Area Product** (L3, planned) is a binary active-crop/no-crop map every 12 days at about 100 m, computed from the **coefficient of variation of HV backscatter** over time. Cropland shows high temporal variability, while forests and urban areas are stable.

4.4 Soil-moisture retrieval with NISAR

Physical basis

Backscatter from bare soil depends on the **dielectric constant** (which rises with volumetric water content, Topp's equation), **surface roughness** (RMS height s, correlation length l), the incidence angle and the wavelength. Vegetation adds its own scattering and attenuates the soil signal, so it has to be corrected for.

$$\text{Topp et al. (1980):}\quad \theta v = -5.3\times10^{-2} + 2.92\times10^{-2}\cdot\varepsilon - 5.5\times10^{-4}\cdot\varepsilon^2 + 4.3\times10^{-6}\cdot\varepsilon^3$$

The Water Cloud Model (Attema & Ulaby 1978)

$$\sigma^0 = A\cdot V_1\cdot\cos\theta\cdot(1-\gamma^2) + \gamma^2\cdot\sigma^0\text{soil}, \qquad \gamma^2 = \exp(-2\cdot B\cdot V_2 / \cos\theta)$$

V_1 and V_2 are vegetation descriptors (LAI, NDVI, vegetation water content or biomass). A and B are crop-specific coefficients fitted against field data. γ^2 is the two-way canopy transmissivity. Once the vegetation term is removed, σ^0soil is inverted for moisture with a soil-scattering model.

Table 6. Soil-moisture retrieval approaches

Approach	Idea	Pros and cons
Change detection (Wagner/TU-Wien)	SM ∝ (σ^0t − σ^0dry)/(σ^0wet − σ^0dry); roughness assumed static over time	Simple and robust; gives relative SM; needs a long time series

Approach	Idea	Pros and cons
Empirical regression	SM = a + b·σ^0 calibrated on field data	Site-specific
Semi-empirical (Oh 1992, Dubois 1995)	Co-pol ratio and cross-pol relations with ε and ks	Works for bare soil; limited validity range
Physical (IEM, AIEM)	Integral Equation Model from EM theory	Accurate but needs roughness inputs
Machine learning (RF, ANN)	Learns σ^0 + NDVI + soil texture + DEM → SM	Flexible; needs many in-situ points
Active–passive fusion (SMAP-style)	Disaggregates coarse radiometer SM using fine SAR σ^0	Fine resolution plus radiometer accuracy

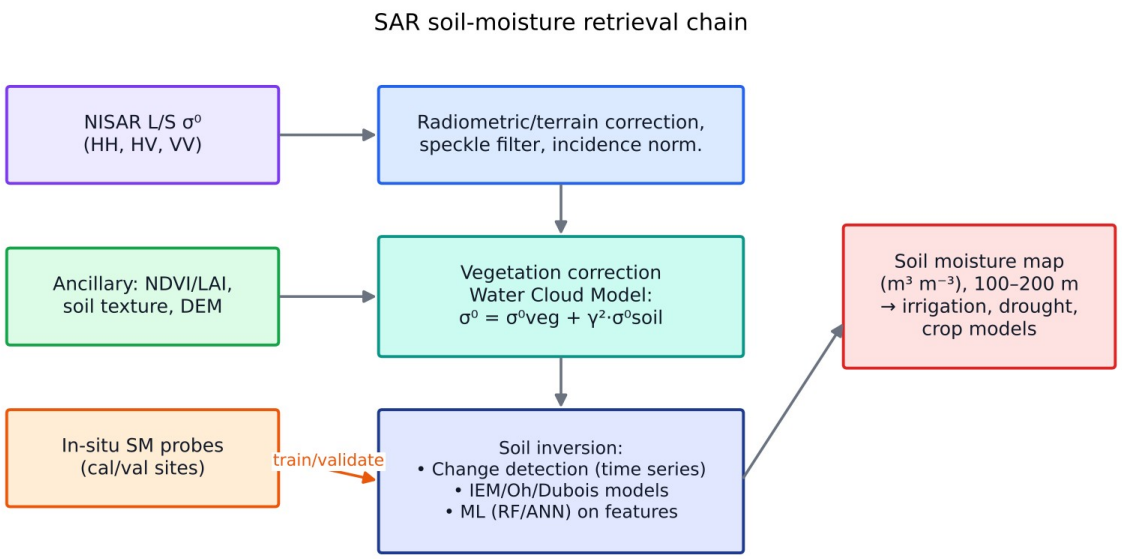


Figure 13. SAR soil-moisture retrieval chain from backscatter to soil-moisture map.

NISAR soil-moisture products

- **NASA L3 global soil moisture:** about 200 m (L-band), 12-day, top ~5 cm, multiple algorithms, requirement of ±0.06 $m^3\ m^{-3}$.
- **ISRO L4 (NRSC): 100 m L- and S-band soil-moisture products** over India's agro-climatic regions, two observations every 12 days, distributed through **Bhoonidhi**. Preliminary products are being evaluated at core cal/val sites, and field-scale products are in progress.
- **L-band** penetrates vegetation better and samples deeper soil. **S-band** gives finer surface sensitivity and detail. Combining the two improves retrieval under crops.

4.5 Other agricultural uses of NISAR

- **Above-ground biomass** (L-band) for sugarcane and agroforestry, relevant to carbon MRV.
- **Flood inundation** and crop-loss assessment for PMFBY localized calamities. Inundation under the canopy shows as an HH double-bounce increase.

- **Tillage, sowing and harvest dates** from InSAR coherence drops (relevant to CA detection and residue burning).
- **Land subsidence** from groundwater depletion in Punjab, Haryana and Gujarat (InSAR), which links irrigation and aquifer health.
- **Freeze/thaw and snow** in Himalayan catchments (irrigation water supply).

4.6 Interview questions on NISAR

Q. Why is L-band better than C-band for soil moisture under crops?

A. At 24 cm wavelength the canopy is more transparent: attenuation is lower and L-band reaches the soil through moderate biomass (up to about 5 kg/m² vegetation water content). C-band saturates over dense crops and mostly measures the canopy.

Q. What does NISAR's S-band add for India?

A. S-band (9.4 cm) is sensitive to leaves and the upper canopy, which helps separate crop types and early growth stages and improves surface-soil-moisture detail. It was built by ISRO, and S-band acquisitions are planned mainly over India.

Q. How will you map kharif rice using NISAR?

A. Use a 12-day HH/HV time series. Look for the flooding minimum at transplanting, followed by a rapid rise in HH (double bounce) and HV (volume scattering). Classify with Random Forest on temporal features and validate with field data. Coherence and the HH/VV ratio help separate rice from other crops.

Q. Explain the Water Cloud Model in one line.

A. It treats the canopy as a uniform cloud of water droplets: total backscatter = vegetation volume scattering + two-way-attenuated soil backscatter, so the soil term can be isolated and inverted for moisture.

Q. Why does change detection work for soil moisture?

A. Over weeks, roughness and vegetation in a pixel change slowly compared with moisture. Scaling each observation between the driest and wettest observed backscatter therefore gives relative saturation, with no need for roughness parameters.

5. Retrieving crop and soil parameters

One-line definition: parameter retrieval is the quantitative estimation of biophysical and biochemical variables (LAI, chlorophyll, nitrogen, water content, phenology, soil moisture, salinity) from remotely sensed radiance. It uses empirical, physical (radiative-transfer) or hybrid machine-learning models.

5.1 Three families of retrieval methods

Family	How it works	Strengths and weaknesses
Parametric / empirical	Regression of a VI against field data (e.g. LAI = a·e^(b·NDVI))	Simple and fast; site-, sensor- and crop-specific; saturation
Non-parametric ML	RF, SVR, GPR, ANN, CNN on bands and VIs	Captures non-linearity; needs large training sets; less transferable
Physically based (RTM inversion)	Invert PROSPECT+SAIL by LUT or optimisation	Generic, transferable; ill-posed (several parameter sets give the same spectrum); needs priors
Hybrid	Train ML (GPR/ANN) on RTM-simulated spectra, then apply to imagery	Generic and fast; basis of the SNAP S2 biophysical processor and ESA/ISRO operational LAI

5.2 Crop parameters

(a) Phenology

- **What:** dates of key stages such as sowing or emergence (SOS), peak or heading, senescence and harvest (EOS), and the length of season (LOS).
- **Data:** NDVI/EVI time series (MODIS, Sentinel-2, HLS), SAR backscatter and coherence (Sentinel-1, NISAR), and SIF.
- **Smoothing:** Savitzky–Golay, Whittaker, harmonic (HANTS) or double-logistic fitting. Use TIMESAT or the phenofit/greenbrown R packages.
- **Extraction:** threshold (e.g. 20 % of amplitude), maximum rate of change (first derivative) or curvature-change methods.
- **Uses:** crop calendars for FASAL, sowing-date-based crop models, insurance (prevented sowing), and detecting late or early sowing in CA fields.

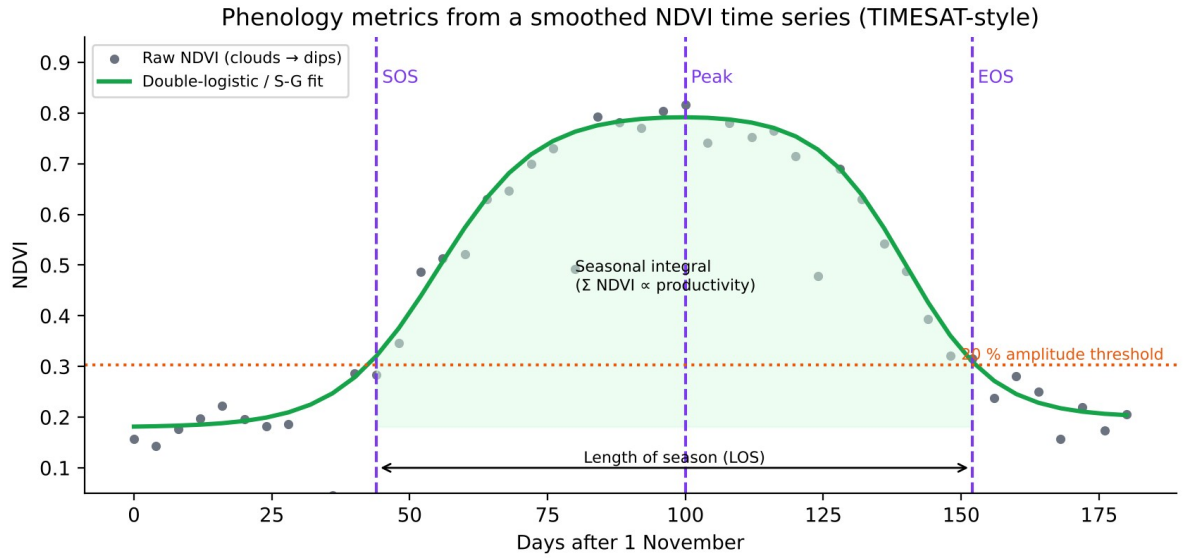


Figure 14. Phenology metrics (SOS, peak, EOS, LOS, seasonal integral) from a smoothed NDVI time series.

(b) Leaf Area Index (LAI)

- **Definition:** one-sided green leaf area per unit ground area ($m^2\ m^{-2}$). It drives light interception (fAPAR = 1 − e^(−k·LAI), Beer–Lambert law, k ≈ 0.5).
- **Ground measurement:** destructive sampling, LAI-2200 plant canopy analyser (gap fraction), AccuPAR ceptometer and digital hemispherical photography.
- **Remote sensing:** NDVI saturates at LAI ≈ 3. Red-edge indices (NDRE, CIre, MTCI), EVI and SAR (L-band) extend sensitivity. RTM-hybrid retrieval (SNAP) gives LAI, fAPAR and fCover from Sentinel-2.
- **Products:** MODIS MCD15 (500 m), Copernicus GLS (300 m) and the ISRO Resourcesat LAI product.

(c) Leaf and canopy water

- **Variables:** EWT/Cw (equivalent water thickness, $g\ cm^{-2}$), FMC and CWC (canopy water content = Cw × LAI).
- **Absorption features:** 970, 1200, **1450** and **1940 nm**. Multispectral indices use SWIR: NDWI(Gao) = (NIR − SWIR1)/(NIR + SWIR1), MSI = SWIR1/NIR and NDII.
- **Thermal:** canopy temperature and CWSI = (Tc − Twet)/(Tdry − Twet), from IR thermometers, UAV thermal cameras or Landsat TIR. Water stress closes stomata, which warms the canopy.
- **Microwave:** vegetation water content from L-band (the τ-ω model in SMAP) and VH backscatter.

(d) Chlorophyll and nitrogen

- **Chlorophyll (Cab, $\mu g\ cm^{-2}$):** red-edge position (REP), MTCI, CIred-edge, TCARI/OSAVI (reduces LAI influence), and ground chlorophyll meters (SPAD-502, atLEAF).

- **Nitrogen:** most leaf N is in Rubisco and chlorophyll, so canopy chlorophyll content (CCC = Cab × LAI) is a strong proxy for canopy N. Use NDRE, CCCI, the leaf colour chart (LCC) and GreenSeeker NDVI to compute the N sufficiency index NSI = NDVI(field)/NDVI(N-rich strip).
- **Hyperspectral:** narrow bands at 550, 705, 720 and 750 nm and PLSR models. Crude protein and N content can be estimated directly from SWIR protein features at 1510, 2054 and 2172 nm.
- **Application:** in-season N top-dressing algorithms (Raun et al. 2002: NFOA = (YP_0 × RI − YP_0) × %N/NUE), i.e. precision N under VRT.

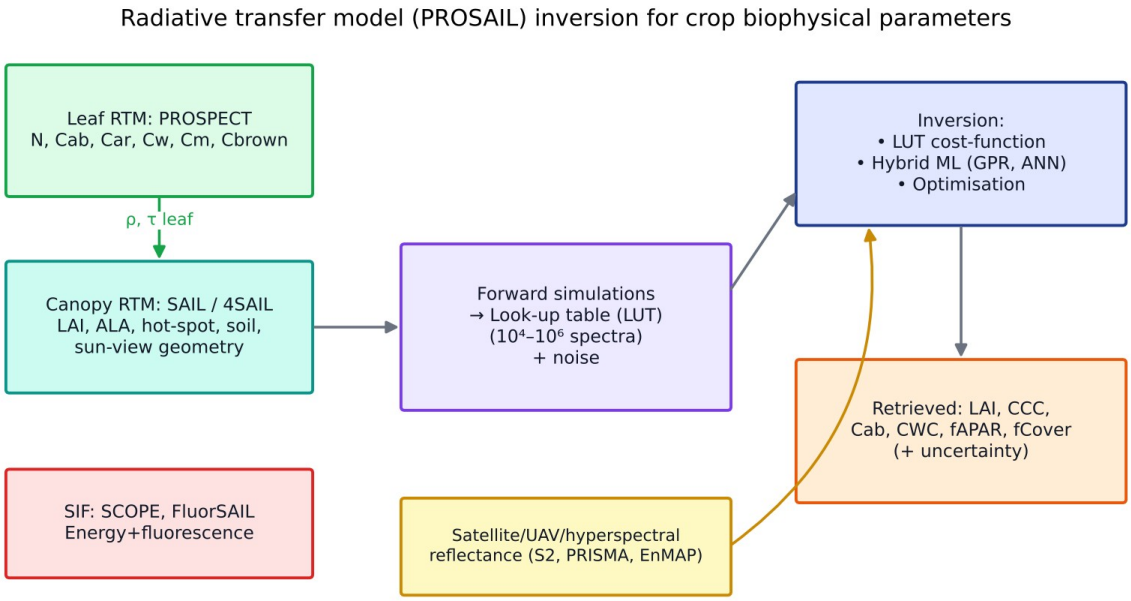


Figure 15. PROSAIL radiative-transfer inversion workflow for crop biophysical and biochemical parameter retrieval (after Jacquemoud et al. 2009; Verrelst et al. 2015).

5.3 Radiative transfer models (RTMs)

Table 7. Common RTMs for crops and soil

Model	What it simulates	Inputs
PROSPECT (-4/-5/-D/-PRO)	Leaf reflectance and transmittance 400–2500 nm (plate model)	N (structure), Cab, Car, Anth, Cw, Cm (dry matter), Cbrown; PROSPECT-PRO separates protein and CBC
SAIL / 4SAIL	Canopy bidirectional reflectance (turbid medium, 4-stream)	LAI, average leaf angle (ALA), hot spot, soil reflectance, solar and view zenith, relative azimuth
PROSAIL	PROSPECT + SAIL coupled; the most widely used crop RTM	All of the above
SCOPE (van der Tol 2009)	Energy balance + photosynthesis + fluorescence + reflectance	Vcmax, meteorology, LAI, Cab; used for FLEX
DART, LESS	3-D ray tracing for heterogeneous canopies	3-D scene; computationally heavy
Hapke, SOILSPECT, MARMIT	Soil reflectance as a function of moisture and roughness	Soil SSA, moisture film thickness

Model	What it simulates	Inputs
WCM, MIMICS, IEM	Microwave scattering from canopy and soil	Biomass, ε, roughness

Inversion strategies and the ill-posed problem

- **LUT inversion:** simulate 10^4–10^6 spectra, then for each pixel pick the n best matches (RMSE cost function) and take their mean or median as the solution.
- **Ill-posedness:** LAI and ALA compensate for each other. Constrain the inversion with prior distributions, add noise, use multi-angle or temporal information, or regularise.
- **Hybrid ML:** Gaussian Process Regression gives an uncertainty estimate with each prediction and is the basis of the ARTMO toolbox (Verrelst et al.) for Sentinel-2/3, PRISMA and EnMAP.

5.4 Soil parameters

(a) Soil moisture

Covered in detail in Chapter 4 (SAR). Other methods: **optical/thermal triangle** (TVDI from Ts–NDVI), **OPTRAM** (STR = $(1 - SWIR)^2/(2 \cdot SWIR)$ vs NDVI), **passive microwave** (SMAP 9–36 km, SMOS), **GNSS-R** (CYGNSS, ISRO EOS-08) and **ground** (TDR, FDR capacitance probes, gravimetric, cosmic-ray neutron COSMOS).

(b) Salinity and alkalinity (sodicity)

- **Definitions:** saline soil has ECe > 4 dS m^{-1}, pH < 8.5 and ESP < 15. Sodic (alkali) soil has ESP > 15, pH > 8.5 and low EC. Saline-sodic soils are both. India has about **6.7 Mha** of salt-affected soils (ICAR-CSSRI), including Haryana, UP, Punjab, Gujarat and Rajasthan.
- **Spectral behaviour:** salt crusts raise VIS–NIR reflectance (white surface). Sodic soils show dispersed, crusted surfaces with high reflectance and poor vegetation. Evaporite minerals (gypsum and halite) have SWIR absorption features at 1.4, 1.9 and 2.2 µm.
- **Indices:** $SI1 = \sqrt{(G \cdot R)}$, $SI2 = \sqrt{(G^2 + R^2 + NIR^2)}$, $SI3 = \sqrt{(B \cdot R)}$, $NDSI = (R - NIR)/(R + NIR)$, the Brightness Index $BI = \sqrt{(R^2 + NIR^2)}$, and the vegetation-based CRSI (canopy response salinity index). For SWIR, (SWIR1 − SWIR2)/(SWIR1 + SWIR2).
- **Approach:** combine indices with EM-38 apparent electrical conductivity surveys, ECe lab data and ML (RF, Cubist) in a digital soil mapping framework. Multi-year maximum-NDVI or yield-gap maps are also used because crops integrate root-zone salinity.
- **Radar:** salt increases the imaginary part of ε, which raises backscatter in wet saline soils.

(c) Soil organic carbon, texture and nutrients

- **Diffuse reflectance spectroscopy (VNIR–SWIR, 350–2500 nm):** SOC absorbs broadly across the visible range (darker soils) and at 1700/2300 nm. Clay minerals show OH features at 1400, 1900 and 2200 nm, and iron oxides at 480 and 900 nm.
- **Models:** PLSR, Cubist, SVR, RF and memory-based learning, using soil spectral libraries (the Indian soil spectral library effort; ICRAF/ISRIC; LUCAS in Europe).

- **Bare-soil composites:** multi-year Sentinel-2 bare-soil composites (SCMaP, GEOS3) to map SOC and texture in cropland.

5.5 Digital soil mapping (DSM): Dr. Bappa Das's line of work

Definition: DSM creates spatial soil information systems by numerical models that infer the spatial and temporal variation of soil properties from observations and environmental covariates (McBratney et al. 2003).

$$\text{scorpan:}\quad S = f(s, c, o, r, p, a, n) + \varepsilon$$

s = soil (existing data), c = climate, o = organisms/vegetation (NDVI), r = relief (DEM derivatives: slope, TWI, curvature), p = parent material, a = age, n = spatial position. The ε term is modelled by kriging the residuals (**regression kriging**).

1. **Soil sampling:** conditioned Latin hypercube (cLHS) or stratified random design. Analyse SOC, pH, EC, N, P, K and texture in the laboratory, or predict them from spectra.
2. **Covariate stack:** Sentinel-2 and Landsat bare-soil and vegetation composites, SRTM/Cartosat DEM derivatives, climate (CHIRPS, IMD gridded data), geology and land use.
3. **Model:** Random Forest, Quantile Regression Forest (for uncertainty), Cubist, XGBoost or regression kriging. In India, regression kriging is the most common, followed by RF and QRF (Geoderma Regional 2022 review).
4. **Validation:** k-fold or spatial CV, reporting RMSE, R^2, Lin's concordance coefficient (CCC) and the prediction interval coverage probability (PICP).
5. **Products:** GlobalSoilMap specification at 6 depths (0–5, 5–15, 15–30, 30–60, 60–100 and 100–200 cm) with 90 % prediction intervals, used for soil-health-card upscaling and fertiliser recommendations.

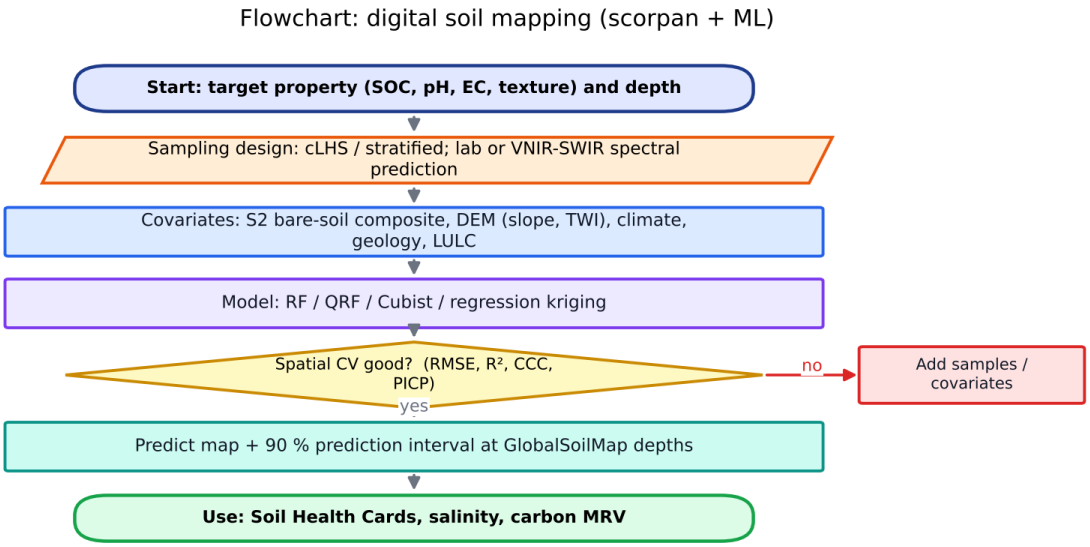


Figure 16. Flowchart of digital soil mapping with the scorpan framework and machine learning.

Dr. Bappa Das (ICAR) works on proximal and remote **spectroscopy plus machine learning** for soil and crop properties, and on DSM. Recent examples include ML-based soil organic carbon mapping in the Indian Himalayan region (Kalambukattu, Kumar, Das, Roy et al. 2025, *Discover Soil*) and VNIR–SWIR spectral prediction of soil properties with ML-model comparison. In the interview, cite the method (spectral library → ML → spatial covariates → uncertainty maps) rather than a specific title unless you have read the paper.

5.6 Where parameter retrieval is heading

- **Hyperspectral missions:** PRISMA (Italy), EnMAP (Germany), EMIT, ISRO's planned hyperspectral missions and **CHIME** (ESA Copernicus expansion) for operational N, protein and soil properties.
- **Thermal missions: TRISHNA** (ISRO–CNES, ~57 m TIR, near-daily), LSTM (ESA) and SBG for ET and water stress at field scale.
- **Fluorescence:** FLEX for photosynthetic efficiency.
- **Foundation models and self-supervised learning** (Prithvi, SatMAE, Clay) fine-tuned with few labels, plus **physics-informed ML** that embeds RTM or crop-model constraints.
- **Data assimilation** of retrieved LAI, SM and SIF into crop models (digital twins).
- **UAV + satellite upscaling** and **uncertainty-aware products** (GPR, QRF, Bayesian inversion).

5.7 Interview questions

Q. Why does NDVI saturate and how do you overcome it?

A. At LAI > 3, red reflectance is almost fully absorbed and stops changing, while NIR keeps increasing, so the normalised ratio flattens. Red-edge indices (NDRE, CIre), EVI, wide-dynamic-range VI, SAR backscatter or RTM inversion stay sensitive at high biomass.

Q. How would you map soil salinity in Haryana?

A. Use multi-date Sentinel-2 in the dry season (bare soil and salt crust) and compute SI, NDSI and BI. Add the multi-year maximum NDVI of crops and DEM-derived wetness. Train RF/QRF on EC measurements from field and EM-38 surveys. Validate with spatial CV and produce maps of salinity classes with uncertainty.

Q. What is the difference between LAI and fAPAR?

A. LAI is a structural quantity (leaf area per ground area). fAPAR is a radiative quantity (fraction of PAR absorbed), related to LAI non-linearly through Beer's law. Production-efficiency models such as LUE use fAPAR.

Q. What does PROSPECT simulate?

A. Leaf hemispherical reflectance and transmittance from 400 to 2500 nm as a function of the structure parameter N, chlorophyll, carotenoids, water, dry matter and brown pigments.

6. Evapotranspiration: SEBAL, METRIC, ALEXI and scaling

One-line definition: evapotranspiration (ET) is the combined loss of water from the soil surface (evaporation) and from plants (transpiration). Energetically it is the latent heat flux λET, which links the water, energy and carbon cycles.

6.1 Definitions you must keep distinct

Term	Meaning
ET_0 (reference ET)	ET from a hypothetical well-watered grass reference (0.12 m height, rs = 70 s m^{-1}, albedo 0.23), computed with FAO-56 Penman–Monteith. Depends on weather only.
ETr	Alfalfa reference ET (ASCE), about 1.2 × ET_0. Used by METRIC.
ETc (crop ET, potential)	ETc = Kc × ET_0 under standard (no-stress) conditions. Kc follows the crop stage (initial, development, mid, late).
AET / ETa (actual ET)	ETc,adj = Ks × Kc × ET_0 under real water or salinity stress. This is what SEB models estimate.
EF (evaporative fraction)	EF = λET/(Rn − G) = λET/(λET + H)
ETrF (= Kc × Ks)	ETrF = ETinst/ETr,inst. METRIC's fraction of reference ET. Your notes write it as AET/ET_0.

$$\text{FAO-56 PM:}\quad ET_0 = [0.408\cdot\Delta\cdot(Rn - G) + \gamma\cdot(900/(T + 273))\cdot u_2\cdot(es - ea)] \,/\, [\Delta + \gamma\cdot(1 + 0.34\cdot u_2)]$$

6.2 Measuring ET on the ground

- **Weighing lysimeter:** the most accurate method (mass balance). It is point-scale and expensive. IARI has lysimeter facilities.
- **Bowen-ratio energy balance (BREB):** β = H/λET = γ·ΔT/Δe, so λET = (Rn − G)/(1 + β).
- **Eddy covariance (EC):** a direct turbulent-flux measurement (Section 6.8).
- **Scintillometer (LAS):** path-averaged H over 0.5–5 km. Matches satellite pixels well.
- **Sap-flow sensors:** transpiration of individual plants.
- **Soil water balance:** ET = P + I − D − R − ΔS, using neutron probes or TDR.

6.3 The surface energy balance (SEB)

$$Rn = G + H + \lambda ET \quad \Rightarrow \quad \lambda ET = Rn - G - H$$

$$Rn = (1 - \alpha)\cdot Rs\downarrow + RL\downarrow - RL\uparrow - (1 - \varepsilon_0)\cdot RL\downarrow, \quad RL\uparrow = \varepsilon_0\cdot\sigma\cdot Ts^4$$

$$H = \rho a\cdot cp\cdot dT / rah, \quad rah = \ln(z_2/z_1) / (k\cdot u^*) \ \text{(corrected for stability with Monin–Obukhov theory)}$$

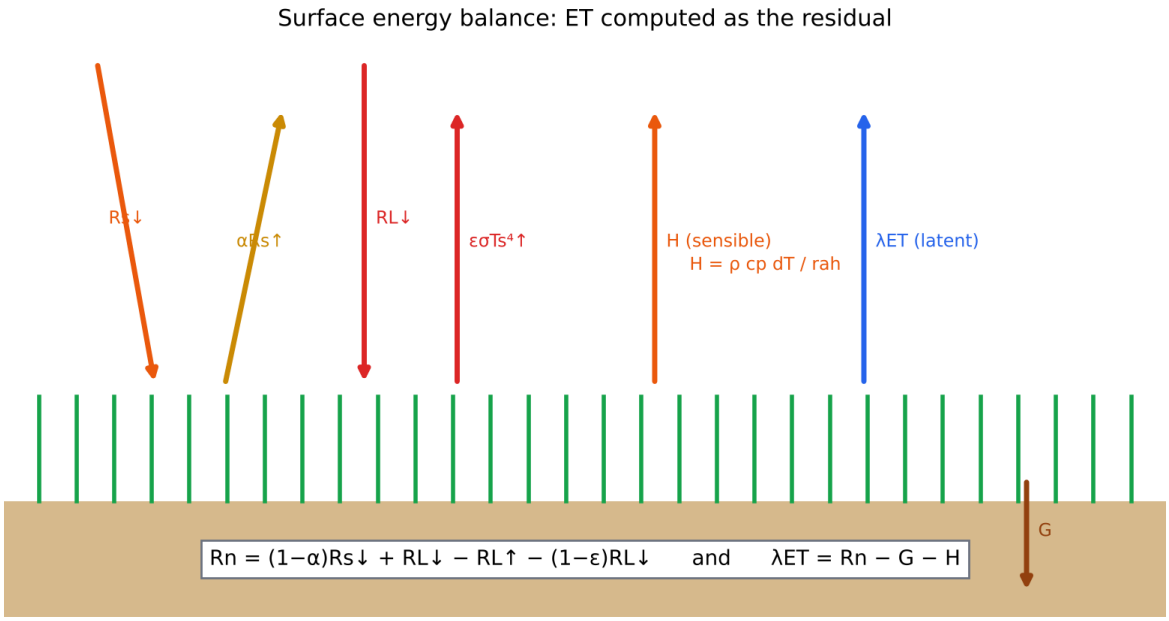


Figure 17. Surface energy balance components; ET is obtained as the residual.

Satellites measure albedo (VNIR) and **land surface temperature Ts (TIR)**. Rn and G are computed fairly reliably. **H is the difficult term** because it needs the near-surface air-temperature gradient dT and the aerodynamic resistance rah. The single-source models (SEBAL, METRIC) and the two-source models (TSEB/ALEXI) differ mainly in how they estimate H.

6.4 SEBAL (Bastiaanssen et al. 1998)

- **Full name:** Surface Energy Balance Algorithm for Land.
- **Key idea:** dT (the surface–air temperature difference) is a **linear function of Ts**, dT = a + b·Ts. The coefficients come from two **anchor pixels** in the scene, so the absolute bias in Ts largely cancels.
- **Cold pixel:** well-watered, full-cover crop (or water body). SEBAL assumes **H ≈ 0**, so λET = Rn − G.
- **Hot pixel:** dry bare field with **λET ≈ 0**, so H = Rn − G.
- **Iteration:** start with neutral rah from the wind speed at a 200 m blending height, compute H, then the Monin–Obukhov length L, stability corrections ψm and ψh, then u^* and rah, and repeat until convergence (5–10 iterations).
- **Advantage:** minimal ground data (wind speed) and self-calibrating within each image.
- **Limitations:** choosing anchor pixels is subjective; it needs both wet and dry extremes in the scene; the assumption of H = 0 at the cold pixel underestimates ET under advection, as in hot and dry NW India.

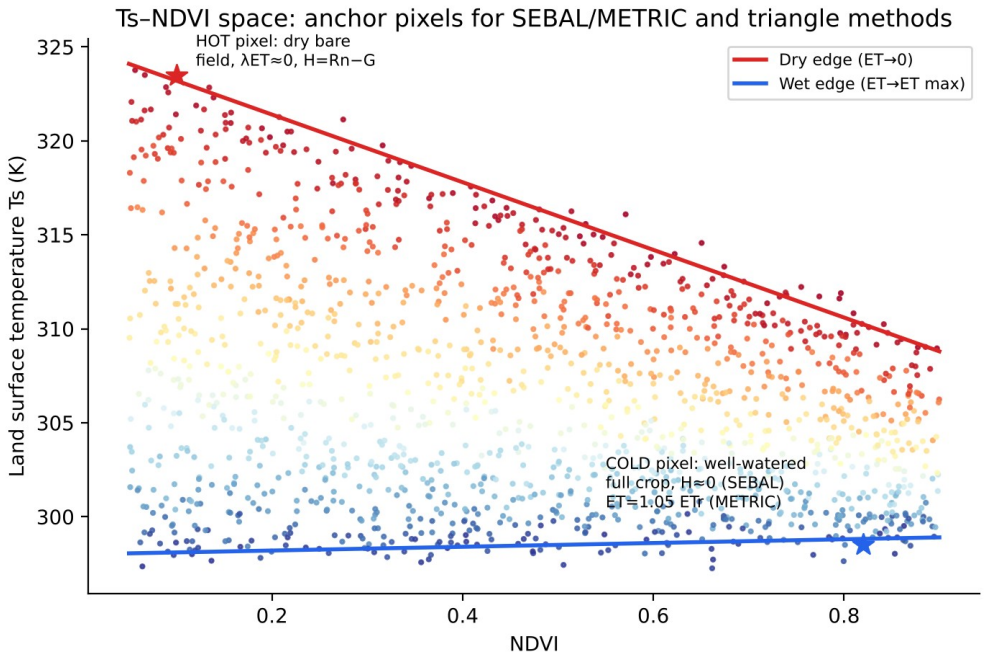


Figure 18. Ts–NDVI scatter showing hot and cold anchor pixels and the dry and wet edges (concept after Bastiaanssen et al. 1998; Allen et al. 2007).

6.5 METRIC (Allen et al. 2007)

- **Full name:** Mapping EvapoTranspiration at high Resolution with Internalized Calibration.
- **Derived from SEBAL**, with two important changes:
- **(1) Cold pixel:** λET = **1.05 × ETr** (alfalfa reference) instead of H = 0. This accounts for advection, and hourly weather data calibrates it.
- **(2) Scaling:** uses **ETrF = ETinst/ETr,inst** (the reference ET fraction) to scale to daily and seasonal ET, instead of SEBAL's EF.
- Hot pixel: λET from a soil water balance (FAO-56) to account for residual soil evaporation after rain.
- Corrects for slope and aspect in mountainous terrain, uses Landsat 30 m data (TIR sharpened from 100 m), and is used for **water-rights and irrigation management** in the USA.
- Implementations: EEFlux, OpenET (Google Earth Engine) and geeSEBAL.

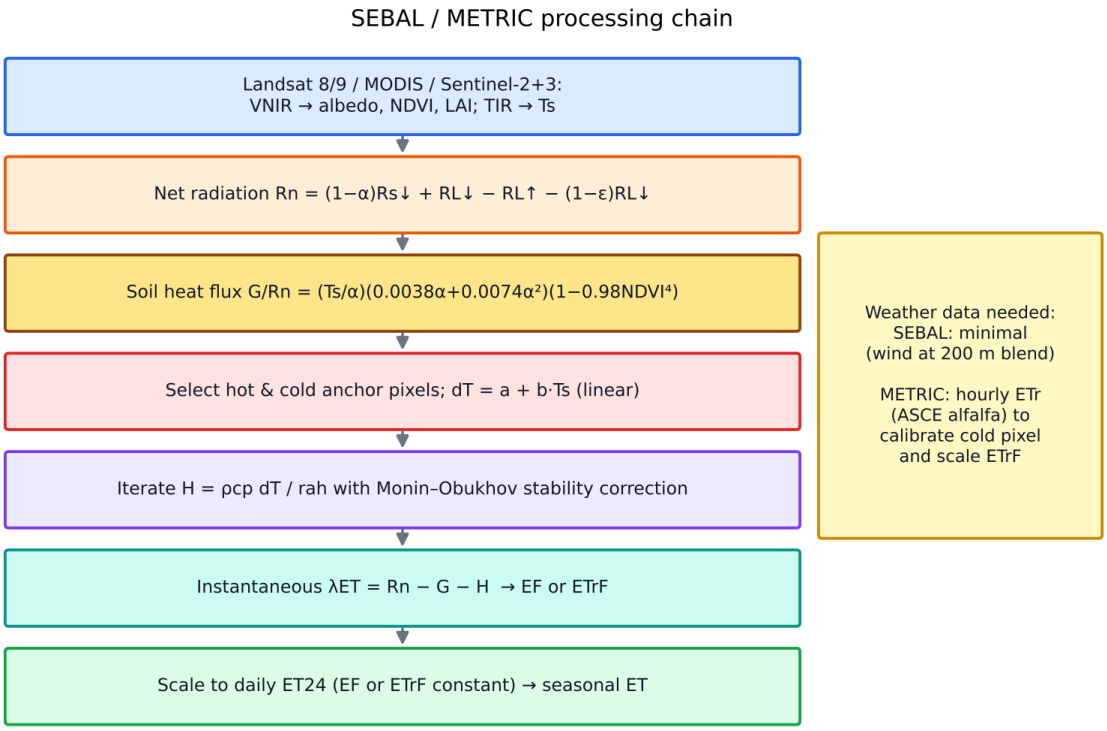


Figure 19. Processing chain of SEBAL and METRIC.

6.6 ALEXI and DisALEXI (Anderson et al. 1997, 2007)

- **Full name:** Atmosphere–Land EXchange Inverse model.
- **Two-source energy balance (TSEB, Norman et al. 1995):** separates soil and canopy, Ts^4 = fc·Tc^4 + (1 − fc)·$Ts,soil^4$. Canopy transpiration is first estimated by **Priestley–Taylor**, λEc = 1.26·fG·Δ/(Δ + γ)·Rnc, and soil evaporation is the residual.
- **Time-differencing:** uses the **morning rise in Ts** between two geostationary observations (about 1.5 h after sunrise and 1.5 h before local noon), from GOES or MSG (INSAT-3D could be used for India).
- **ABL coupling:** the rise in Ts drives sensible-heat growth of the atmospheric boundary layer (a slab model, McNaughton & Spriggs). This closes the equations **without air-temperature measurements**, which are the main uncertainty in single-source models.
- Robust to absolute Ts errors (emissivity, atmospheric correction) because it uses the **temporal change**.
- **DisALEXI:** disaggregates the coarse (3–10 km) ALEXI flux to 30 m (Landsat) or 1 km (MODIS) using local TSEB and an ALEXI-derived air temperature.
- **STARFM fusion:** daily 30 m ET by fusing Landsat and MODIS (Cammalleri et al. 2014).
- **ESI (Evaporative Stress Index)** = 1 − AET/PET anomaly, used for flash-drought early warning (the US Drought Monitor uses it).

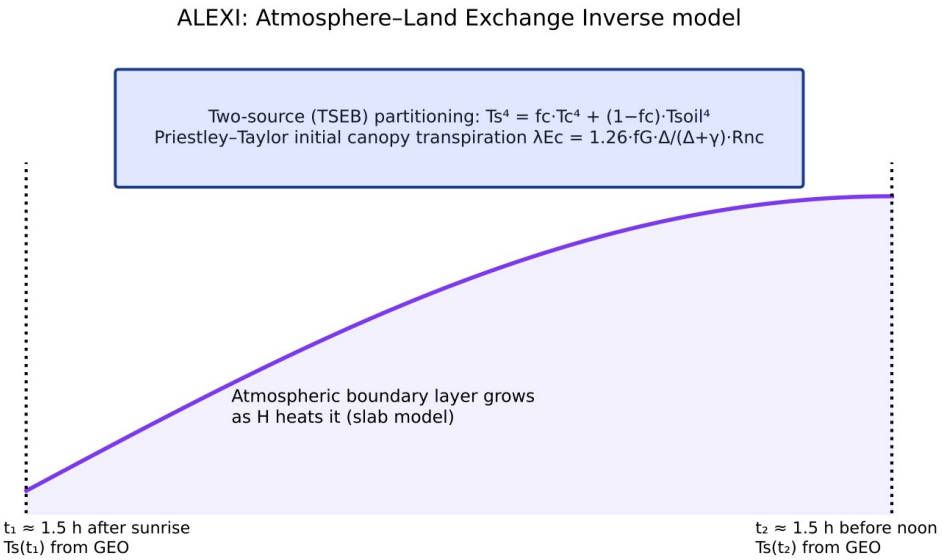


Figure 20. ALEXI concept: two-source partitioning and boundary-layer growth driven by the morning rise in surface temperature (after Anderson et al. 1997).

Table 8. SEB-based ET models compared

Aspect	SEBAL	METRIC	ALEXI / DisALEXI	SSEBop / SEBS
Source	Single	Single	Two (soil + canopy)	Single
H estimation	dT = a + bTs; anchor pixels	Same, cold pixel calibrated to ETr	Time-rate of Ts + ABL model	SSEBop: predefined dT from clear-sky energy; SEBS: kB^{-1} roughness physics
Weather needs	Minimal	Hourly ETr	Morning sounding, GEO Ts	Ta grid (SSEBop) / full meteorology (SEBS)
Daily scaling	EF constant	ETrF constant	Rs (insolation) ratio	ETf × ET_o
Resolution	Landsat/MODIS	Landsat 30 m	3–10 km → 30 m	MODIS/Landsat
Weakness	Anchor choice, advection	Anchor choice	Complexity, GEO data	Ta and roughness errors

6.7 Scaling instantaneous ET to daily ET

Polar orbiters such as Landsat (≈10:30) and MODIS Terra (10:30) and Aqua (13:30) give a **single snapshot** of λETinst (W m^{-2}). Irrigation and crop-water-use studies need ET24 (mm day^{-1}) and seasonal ET.

> **Your notes' answer:** The ratio of actual ET to reference ET (AET/ET_o, or ETrF) for a field stays **nearly constant through the day**, because both respond in the same way to the diurnal cycle of radiation and vapour-pressure deficit. So the ratio at overpass time is multiplied by the **24-hour reference ET from a weather observatory** to give daily ET.

$$\text{METRIC:}\quad ET24 = ETrF_inst \times ETr,24 \qquad \text{where}\quad ETrF_inst = ETinst \,/\, ETr,inst$$

$$\text{SEBAL (EF method):}\quad ET24 = 86400 \cdot EF \cdot (Rn24 - G24) / \lambda, \qquad G24 \approx 0$$

Table 9. Instantaneous-to-daily scaling methods

Method	Assumption	Comment
Evaporative fraction (EF)	EF = λET/(Rn − G) is constant through daytime (Shuttleworth 1989; Crago 1996, 'self-preservation')	Underestimates ET by 5–10 % under advection; EF has a slight U-shape through the day
Reference ET fraction (ETrF / Kc)	ETa/ETr is constant	Captures advection and wind effects; needs hourly weather; best in arid irrigated regions
Solar radiation ratio	ET24/ETinst = Rs24/Rs,inst	ALEXI uses it; simple, needs only insolation
Sine-function / Jackson	Diurnal ET follows a sine curve between sunrise and sunset	Good on clear days only
Temporal interpolation between overpasses	Linear or spline interpolation of ETrF, cloud gaps filled with water balance	Seasonal ET (METRIC)

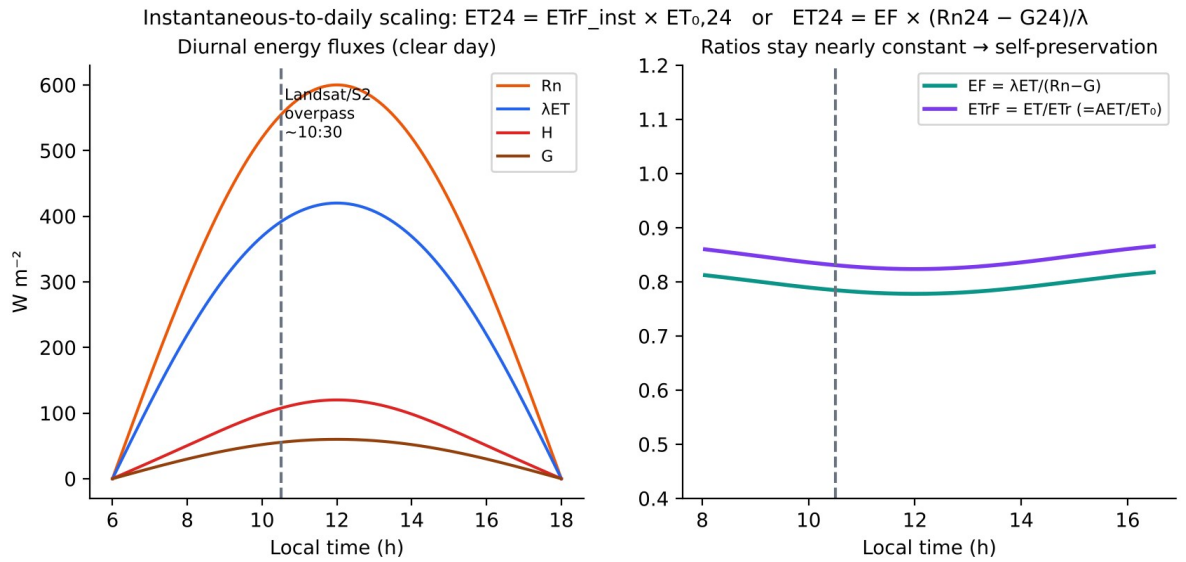


Figure 21. Diurnal course of energy fluxes and the near-constancy of EF and ETrF, the basis of daily scaling (after Crago 1996; Allen et al. 2007). Illustrative simulated curves.

6.8 Eddy covariance (EC) and NRSC's work

Principle: turbulent eddies carry scalars (heat, water vapour, CO_2) vertically. Using Reynolds decomposition ($x = \bar{x} + x'$), the vertical flux is the covariance of fluctuations in vertical wind (w') and the scalar (c'), averaged over 30 minutes.

$$Fc = \rho a \cdot \langle w'c' \rangle \ (CO_2 = NEE), \quad \lambda E = \lambda \cdot \rho a \cdot \langle w'q' \rangle, \quad H = \rho a \cdot cp \cdot \langle w'T' \rangle$$

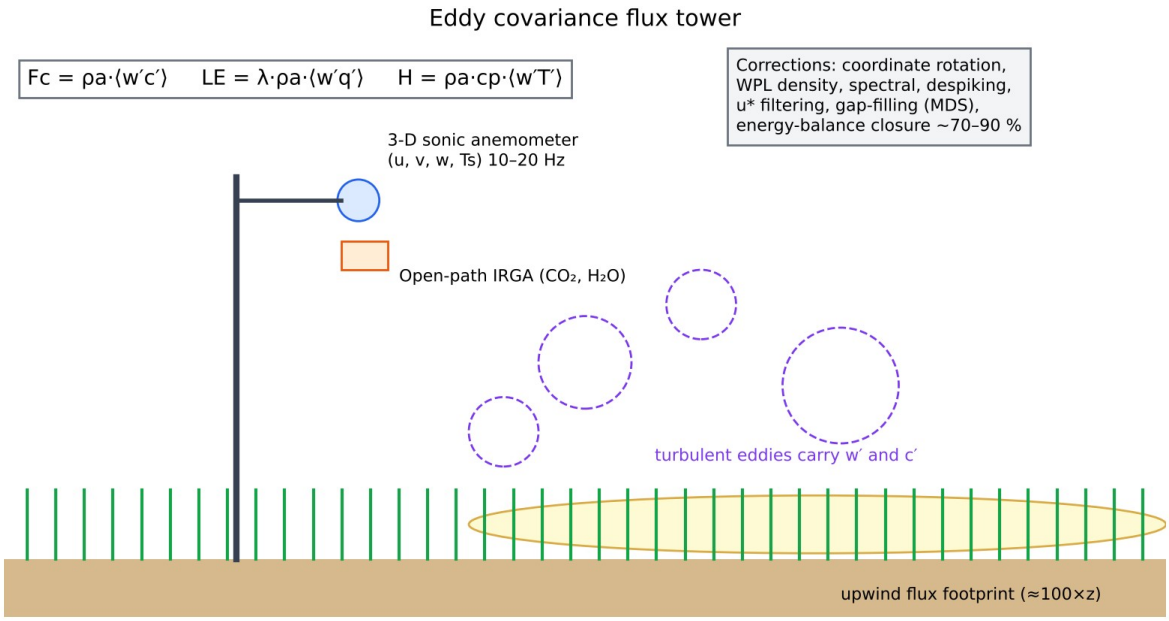


Figure 22. Eddy covariance flux tower: sonic anemometer, gas analyser, turbulent eddies and flux footprint.

- **Instruments:** 3-D sonic anemometer (Campbell CSAT3, Gill WindMaster) at 10–20 Hz; open-path (LI-7500) or closed-path (LI-7200) IRGA for CO_2 and H_2O; radiation (net radiometer), soil heat-flux plates, soil moisture and temperature, and PAR sensors.
- **Processing (EddyPro, TK3):** despiking, coordinate rotation (double/planar fit), time-lag correction, **WPL (Webb–Pearman–Leuning) density correction**, spectral corrections, stationarity and integral turbulence tests (Foken QC flags), **u* filtering **of night-time data, gap-filling (MDS, Reichstein 2005) and** flux partitioning** of NEE into GPP and Reco (night-time Lloyd–Taylor or daytime Lasslop).
- **Energy-balance closure:** $(H + \lambda E)/(Rn - G)$ is typically 0.7–0.9. Causes of non-closure include large eddies, storage terms and footprint mismatch. Bowen-ratio closure forcing is used for model validation.
- **Footprint:** the upwind source area, roughly 100 × measurement height (Kljun 2015 FFP model). It must sit within a homogeneous field.
- **Uses:** validation of satellite ET (SEBAL/METRIC/MOD16) and GPP (SIF, LUE models), calibration of crop models, and carbon budgets for MRV.

Indian EC efforts: NRSC and partners

- **ISRO Geosphere-Biosphere Programme, National Carbon Project (NCP)**, coordinated by **NRSC**, established EC flux towers over representative ecosystems: forest (e.g. tropical deciduous and sal forests), mangroves (Sundarbans) and agricultural systems. The towers quantify CO_2, H_2O and energy fluxes and validate satellite GPP/NPP and ET products.
- NRSC uses the flux data to **calibrate LUE parameters** (εmax) for crop productivity models, to validate **INSAT-3D ET and insolation products** (MOSDAC), and to develop soil and vegetation carbon-flux models.

- **ICAR-NRRI Cuttack** runs a long-term EC system over flooded rice (CH_4 and CO_2 fluxes). Other EC sites are at universities (IIT Kharagpur, ICRISAT) and in the IITM/MoES flux network.
- **Gaps and opportunities:** few cropland towers in the rice–wheat system of the IGP, no SIF instruments co-located with EC, and a need for an Indian FLUXNET-type open data portal.

> **Model answer:** If asked about NRSC's EC work: "NRSC, under the ISRO-GBP National Carbon Project, operates a network of eddy-covariance flux towers across major Indian ecosystems. They give direct continuous measurements of net ecosystem CO_2 exchange, latent and sensible heat. NRSC uses them to calibrate light-use-efficiency GPP models and to validate satellite ET and carbon products."

6.9 Satellite ET products to know

Product	Basis	Resolution / remarks
MOD16 (MODIS)	Penman–Monteith with MODIS LAI, albedo and meteorology	500 m, 8-day; underestimates ET in irrigated areas
SSEBop (USGS FEWS NET)	Simplified SEB with predefined dT	1 km global; Landsat version 30 m
GLEAM	Priestley–Taylor + soil-moisture stress	0.1–0.25°, daily
WaPOR (FAO)	ETLook (two-source PM)	100–250 m Africa/Near East; L1 global
OpenET	Ensemble of 6 models (METRIC, SSEBop, SIMS, PT-JPL, DisALEXI, geeSEBAL)	30 m, USA
INSAT-3D ET (MOSDAC)	SEB/PM with INSAT LST and insolation	Daily, about 4 km, India
ECOSTRESS	PT-JPL on ISS TIR	70 m, variable overpass time (samples the diurnal cycle)
TRISHNA (ISRO–CNES, upcoming)	TIR 57 m, near-daily	Will transform field-scale ET and irrigation monitoring in India

6.10 Interview questions on ET

Q. What is the difference between SEBAL and METRIC?

A. Both are single-source residual SEB models using hot and cold anchor pixels. METRIC calibrates the cold pixel to 1.05 × ETr from ground weather data instead of assuming H = 0, and scales instantaneous to daily ET with ETrF rather than EF. This makes it better under advection in arid, irrigated areas.

Q. Why does ALEXI not need air temperature?

A. It uses the time rate of change of surface temperature in the morning together with an atmospheric boundary-layer growth model. The ABL height and temperature become part of the solution, so no measured air temperature is needed, and absolute Ts errors cancel.

Q. Why is EF assumed constant through the day?

A. Under clear-sky, fair-weather conditions, available energy and latent heat follow similar diurnal shapes, so their ratio is self-preserving (Crago 1996). The assumption fails on cloudy days, with advection, and in the early morning and late evening.

Q. How do you validate satellite ET?

A. Compare with EC or lysimeter ET at overpass and daily scale, after correcting EC for energy-balance closure and matching the footprint to the pixel. Report RMSE, bias and R^2. Check seasonal totals against the soil water balance.

Q. What is the WPL correction?

A. The Webb–Pearman–Leuning correction removes apparent CO_2 and H_2O fluxes caused by air-density fluctuations from heat and water-vapour transfer. It is essential for open-path analysers.

7. Meteorology for agriculture, including El Niño and "Super El Niño"

One-line definition: agricultural meteorology studies how weather and climate interact with crops, soil, livestock and pests, and turns that knowledge into forecasts, advisories and risk tools for farming.

7.1 Core concepts

- **Weather vs climate:** weather is the state of the atmosphere over hours to days. Climate is the long-term statistics of weather, conventionally over a 30-year normal (e.g. 1991–2020).
- **Radiation balance:** shortwave in (≈ 0.48 of Rs is PAR), albedo, longwave exchange and the greenhouse effect.
- **Humidity:** vapour pressure e and saturation vapour pressure es(T) = 0.6108·exp(17.27T/(T + 237.3)) kPa. VPD = es − ea drives transpiration.
- **Stability and lapse rates:** the dry adiabatic rate is 9.8 K km^{-1}. Inversions trap smoke, which is why residue-burning smoke stagnates over Delhi in Oct–Nov under calm, cool conditions with a shallow boundary layer.
- **Growing degree days:** GDD = Σ[(Tmax + Tmin)/2 − Tbase]. Tbase is about 5 °C for wheat and 10 °C for rice and maize. GDD drives phenology in crop models.
- **Heat stress:** terminal heat in wheat (Tmax > 34 °C during grain filling) shortens the grain-filling period and reduces yield by about 3–5 % per °C.
- **Agromet indices:** SPI (standardised precipitation index), SPEI, the aridity index, the moisture adequacy index (AET/PET), the temperature-humidity index (THI) and the crop water stress index (CWSI).

7.2 Indian summer monsoon (ISM)

- June–September (JJAS) brings about 75 % of annual rainfall. The **long-period average (LPA) is 87 cm** (1971–2020 base).
- IMD categories: deficient < 90 % of LPA, below normal 90–95 %, normal 96–104 %, above normal 105–110 %, excess > 110 %.
- **Drivers:** land–sea thermal contrast, the ITCZ (monsoon trough) migration, the Somali jet (low-level westerly jet), the tropical easterly jet, the Tibetan high, and **ENSO, IOD and MJO** variability.
- **Intra-seasonal behaviour:** active and break spells (the 30–60-day Boreal Summer Intra-Seasonal Oscillation), and lows and depressions from the Bay of Bengal.
- **Normal onset over Kerala is 1 June.** In 2026 it was 4 June. Normal withdrawal starts from western Rajasthan around 17 September.
- **Northeast monsoon** (Oct–Dec) is important for Tamil Nadu and coastal Andhra Pradesh.

- **Western disturbances** bring winter rain over NW India that benefits rabi wheat. Untimely hail and rain at harvest cause losses that can trigger PMFBY claims.

7.3 ENSO, El Niño, La Niña and "Super El Niño"

- **ENSO (El Niño–Southern Oscillation)** is a coupled ocean–atmosphere oscillation of the tropical Pacific with a 2–7-year period.
- **El Niño:** weak trade winds, warm water spreading east, a deeper thermocline in the east and suppressed upwelling off Peru. The Walker circulation weakens and shifts east, giving **subsidence over India and a weaker monsoon**.
- **La Niña:** the opposite phase, usually bringing a normal or above-normal monsoon.
- **Measurement:** the **Oceanic Niño Index (ONI)** is the 3-month running mean SST anomaly in the **Niño 3.4 region** (5° N–5° S, 170°–120° W). El Niño means ONI ≥ +0.5 °C for at least 5 overlapping seasons. Atmospheric partner: the Southern Oscillation Index (Tahiti − Darwin pressure).
- **Strength categories:** weak (0.5–0.9), moderate (1.0–1.4), strong (1.5–1.9) and **very strong (≥ 2.0 °C), informally called a "Super El Niño"**. Examples: 1982–83, 1997–98, 2015–16 and 2023–24 (peak about +2.0).
- **Super El Niño is not an official WMO category**, but the media and many scientists use the term for ONI ≥ +2 °C events.
- **2026 status:** NOAA gave an 82 % chance of El Niño developing in May–July 2026, and many models projected a strong to very strong event in 2026–27. IMD cut its monsoon forecast from 92 % to **90 % of LPA**, and Skymet put the chance of drought at about 70 %. For comparison, monsoon rainfall in 2015, a super El Niño year, was 86 % of LPA. **Check the latest IMD end-of-season figures before the interview.**
- **Not every El Niño causes drought:** in 1997, a very strong El Niño, India's monsoon was normal because of a positive IOD. Roughly 60 % of Indian droughts are El Niño years.

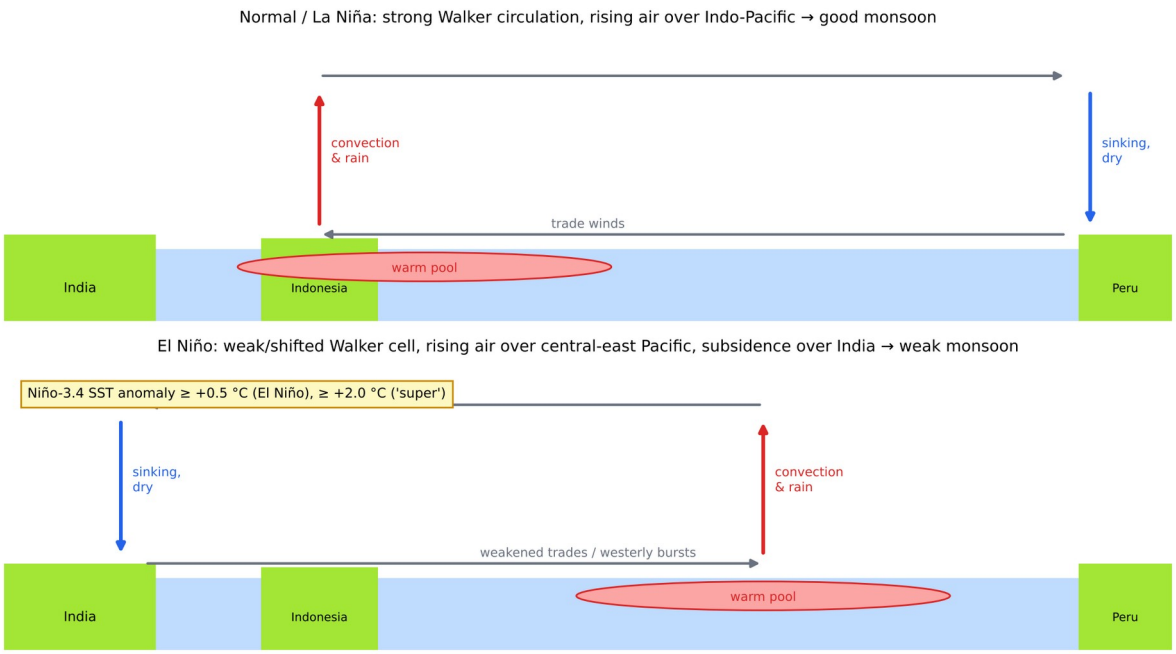


Figure 23. Walker circulation in normal/La Niña and El Niño conditions, and why El Niño weakens the Indian monsoon.

7.4 Indian Ocean Dipole (IOD) and MJO

- **IOD:** the SST difference between the western (50–70° E, 10° S–10° N) and south-eastern (90–110° E, 10° S–0°) Indian Ocean, measured by the Dipole Mode Index (DMI). A **positive IOD** (warm west) strengthens the monsoon and can offset El Niño, as in 1997, 2006 and 2019. IMD forecast neutral IOD conditions for 2026.
- **MJO (Madden–Julian Oscillation):** an eastward-moving envelope of convection with a 30–60-day period. Its phases 3–5 over the Indian Ocean favour active monsoon spells. IMD uses it in extended-range forecasts.

7.5 How El Niño affects Indian agriculture

Impact	Details
Kharif sowing and yield	Delayed onset and long breaks reduce sowing area of rainfed rice, pulses, soybean, cotton and groundnut. Rainfed areas (about 51 % of net sown area) are the most affected.
Reservoirs and groundwater	Lower storage limits rabi irrigation; more groundwater pumping.
Heat	El Niño years tend to bring hotter pre-monsoon and post-monsoon periods, terminal heat stress in wheat, and livestock heat stress.
Food inflation	Pulses, vegetables and edible oils; government stock management and export restrictions (rice in 2023).
Insurance	Higher PMFBY claims (prevented sowing, mid-season adversity, yield loss); weather-index products (RWBCIS) are triggered.
Response	District contingency plans (ICAR-CRIDA), short-duration varieties, the Crop Weather Watch Group, seed banks, MGNREGA water conservation, and GKMS advisories.

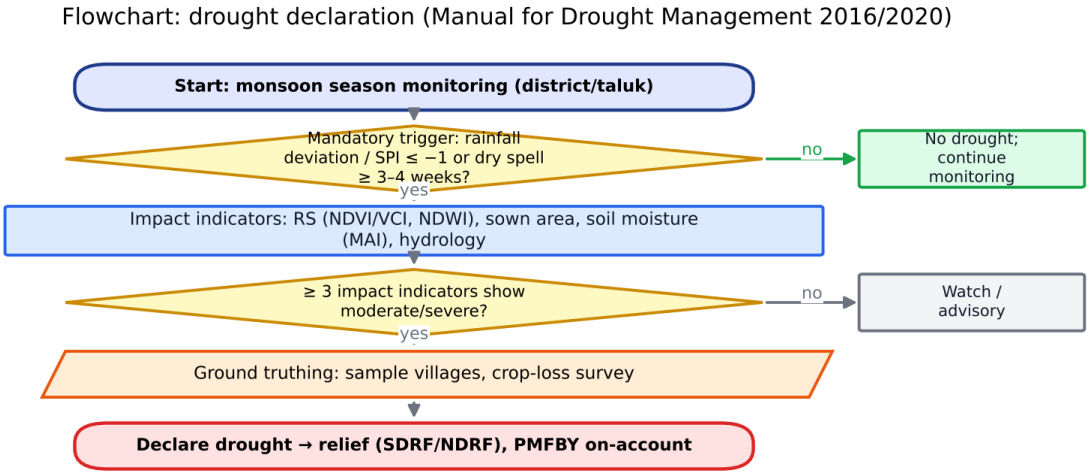


Figure 24. Flowchart of drought declaration under the Manual for Drought Management (2016, revised 2020).

7.6 Forecasts and agro-advisories in India

- **IMD forecasting system:** nowcast (0–3 h), short range (1–3 days), medium range (3–10 days, GFS/NCUM models at about 12 km), extended range (2–4 weeks, the MME of CFSv2) and seasonal forecasts (MMCFS, issued in April and June).
- **GKMS (Gramin Krishi Mausam Sewa):** 130 Agromet Field Units (AMFUs) at SAUs and ICAR institutes and District Agromet Units (DAMUs) at KVKs (DAMUs were discontinued in 2024; check whether they have been restored). They issue twice-weekly district and block-level advisories through **Meghdoot**, Mausam and Kisan SMS.
- **Satellite meteorology:** INSAT-3D/3DR/3DS provide imager and sounder data, insolation, LST, rainfall (IMSRA/HE), fog, outgoing longwave radiation and cloud-motion winds (MOSDAC). Oceansat-3 gives SST and SCATSAT-type winds. **Bharat Forecast System (BharatFS)**, launched in 2025, runs at about 6 km resolution.
- **Weather in crop insurance:** reference automatic weather stations (AWS/ARG) for RWBCIS triggers, and the **WINDS** programme (a network of AWS and rain gauges at taluk and GP level).

7.7 Interview questions

Q. What is a Super El Niño and how does it affect India?

A. A very strong El Niño with the Niño-3.4 SST anomaly at or above +2 °C, such as 1997–98, 2015–16 and the event projected for 2026–27. It weakens the Walker circulation and causes subsidence over India. This tends to bring a deficient monsoon, delayed kharif sowing, heat stress and higher food inflation, unless a positive IOD offsets it.

Q. Why didn't the 1997 El Niño cause drought?

A. A strong positive Indian Ocean Dipole developed at the same time. It enhanced convection over the western Indian Ocean and moisture transport to India, which offset El Niño's suppression.

Q. How can remote sensing help during an El Niño year?

A. Through near-real-time sowing progress from SAR, crop condition (NDVI/VCI), soil moisture (NISAR, SMAP), ET deficit and ESI, SIF-based stress detection, and reservoir area monitoring. All of these feed drought declaration (the Manual for Drought Management 2016/2020) and insurance.

Q. Define SPI.

A. The Standardised Precipitation Index fits a gamma distribution to long-term precipitation totals over a chosen time scale (1, 3, 6 or 12 months) and transforms the result to a standard normal variate. SPI ≤ −1 is moderately dry, ≤ −1.5 severely dry and ≤ −2 extremely dry.

8. Statistics and machine learning for agricultural remote sensing

One-line definition: statistics quantifies uncertainty and relationships in data. Machine learning (ML) builds models that learn patterns from data to predict or classify without explicitly programmed rules. In agricultural remote sensing both are used for classification (crop type), regression (yield, LAI, SM) and forecasting.

8.1 Statistical foundations

Table 10. Statistics to know

Concept	What to say
Mean, median, SD, CV	CV = SD/mean × 100. CV of CCE yields drives the required sample size.
Correlation vs regression	Pearson r measures linear association. Regression models the dependence $Y = \beta_0 + \beta_1 X + \varepsilon$ (OLS minimises $\Sigma\varepsilon^2$).
R^2 and adjusted R^2	$R^2 = 1 - SSres/SStot$. Adjusted R^2 penalises extra predictors.
Error metrics	$RMSE = \sqrt{(\Sigma(P - O)^2/n)}$; MAE; bias = mean(P − O); nRMSE = RMSE/mean(O) (< 10 % excellent, 10–20 % good, 20–30 % fair); MAPE
Hypothesis testing	H_0 vs H_1; p-value; Type I (α) and Type II (β) errors; t-test, ANOVA, χ^2
Multicollinearity	VIF > 10 is problematic. Fix with ridge/LASSO, PCA or variable selection.
Autocorrelation	Spatial (Moran's I, variogram) and temporal (ACF). It violates the independence assumption and inflates validation scores.
Sampling	Simple random, stratified, cluster, systematic, PPS. Stratification by RS crop vigour increases CCE efficiency.
Sample size	$n = (z \cdot CV/E)^2$, e.g. CV 30 %, error 10 %, 95 % confidence gives n ≈ 35
Geostatistics	Semi-variogram γ(h) (nugget, sill, range), ordinary kriging, regression kriging, co-kriging
Time series	Trend (Mann–Kendall, Sen's slope), ARIMA, seasonal decomposition; yield detrending (technology trend)
Bayesian	Posterior ∝ likelihood × prior. Used in data assimilation, BMA ensembles and uncertainty quantification.

8.2 Machine-learning algorithms

Table 11. Common ML algorithms in agricultural remote sensing

Algorithm	Idea	Agri use / note
Random Forest (Breiman 2001)	Ensemble of decision trees on bootstrap samples (bagging) with random feature subsets; majority vote or mean; OOB error; feature importance	The default for crop classification and yield; robust and little tuning needed
Gradient boosting (XGBoost,	Sequential trees fitted to residuals with regularisation and learning rate	Often the best on tabular yield data; SHAP for interpretation

Algorithm	Idea	Agri use / note
LightGBM, CatBoost)		
SVM / SVR	Maximum-margin hyperplane; kernel trick (RBF)	Small training sets; hyperspectral data
Gaussian Process Regression	Bayesian non-parametric; gives mean and variance	LAI/Cab retrieval with uncertainty (ARTMO)
k-NN	Prediction from the nearest neighbours in feature space	LUT inversion and gap-filling
ANN / MLP	Layers of weighted sums plus non-linear activations; backpropagation	RTM-hybrid inversion
CNN	Convolutions extract spatial features	Field-boundary delineation (U-Net), pest and disease images
RNN / LSTM / GRU	Memory gates for sequences	Crop type from time series; in-season yield forecasting
Transformers / attention	Self-attention over time and space	SITS-BERT, TempCNN vs Transformer for crop mapping; foundation models (Prithvi)
Unsupervised (k-means, ISODATA)	Clustering without labels	Management zones in precision agriculture

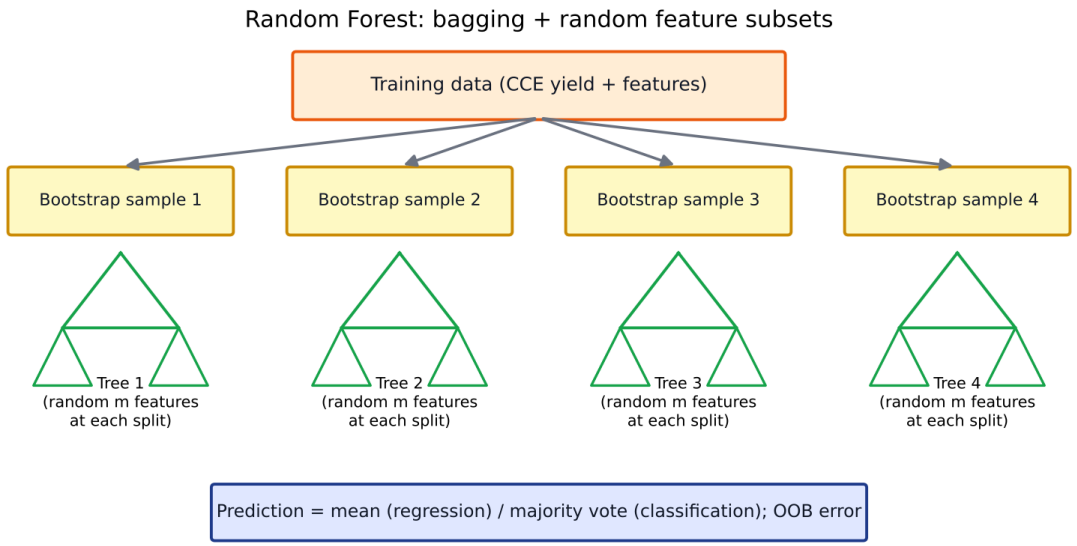


Figure 25. Random Forest: bootstrap aggregation of decision trees with random feature subsets.

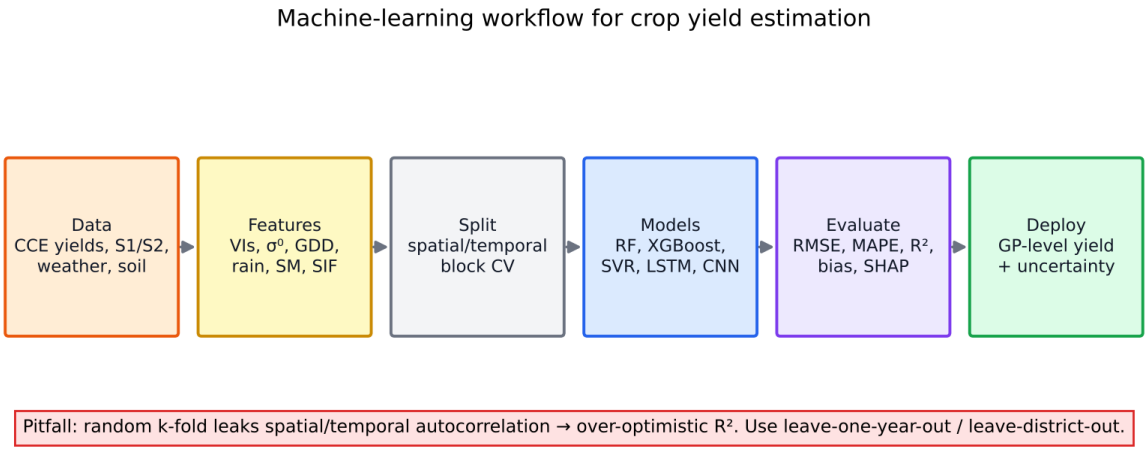


Figure 26. End-to-end ML workflow for yield estimation, including the spatial cross-validation pitfall.

8.3 The bias–variance trade-off, overfitting and validation

- **Bias** is error from overly simple assumptions (underfitting). **Variance** is sensitivity to the training sample (overfitting). Total error = bias² + variance + irreducible noise.
- **Regularisation:** L1 (LASSO, which selects variables), L2 (ridge), dropout, early stopping and tree depth limits.
- **Cross-validation:** k-fold, leave-one-out, and especially **spatial block CV / leave-one-district-out** and **leave-one-year-out** for yield. These test the ability to predict a new year or region, which is the operational need.
- **Hyperparameter tuning:** grid or random search and Bayesian optimisation, done inside nested CV.
- **Class imbalance:** stratified sampling, SMOTE and class weights; report F1, precision and recall rather than OA alone.
- **Interpretability:** permutation importance, partial dependence plots, **SHAP values** and attention maps. Explaining models matters for insurance, where farmers and insurers dispute outcomes.
- **Uncertainty:** quantile regression forests, prediction intervals, ensembles and Monte-Carlo dropout.

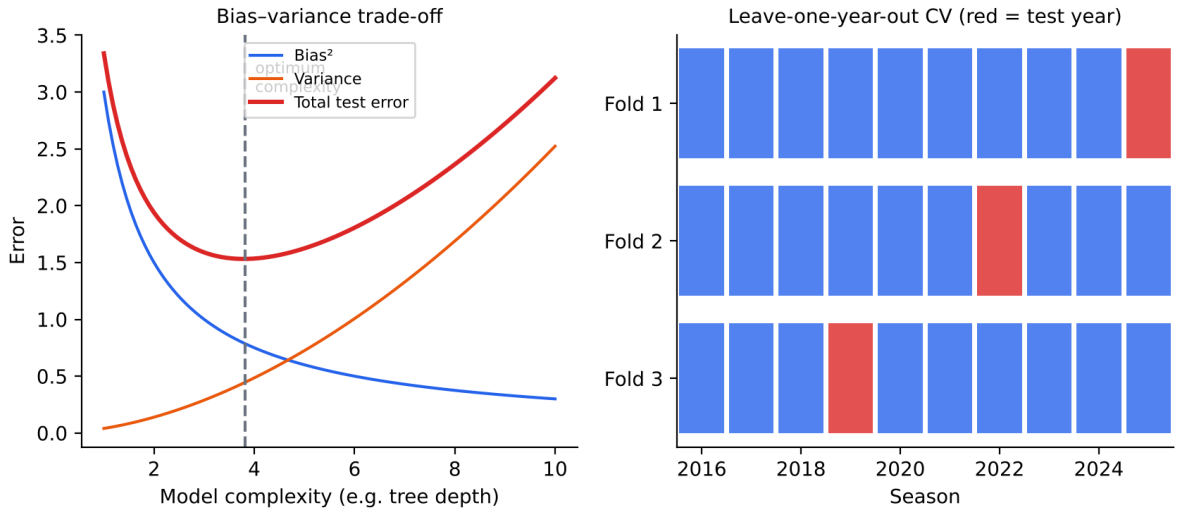


Figure 27. Left: the bias–variance trade-off. Right: leave-one-year-out cross-validation for yield models.

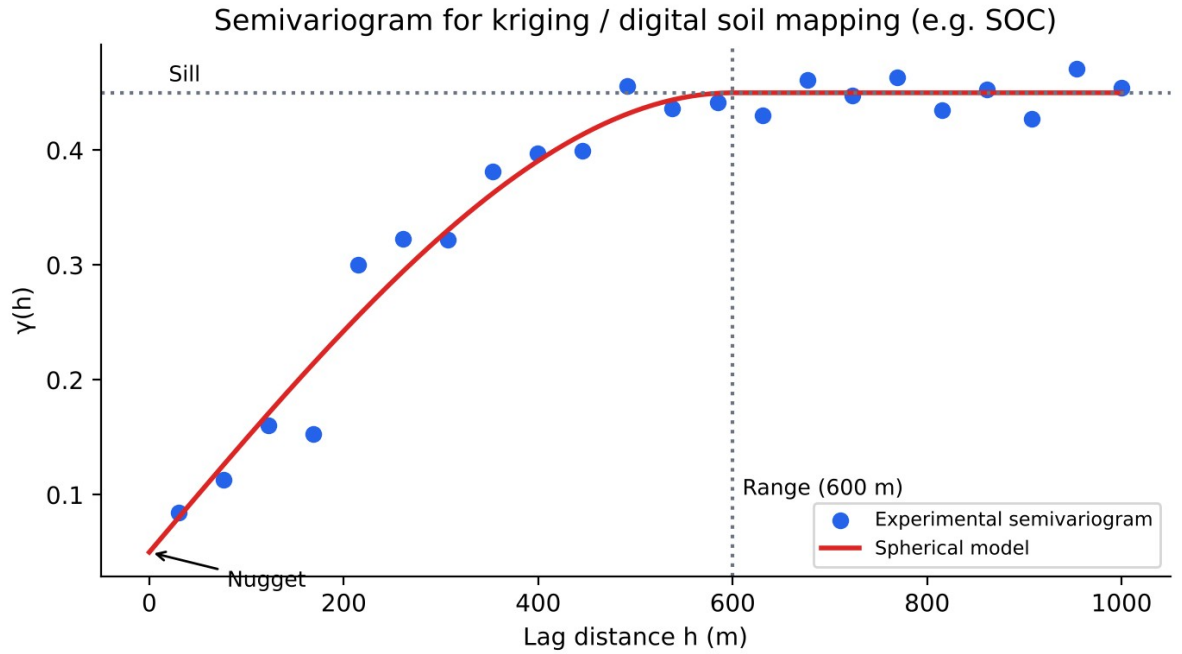


Figure 28. Semivariogram (nugget, sill, range) used in kriging and digital soil mapping.

8.4 ML vs process-based models

Aspect	Process (crop simulation) models	ML models
Basis	Physiology: photosynthesis, phenology, water and N balance	Statistical patterns in data
Data need	Detailed inputs: soil, management, cultivar coefficients	Large labelled datasets
Extrapolation	Better for new climates and scenarios	Poor outside the training domain
Interpretability	High (mechanistic)	Lower (needs XAI)
Scale	Point-based; regional runs need gridded inputs	Scales easily with RS data
Hybrid	Assimilate RS data, use ML emulators,	Use crop-model outputs as features

Aspect	Process (crop simulation) models	ML models
	physics-informed ML	

8.5 Interview questions

Q. Why can't you use random k-fold CV for yield models?

A. Neighbouring fields and the same year share weather and soils, so a random split puts near-duplicates in training and test sets. This leakage inflates R^2. Operational forecasting predicts an unseen year, so use leave-one-year-out or spatial block CV.

Q. What is out-of-bag (OOB) error in Random Forest?

A. Each tree is trained on a bootstrap sample that leaves out about 37 % of the data. Those left-out samples are predicted by the trees that did not see them, which gives a built-in, nearly unbiased estimate of generalisation error.

Q. How do you choose between RF and LSTM for crop classification?

A. RF works well with engineered temporal features (monthly composites, phenological metrics) and modest data. LSTMs and Transformers learn directly from irregular raw time series, need more labels and compute, and are usually better for large areas and early-season classification. Benchmark both under the same spatial CV.

Q. What is SHAP?

A. SHapley Additive exPlanations: a game-theory method that attributes each prediction to the contributions of its features, consistently and locally. For example, it can show that VH backscatter in August and rainfall in September drove a low yield prediction.

Q. Kriging vs IDW?

A. IDW is a deterministic distance-weighted average. Kriging is a best linear unbiased predictor that uses the modelled spatial autocorrelation (variogram) and gives a prediction variance.

9. Digital twins vs crop simulation models

One-line definitions: a **crop simulation model** is a set of mathematical equations representing crop growth and development as functions of weather, soil, genotype and management. An **agricultural digital twin** is a living virtual replica of a real farm or field, continuously synchronised with observations from sensors and satellites. It is used to monitor, predict and prescribe actions in near-real time, in a closed loop with the physical system.

9.1 Crop simulation models: a refresher

Table 12. Major crop simulation models

Model	Origin	Key features
DSSAT (CERES, CROPGRO)	IBSNAT/ICASA, USA	42+ crops; daily step; genetic coefficients (P1, P5, G1…); soil water and N; widely used in India
APSIM	CSIRO/Queensland, Australia	Modular; farming systems, rotations and residue; strong soil processes
InfoCrop	ICAR-IARI (Aggarwal et al. 2006)	Indian crops; pests, GHG emissions and climate-change impact; generic crop model
WOFOST	Wageningen, Netherlands	Used in the EU MARS and MCYFS forecasting systems and in FASAL research; potential and water-limited yield
AquaCrop	FAO	Water-driven (WP × Tr); few parameters; irrigation planning
ORYZA	IRRI	Rice; flooded and aerobic conditions
STICS, CropSyst, EPIC	France, USA	Soil–crop–environment

Core processes

- **Phenology:** thermal time or GDD plus photoperiod and vernalisation.
- **Carbon assimilation:** radiation-use efficiency (biomass = RUE × intercepted PAR), or leaf photosynthesis (Farquhar) integrated over the canopy.
- **Partitioning:** dry matter to leaves, stems, roots and grain with stage-dependent coefficients. Yield = HI × biomass, or grain number × grain weight.
- **Water balance:** infiltration, runoff (SCS curve number), drainage (tipping bucket), ET (Priestley–Taylor or Penman–Monteith), and a stress factor from the supply/demand ratio.
- **Nitrogen balance:** mineralisation, nitrification, leaching, volatilisation, uptake and N stress.
- **Uses:** yield-gap analysis (potential vs actual), climate-change impact, sowing-date and irrigation optimisation, and forecasting (FASAL).

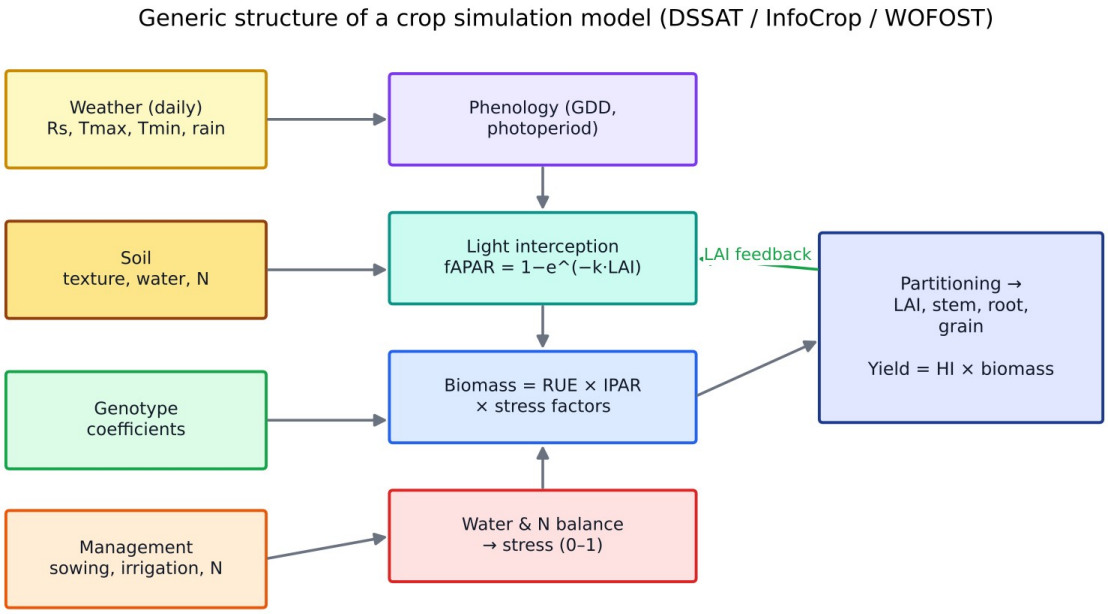


Figure 29. Generic structure of a crop simulation model (DSSAT / InfoCrop / WOFOST).

9.2 What makes something a digital twin

1. **Physical entity:** the field, farm, orchard or greenhouse.
2. **Virtual entity:** a model, which can be a crop simulation model, an ML model or a hybrid.
3. **Data connection (physical → virtual):** a continuous stream from IoT soil-moisture probes, weather stations, UAVs, satellites (Sentinel, NISAR, FLEX) and farmer logs.
4. **Data assimilation:** observations update model states and parameters, using the Ensemble Kalman Filter (EnKF), particle filters or variational (4D-Var) methods, or forcing and calibration.
5. **Feedback (virtual → physical):** recommendations or automatic actions such as irrigation valves, VRT prescriptions, pest alerts or insurance triggers.
6. **Persistence and fidelity:** the twin exists for the whole life cycle (the season and beyond) and its level of fidelity is fit for purpose.

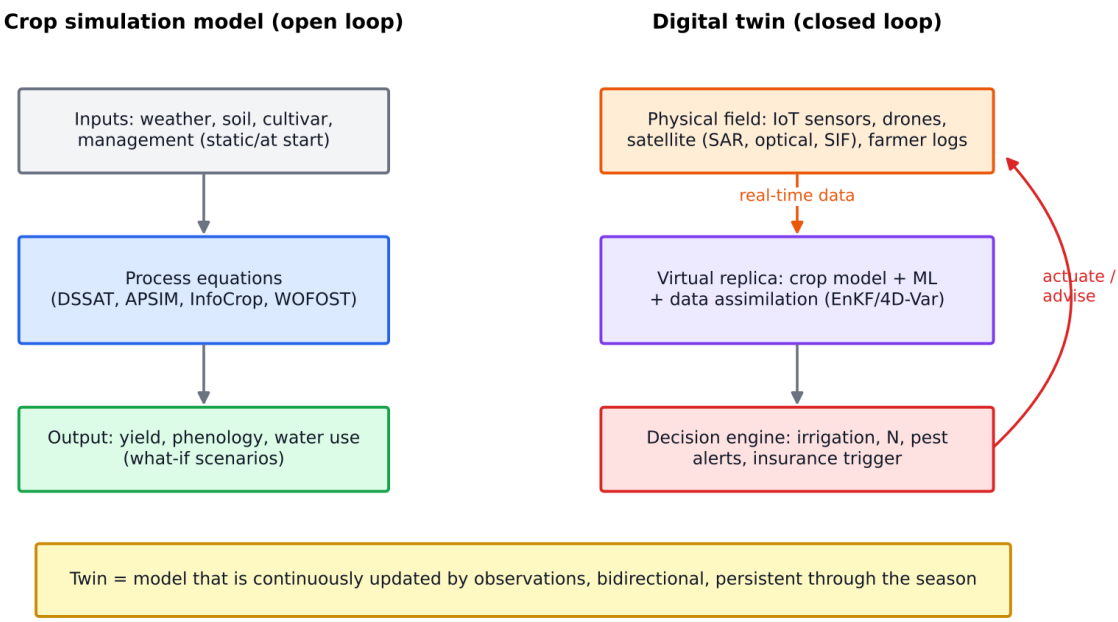


Figure 30. Open-loop crop simulation model compared with a closed-loop agricultural digital twin.

Table 13. Crop model vs digital twin

Aspect	Crop simulation model	Digital twin
Nature	Static tool; runs offline	Dynamic, live and persistent
Data flow	One-way, inputs set at the start	Two-way; continuous real-time data
State updating	None (open loop)	Data assimilation keeps the model on track
Purpose	Scenario analysis, research	Operational monitoring, prediction and control
Components	Equations only	Sensors + IoT + model + AI + visualisation + actuation
Spatial unit	Point or homogeneous field	Field to farm to region; GIS-integrated
Uncertainty	Usually deterministic	Probabilistic (ensembles) and updated
Example	DSSAT run for a 2050 climate scenario	Wheat field twin that assimilates Sentinel-2 LAI and probe soil moisture to schedule irrigation and forecast yield

> **One-line distinction:** A crop model can be the engine inside a digital twin. The twin adds the live data connection, assimilation, feedback and persistence. **Every digital twin needs a model, but a model on its own is not a twin.**

9.3 Data assimilation strategies

- **Forcing:** replace a simulated state (e.g. LAI) with the observed value at each observation date. Simple, but it ignores observation error.

- **Calibration / re-initialisation:** adjust parameters or initial conditions (sowing date, soil water) so the simulated LAI matches the observed trajectory. Optimisers include SCE-UA, PSO and MCMC.
- **Sequential updating (EnKF):** run an ensemble of model runs, and at each observation update states in proportion to the model and observation error covariances (the Kalman gain).
- **Variational (4D-Var):** minimise a cost function over a time window.
- **Variables assimilated:** LAI, fAPAR, soil moisture (SAR), ET, SIF and canopy cover.
- **Research example:** assimilating Sentinel-2 LAI and Sentinel-1/NISAR soil moisture into WOFOST/DSSAT improves district wheat-yield RMSE by about 20–40 % over the open loop (Huang et al. 2019 review).

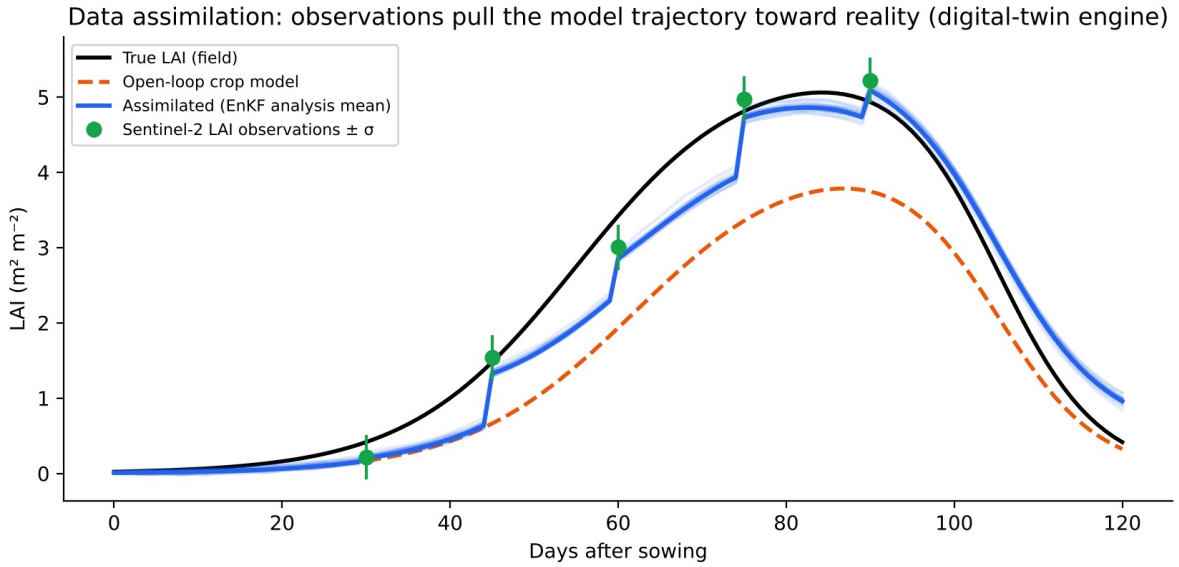


Figure 31. Data assimilation of Sentinel-2 LAI into a crop model (EnKF), the engine of a digital twin. Illustrative simulation.

9.4 Digital-twin initiatives

- **Destination Earth (EU):** Earth-system digital twins for climate adaptation and extremes.
- **NVIDIA Earth-2 and Google Weather/AI** models used for digital-twin weather.
- **Agriculture:** Wageningen's Digital Twins programme, the USDA/NASA Harvest crop twin concepts, and the commercial platforms Cropin (India) and xarvio.
- **Possible ISRO and India direction:** a national **"Krishi Digital Twin"** combining Bhuvan/Bhoonidhi EO data, the AgriStack farmer registry, Digital Crop Survey geo-tagged crop data, Soil Health Cards, IMD forecasts and crop models. It could serve FASAL, YES-TECH and advisory at GP level.

9.5 Interview questions

Q. Is DSSAT a digital twin?

A. No. DSSAT is a process model that runs open loop with fixed inputs. It becomes the engine of a digital twin only when it is continuously updated with real-time observations through data assimilation and feeds decisions back to the field.

Q. What role can ISRO data play in an agricultural digital twin?

A. ISRO data provide the observation layer for the twin. That includes crop type and sowing date (EOS-04/NISAR), LAI and chlorophyll (Resourcesat/Sentinel), soil moisture (NISAR L4 100 m), ET (INSAT, TRISHNA), weather (INSAT-3DS) and photosynthesis (FLEX SIF via international data). All of these can be assimilated.

Q. What is the Ensemble Kalman Filter in one line?

A. A Monte-Carlo approach to the Kalman filter. An ensemble of model runs represents the forecast uncertainty, and each observation updates every member in proportion to the relative model and observation error.

10. Crop-residue burning: detection, FRP, burnt area and management

One-line definition: crop-residue burning is the deliberate in-field burning of the straw and stubble left after harvest, mainly paddy straw in Oct–Nov in Punjab and Haryana and wheat stubble in Apr–May. Farmers burn residue to clear fields quickly for the next crop. It is a major source of air pollution, GHG emissions and nutrient loss.

10.1 Why farmers burn and why it matters

- **Narrow window:** combine-harvested paddy (late October) leaves only 15–20 days before wheat sowing. The **Punjab Preservation of Subsoil Water Act 2009** delayed paddy transplanting to mid-June to save groundwater, which pushed harvest later and squeezed the window further.
- **Combine harvesting** leaves loose straw and 30–40 cm stubble that is hard to manage without machinery.
- **Low economic value:** high-silica paddy straw is unsuitable as fodder (unlike wheat straw, used as *bhusa*).
- **Scale:** India produces about 500 Mt of crop residue per year, with about 90–140 Mt surplus, of which roughly 85–90 Mt is burnt (MNRE and various estimates). Punjab produces about 20 Mt of paddy straw a year.
- **Impacts:** PM2.5 and PM10, CO, black carbon, CO_2, CH_4 and N_2O. Burning contributes up to 30–40 % of Delhi's PM2.5 on peak days (SAFAR/DSS estimates). Each tonne of straw burnt loses about 5.5 kg N, 2.3 kg P, 25 kg K and 1.2 kg S, and burning kills soil microbes and raises topsoil temperature.
- **Policy:** the **Commission for Air Quality Management (CAQM) Act 2021**, National Green Tribunal orders, environmental compensation fines, and the **Crop Residue Management (CRM) scheme** (since 2018), which subsidises machinery at 50 % for individuals and 80 % for CHCs and FPOs.

10.2 Active-fire detection from thermal data

Physics

A fire at 800–1200 K emits strongly in the **mid-infrared (3.7–4 µm)**, as Wien's law predicts, while the 300 K background peaks near 10 µm. Even a fire occupying 0.01 % of a 1 km pixel raises the 4 µm brightness temperature by many kelvin, much more than the 11 µm temperature. The **difference T4 − T11** is the basis of detection.

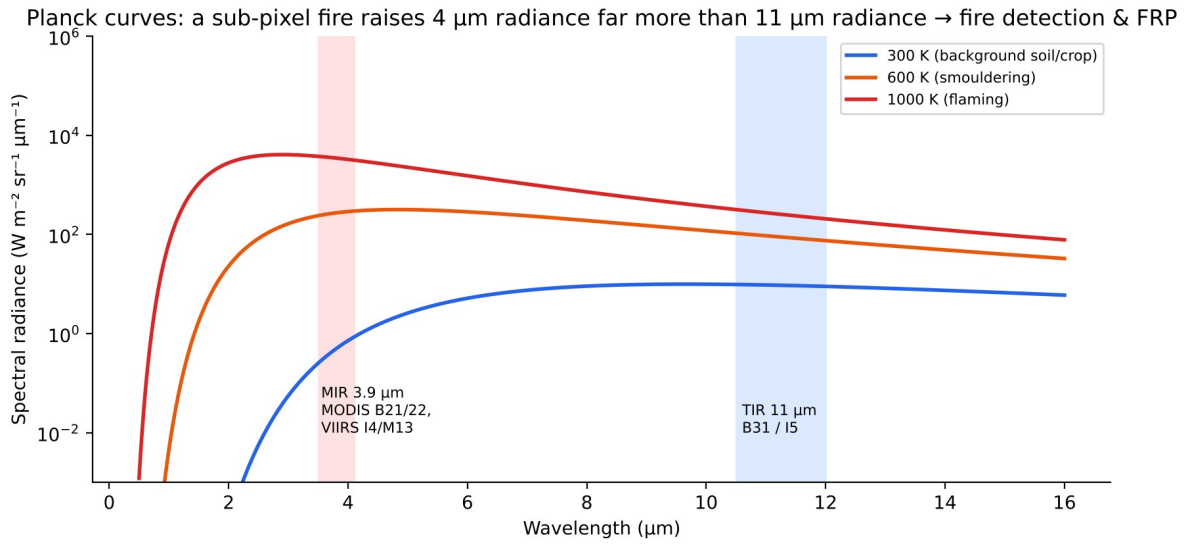


Figure 32. Planck curves for background and fire temperatures, showing why the 3.9 µm channel is sensitive to sub-pixel fires.

Sensors and algorithms

Table 14. Active-fire sensors used for India

Sensor	Platform / overpass	Resolution	Notes
MODIS (MOD14/MYD14, Collection 6.1)	Terra 10:30 / Aqua 13:30	1 km	Contextual algorithm (Giglio 2016); bands 21/22 (3.9 µm) and 31 (11 µm); FRP from band 21/22
VIIRS (VNP14IMG)	S-NPP, NOAA-20, NOAA-21, about 13:30	375 m (I-band)	I4 (3.74 µm), I5 (11.45 µm); about 3–4× more small agricultural fires than MODIS; M13 for FRP
INSAT-3D/3DR	Geostationary, every 15–30 min	4 km MIR	Diurnal sampling of afternoon fires; low sensitivity
Sentinel-3 SLSTR	10:00	1 km	Night-time fire product
Landsat 8/9, Sentinel-2	10:30	30 m / 20 m SWIR	Active flame fronts visible in SWIR; mainly used for burnt area

- **Contextual fire algorithm (MODIS/VIIRS):** (1) cloud and water masking; (2) potential-fire test, e.g. T4 > 310 K and ΔT = T4 − T11 > 10 K by day; (3) compare with neighbouring background pixels (mean and MAD of T4 and ΔT); (4) reject false alarms from sun-glint, desert boundaries and hot bare soil; (5) assign a confidence class (low, nominal, high).
- **Data delivery:** NASA **FIRMS** (Fire Information for Resource Management System) provides near-real-time data within 3 hours.

10.3 Fire radiative power (FRP)

Definition: FRP is the rate of radiant energy emitted by a fire (MW). It is proportional to the fuel combustion rate, so integrating FRP over time gives **fire radiative energy (FRE, MJ)**, and FRE converts directly to **biomass burnt** and emissions.

MIR radiance method (Wooster et al. 2003, 2005): $FRP = (A_{sampl} \cdot \sigma / a) \cdot (L4 - L4,bg)$

Biomass burnt (kg) = 0.368 kg MJ^{-1} × FRE (MJ) $Emission_x = Biomass \times EF_x$ (g/kg)

- Asampl is the pixel area, a is a sensor-specific constant (about 3×10^{-9} W m^{-2} sr^{-1} μm^{-1} K^{-4}), and L4 and L4,bg are the MIR radiances of the fire and background pixels.
- **Typical values:** small agricultural fires in Punjab are 1–30 MW per VIIRS pixel. Forest or large fires can exceed 100 MW.
- **Why it matters:** fire **counts** do not show the size or intensity of fires. FRP and FRE quantify emissions. The **GFAS** (CAMS, ECMWF) emission inventory is built on FRP.
- **Limitations:** a polar orbiter gives one or two snapshots a day, so diurnal integration needs GEO data or a fitted diurnal-cycle model. Small or smouldering fires are missed, and so are fires under cloud or smoke.

10.4 The daily fire bulletin (CREAMS-IARI / NRSC)

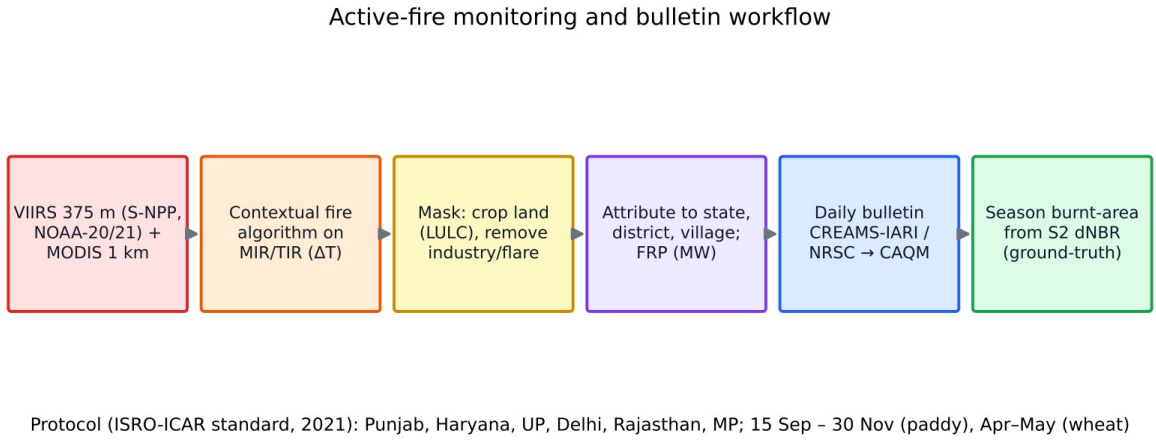


Figure 33. Workflow from satellite fire detection to the daily residue-burning bulletin.

- **Who:** the **CREAMS laboratory, Division of Agricultural Physics, ICAR-IARI**, issues the daily bulletin under the **ISRO–ICAR standard protocol** (developed with ISRO-NRSC and released in 2021). Bulletins go to MoA&FW, CAQM, MoEFCC and the state agriculture and pollution-control departments.
- **Coverage: Punjab, Haryana, Uttar Pradesh, Delhi, Madhya Pradesh and Rajasthan**, for paddy residue from **15 September to 30 November**. Wheat-residue burning is also monitored.
- **Content:** daily and cumulative fire-event counts by state and district, a comparison with previous years (2020–2025), maps, and **FRP-based intensity classes**.
- **Protocol rules:** use VIIRS 375 m and MODIS 1 km; mask to agricultural land; remove persistent industrial heat sources; merge detections within a spatial and temporal window to avoid double counting; standardise the season and region definitions so that states and agencies report the same numbers.
- **Trend:** official counts fell sharply in 2023–2025. Government reports claim about a 90 % reduction in Punjab and Haryana in 2025 compared with 2022.

- **Criticism (iFOREST 2025):** burning has shifted to the late afternoon, after the polar-orbiter overpasses at about 10:30 and 13:30, so a large share of fires may go undetected. **Burnt-area analysis** (Sentinel-2) shows a much smaller decline than fire counts. This is why the board may ask about **geostationary monitoring** and **burnt-area mapping**.

> **Remember:** Fire counts measure active flames at the moment of overpass. Burnt area measures the cumulative scar. Use both: counts for daily enforcement, burnt area for season totals and emission estimates. Geostationary sensors (INSAT-3DS, GEO-KOMPSAT, Himawari) help capture late-afternoon burning.

10.5 Burnt-area mapping with Sentinel-2's two SWIR bands

Burning removes green vegetation (NIR drops sharply) and leaves char, ash and exposed dry soil (**SWIR2 at 2.19 µm rises**). The opposite movement of NIR and SWIR2 is captured by the **Normalised Burn Ratio**. Sentinel-2 has two SWIR bands: **B11 (1610 nm, 20 m)** and **B12 (2190 nm, 20 m)**.

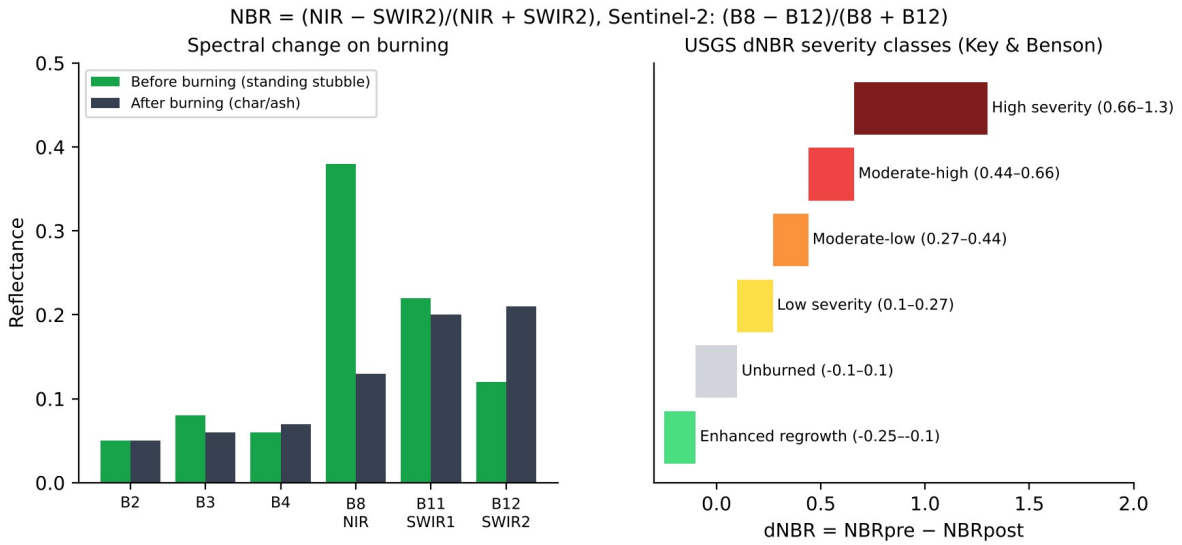


Figure 34. Spectral change from burning, and the USGS dNBR burn-severity classes (thresholds from Key & Benson 2006). Reflectance values are illustrative.

Table 15. Burn-sensitive indices

Index	Formula (Sentinel-2)	Remarks
NBR	(B8 − B12)/(B8 + B12)	High for vegetation, low or negative for burnt surfaces
dNBR	NBRpre − NBRpost	Severity: > 0.1 burnt; > 0.27 moderate; > 0.66 high (Key & Benson 2006)
RdNBR / RBR	dNBR/√ NBRpre ; dNBR/(NBRpre + 1.001)	Normalises for low pre-fire biomass (stubble)
NBR2	(B11 − B12)/(B11 + B12)	Uses both SWIR bands; sensitive to moisture and char; good for post-harvest stubble

Index	Formula (Sentinel-2)	Remarks
BAI	$1/[(0.1 - B4)^2 + (0.06 - B8)^2]$	Distance to the charcoal spectral point (Chuvieco 2002)
BAIS2	$(1 - \sqrt{(B6 \cdot B7 \cdot B8A/B4)}) \cdot ((B12 - B8A)/\sqrt{(B12 + B8A)} + 1)$	Sentinel-2 specific, uses the red edge (Filipponi 2018)
MIRBI	$10 \cdot B12 - 9.8 \cdot B11 + 2$	SWIR-only; robust for agricultural burns (Trigg & Flasse 2001)
CSI / NDVI / SAVI	NIR/SWIR; NDVI drop	Vegetation indices that respond to burning (loss of green cover)

Workflow for paddy-season burnt area (Punjab)

1. Make a **paddy mask** from Sentinel-1 VH time series (flooding signature) or a crop map.
2. Build cloud-masked Sentinel-2 composites at 5–10 day intervals from October to November, using Google Earth Engine and the s2cloudless cloud mask.
3. Compute NBR, NBR2, MIRBI and BAIS2. Compare each date with the post-harvest stubble date to get dNBR. **Important:** the right comparison is harvested stubble vs burnt stubble, not green crop vs burnt.
4. Classify burnt, unburnt stubble, harvested-and-removed, and sown wheat with Random Forest trained on field or VIIRS-guided samples, or use thresholds (e.g. dNBR > 0.1 and NBR2 drop).
5. Validate with field photos, PlanetScope imagery and active-fire points (commission and omission errors).
6. Aggregate by village and district and compare with fire counts. The ratio of burnt area to fire count shows detection gaps.

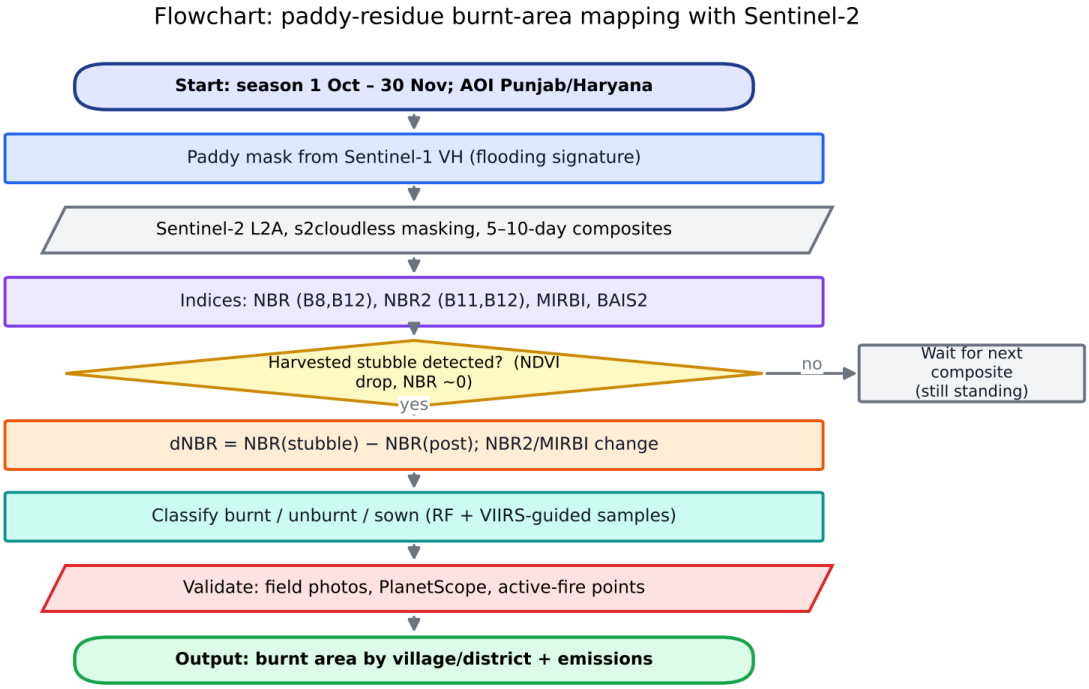
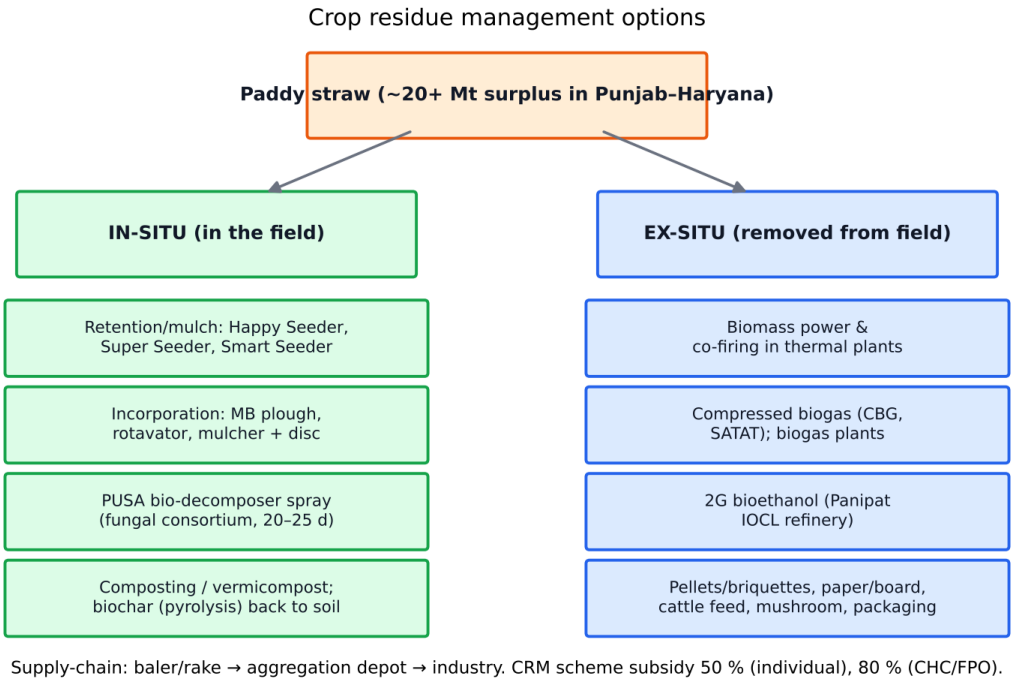


Figure 35. Flowchart of paddy-residue burnt-area mapping with Sentinel-2.

10.6 Residue management options



Supply-chain: baler/rake → aggregation depot → industry. CRM scheme subsidy 50 % (individual), 80 % (CHC/FPO).

Figure 36. Classification of crop-residue management options.

In-situ management (residue stays in the field)

Option	How	Pros and cons
Retention / mulching	Residue kept on the surface; wheat sown directly with the Happy Seeder (PAU/ACIAR), Super Seeder or Smart	Conserves moisture, moderates soil temperature

Option	How	Pros and cons
	Seeder. Stubble is cut, a slot is drilled and straw is placed as mulch.	against terminal heat, suppresses weeds and builds SOC. Needs a high-HP tractor; slow field capacity.
Incorporation	Straw is chopped (mulcher, Super Straw Management System on combines), then mixed into soil with a reversible MB plough, rotavator or disc harrow.	Adds organic matter; but needs high energy and time, can immobilise N early (wide C:N about 60–80:1) and releases more CO_2 quickly.
PUSA bio-decomposer	A fungal consortium developed by ICAR-IARI (capsule or liquid formulation), sprayed on stubble at about 10 L concentrate in 500 L water per ha; decomposes residue in about 20–25 days with adequate moisture.	Cheap; but performance depends on moisture and temperature (November cold slows it), and the time window is tight.
Composting	Pit or windrow composting with cow dung and urea to correct C:N; vermicomposting.	Produces valuable manure; needs labour and transport.
Biochar	Pyrolysis of straw at 350–600 °C with limited oxygen; biochar returned to soil.	Stable C (half-life of centuries), counts for carbon credits, improves CEC and water retention; kilns cost money.

Residue incorporation vs residue retention

Aspect	Incorporation (mixed into soil)	Retention (surface mulch, zero-till)
Tillage	Intensive (plough or rotavator)	None or minimum
Decomposition	Faster (soil contact, microbes)	Slower; stays as mulch through the season
Soil moisture	Moderate benefit	High: less evaporation and better infiltration
Soil temperature	Little effect	Lower soil temperature in spring; protects against terminal heat
Weeds	Buries weed seeds	Mulch suppresses Phalaris minor
N dynamics	Early N immobilisation	Gradual N release
SOC	Increases, but tillage oxidises some C	Greater long-term SOC build-up in the surface layer
Energy and cost	High fuel, about ₹3,000–5,000/ha extra	Lower operational cost; machine purchase is the cost
GHG	More CO_2 from tillage	Lower fuel CO_2; possible N_2O trade-offs

Ex-situ management (residue removed)

- **Collection:** rake + **baler** (rectangular or round bales), then aggregation at depots. The supply chain and storage cost about ₹1,500–2,500 per tonne.

- **Biomass power and co-firing:** the Ministry of Power mandates 5–7 % co-firing of biomass pellets in thermal power plants (SAMARTH mission). Punjab has biomass power plants of about 10–18 MW.
- **Compressed biogas (CBG):** under **SATAT** (Sustainable Alternative Towards Affordable Transportation), straw goes through anaerobic digestion to raw biogas (about 55–60 % CH_4), which is upgraded to more than 90 % CH_4 as CBG. The by-product is fermented organic manure. An example is the Verbio plant at Lehragaga, Sangrur, Punjab.
- **2G bioethanol:** lignocellulose is pre-treated, hydrolysed enzymatically and fermented into ethanol. An example is the **IOCL Panipat 2G ethanol refinery** (inaugurated 2022; about 100 kL/day from about 2 lakh t of straw a year), which supports the **E20 ethanol blending** target.
- **Other uses:** pellets and briquettes for industrial boilers, paper and cardboard, packaging, mushroom cultivation, animal bedding, and fodder after urea-molasses enrichment.

10.7 Interview questions

Q. What is FRP and why is it better than fire counts?

A. Fire radiative power is the radiant energy release rate of a fire in MW, derived from the excess MIR radiance. It scales with the combustion rate, so integrating it over time gives the biomass burnt and the emissions. Counts treat a small fire and a large fire equally.

Q. Why use SWIR2 in NBR rather than SWIR1?

A. Charred surfaces and ash raise reflectance most strongly at about 2.2 µm, and green vegetation absorbs there because of leaf water. The NIR–SWIR2 contrast is therefore the largest. SWIR1 is used in NBR2 to separate burnt from merely dry stubble.

Q. Why do official fire counts fall faster than burnt area?

A. Possible causes: burning shifted after satellite overpass times; fires are smaller and short-lived; there is cloud or haze; and fewer detections at the same time as some genuine reduction. Burnt-area mapping from Sentinel-2, geostationary detection and ground verification are needed to confirm the trend.

Q. Which crops' residues are burnt, and when?

A. Paddy straw in Punjab, Haryana and western UP (Oct–Nov, before wheat), wheat stubble (Apr–May, before the kharif crop or green manure), and sugarcane trash in UP and Maharashtra.

Q. Recommend a residue strategy for a 2-acre Punjab farmer.

A. Hire a Super Seeder or Happy Seeder through a CHC (subsidised under CRM) for in-situ mulching. Use the PUSA decomposer where there are 20+ days before sowing. Where a CBG plant or baler aggregator is nearby, sell the bales (about ₹1,500–2,000/t). Carbon-credit aggregation for no-burn plus zero-tillage can add income.

11. Conservation agriculture, regenerative agriculture and biogeochemical cycles

Conservation agriculture (CA), one line (FAO): a farming system that rests on three linked principles: **(1) minimum mechanical soil disturbance** (zero or reduced tillage), **(2) permanent soil organic cover** (residue or cover crops, at least 30 %) and **(3) species diversification** (rotations and intercrops).

Regenerative agriculture (RA), one line: an outcome-oriented approach that aims to **restore soil health, biodiversity, water cycles and the carbon cycle**. It combines CA principles with integrated livestock, organic inputs, reduced synthetic chemicals and agroforestry. It is defined more by outcomes (rising SOC, biodiversity) than by fixed practices.

Table 16. Conservation vs regenerative agriculture

Aspect	Conservation agriculture	Regenerative agriculture
Definition	Practice-based (3 principles)	Outcome-based (soil health, carbon, biodiversity)
Tillage	Zero or minimum	Minimum; tillage allowed if it serves the outcomes
Inputs	Herbicides often used for weed control	Reduced synthetic inputs; compost, biologicals
Livestock	Not central	Integrated grazing encouraged
Origins	FAO, CIMMYT, the Rice-Wheat Consortium (IGP)	Rodale Institute, the corporate sustainability movement
Carbon credits	Eligible under VM0042 (reduced tillage, residue retention)	Same; plus agroforestry and cover crops

11.1 CA in India: status and benefits

- Zero-tillage wheat after rice in the IGP covers about 2–3 Mha (estimates vary). Adoption is driven by the Happy Seeder, the Super Seeder and CRM subsidies.
- **Benefits:** wheat sown 1–2 weeks earlier (avoids terminal heat), savings of about ₹2,500–3,000/ha in tillage costs, 20–30 % irrigation water saving, less *Phalaris minor* weed pressure, higher SOC over the long term, and lower diesel CO_2.
- **Long-term trials:** CIMMYT–BISA (Karnal, Ladhowal), ICAR-CSSRI and ICAR-IARI show SOC gains of 0.2–0.5 t C/ha/yr in the 0–15 cm layer under ZT plus residue retention, compared with conventional puddled transplanted rice followed by conventionally tilled wheat (results depend on depth).
- **Constraints:** machinery cost and availability, herbicide-based weed management, rodent and termite pests in mulch, farmer mindset, and fragmented holdings.

11.2 Identifying CA wheat fields from satellite data (the paper by Dr. Rajkumar)

Context: Dr. Rajkumar Dhakar and Dr. V. K. Sehgal of the CREAMS lab (Division of Agricultural Physics, IARI) developed a **Zero-Tillage Mapper**. It tracked the **time series of ZT adoption in wheat across about 63,000 fields in Fazilka, Punjab (2018–2023)**, and it was presented at Geospatial World Forum/GSI 2025. Related international studies map tillage in the IGP with Sentinel-2 and Random Forest on Google Earth Engine (e.g. Zhou et al., *Remote Sensing* 2021). Read the exact published paper before quoting numbers. The method logic below is what the board will test.

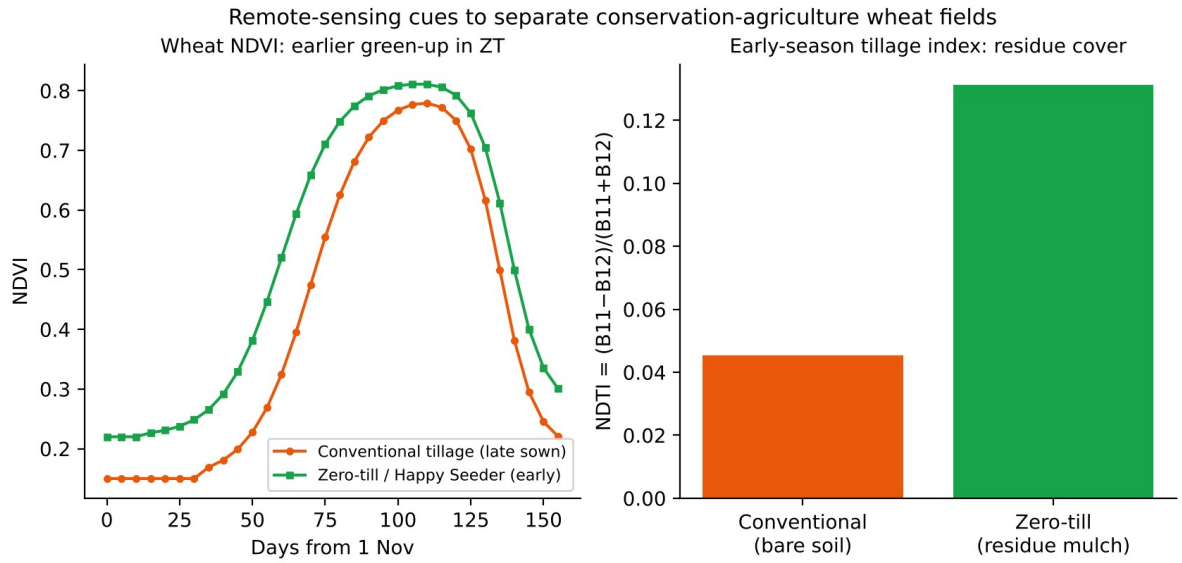


Figure 37. Remote-sensing signatures that separate zero-till (CA) wheat from conventionally tilled wheat. Illustrative curves based on the detection logic of Zhou et al. (2021) and the IARI CREAMS ZT Mapper.

Detection principles

1. **Residue cover at sowing:** ZT fields keep paddy straw or anchored stubble on the surface, while conventional fields are ploughed to bare soil. Cellulose and lignin absorb at about 2.1 µm, so **NDTI = (B11 − B12)/(B11 + B12)** and the **cellulose absorption index (CAI)** are higher on residue than on bare soil.
2. **No burning:** CA fields show no fire detection and no burn scar (dNBR) between paddy harvest and wheat sowing. Fire and burnt-area products are used as a negative filter.
3. **Sowing date and green-up:** ZT and Happy Seeder fields are sown immediately after harvest, so **NDVI rises 7–15 days earlier** than in conventional fields, which need ploughing and planking.
4. **Surface roughness and tillage operations:** conventional fields show a sudden **InSAR coherence loss** or a VV backscatter change (ploughing). ZT fields stay coherent and smooth.
5. **Classification:** features (NDTI, NBR2 and SWIR brightness at sowing, green-up date, VH/VV change, coherence) feed a Random Forest or SVM. Training uses field surveys and CRM machine-usage records. Validation reports OA, F1 and kappa.

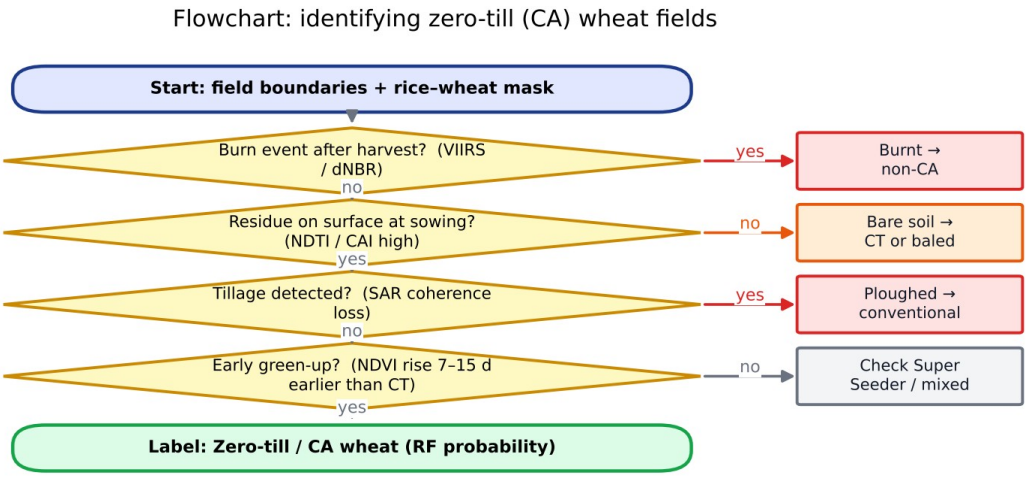


Figure 38. Decision flowchart for identifying zero-till (CA) wheat fields from satellite data.

Challenges

- Clouds and fog in November–December reduce Sentinel-2 availability. Use Sentinel-1/NISAR as a fallback.
- **Super Seeder fields** partly incorporate residue (rotary action), which gives a mixed residue signal.
- Residue removed by baling followed by zero-till sowing (no mulch) looks like conventional tillage.
- Soil moisture changes SWIR reflectance and the NDTI dynamic range. Use the moisture-robust indices (NDTI) or the dry-date composites.
- Small fields (0.5–2 ha) and mixed pixels. Sentinel-2's 10–20 m is adequate for most Punjab fields.

11.3 Carbon cycle in agroecosystems

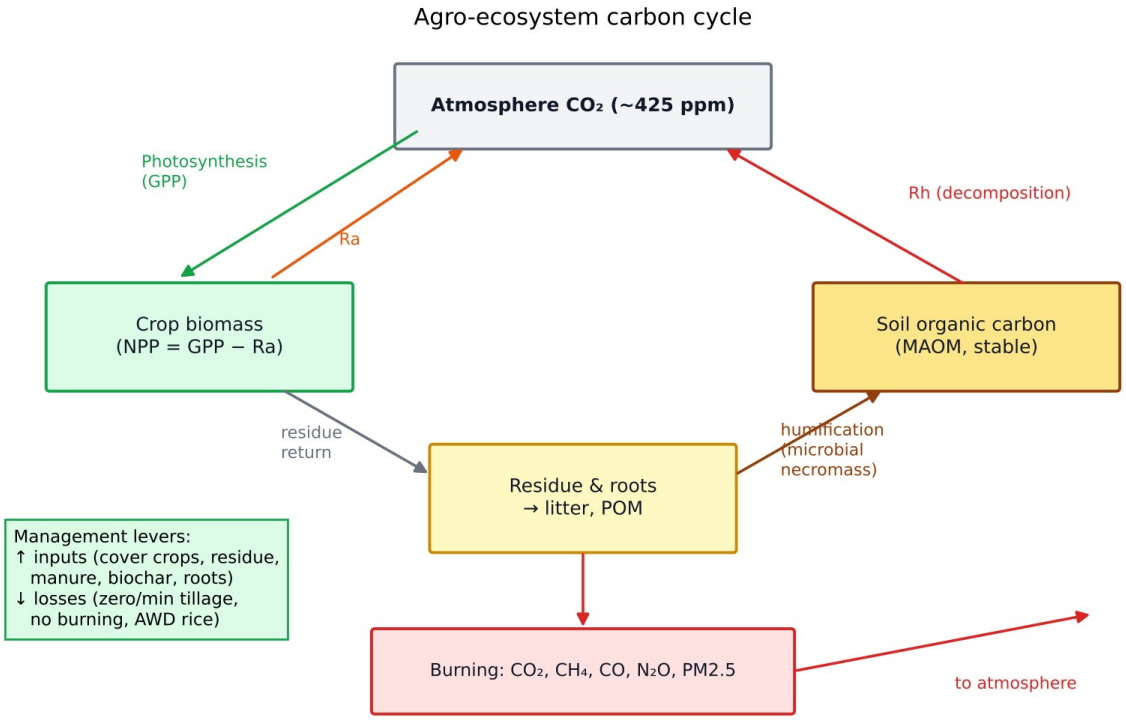


Figure 39. The agroecosystem carbon cycle and the management levers that build soil organic carbon.

- **Carbon pools:** atmosphere (about 875 GtC, about 425 ppm CO_2 in 2026), vegetation (about 450–650 GtC), **soils (about 1,500–2,400 GtC to 1 m)** and oceans (about 38,000 GtC).
- **Fluxes:** GPP → NPP (= GPP − autotrophic respiration Ra) → litter and roots → **SOC**, which is lost by **heterotrophic respiration (Rh)**. NEP = NPP − Rh. NEE = −NEP (measured by eddy covariance).
- **SOC fractions:** particulate organic matter (POM, fast turnover) and **mineral-associated organic matter (MAOM, stable, protected on clay surfaces)**. The current paradigm (Lehmann & Kleber 2015; Cotrufo 2013 MEMS) holds that microbial necromass forms MAOM.
- **Factors:** climate (warm, moist conditions speed decomposition), clay content, tillage (breaks aggregates and exposes C), residue inputs, and nitrogen.
- **Equilibrium:** SOC rises after a practice change until a new steady state is reached (about 20–30 years). This is why **permanence** is a concern in carbon credits.
- **4 per 1000 initiative (COP21):** a 0.4 % annual increase in global soil C would offset the annual increase in atmospheric CO_2.

11.4 Other biogeochemical cycles

Cycle	Key processes	Agricultural relevance
Nitrogen	Fixation (biological and Haber–Bosch), mineralisation, nitrification ($NH_4^+ \rightarrow NO_3^-$),	Urea NUE is only 30–40 %; N_2O has a GWP of 273 (AR6); neem-coated urea and nano-urea;

Cycle	Key processes	Agricultural relevance
	denitrification ($NO_3^- \rightarrow N_2O/N_2$), volatilisation ($NH_3$), leaching	precision N
Phosphorus	Weathering, fixation by Fe/Al (acid soils) and Ca (alkaline soils), microbial solubilisation; no gaseous phase	Legacy P, PSB biofertilisers, eutrophication
Sulphur	Mineralisation, oxidation (SO_4^{2-}), reduction (H_2S) in flooded soils	Oilseed nutrition
Methane (C)	Methanogenesis in anaerobic flooded rice (archaea); oxidation by methanotrophs	Rice CH_4 is about 10 % of India's agricultural GHG; AWD cuts CH_4 by 30–50 %; DSR
Water	Precipitation, infiltration, ET, runoff, groundwater recharge	Linked to the energy balance and the carbon cycle (water-use efficiency)

11.5 Interview questions

Q. How would you distinguish a zero-till wheat field from a conventional one using satellites?

A. At sowing, ZT fields have surface residue (high NDTI/CAI in Sentinel-2 SWIR bands), no fire or burn scar, and an earlier NDVI green-up. Conventional fields show bare-soil spectra after ploughing, a coherence loss in SAR and later emergence. A Random Forest on these temporal features, trained with field data, maps CA adoption.

Q. Is regenerative agriculture the same as organic farming?

A. No. Organic farming is defined by prohibited inputs (no synthetic fertilisers or pesticides) and by certification. Regenerative agriculture is defined by soil-health outcomes and can use judicious synthetic inputs. Practices overlap.

Q. Why does tillage reduce SOC?

A. Tillage breaks soil aggregates, exposes physically protected organic matter to microbes and oxygen, and raises soil temperature and aeration. All of these accelerate decomposition and CO_2 loss, and tillage also increases erosion.

12. Carbon credits and MRV (monitoring, reporting, verification)

One-line definition: a **carbon credit** is a tradable certificate representing **one tonne of CO_2-equivalent (tCO_2e)** either removed from the atmosphere or prevented from being emitted. It is verified by an independent third party and issued on a registry, so a buyer can claim it against their own emissions.

12.1 Types of carbon credits

Basis	Avoidance / reduction credits	Removal / sequestration credits
Mechanism	Emissions that would have occurred are prevented	CO_2 is physically taken out of the atmosphere and stored
Agriculture examples	No residue burning; AWD in rice (CH_4 ↓); efficient N (N_2O ↓); less diesel through ZT; biogas replacing firewood	SOC build-up (ZT, cover crops, residue retention); agroforestry; biochar
Permanence	Permanent once avoided	Reversible (SOC can be lost if practices stop); needs a buffer pool
Price (2024–26)	Lower (US$2–10/t for many nature-based avoidance credits)	Higher (US$15–40/t for SOC; US$100–200/t for durable biochar)
Market preference	Declining for net-zero claims	Preferred (Oxford Offsetting Principles; SBTi net-zero)

> **Sequestration vs reduction: Carbon sequestration** removes CO_2 already in the air and locks it into soil, biomass or biochar. **Emission reduction** prevents new CO_2, CH_4 or N_2O from entering the air. Both earn credits, but removals are more valuable for net-zero claims and harder to guarantee because they can be reversed.

12.2 How credits are assigned: the project cycle

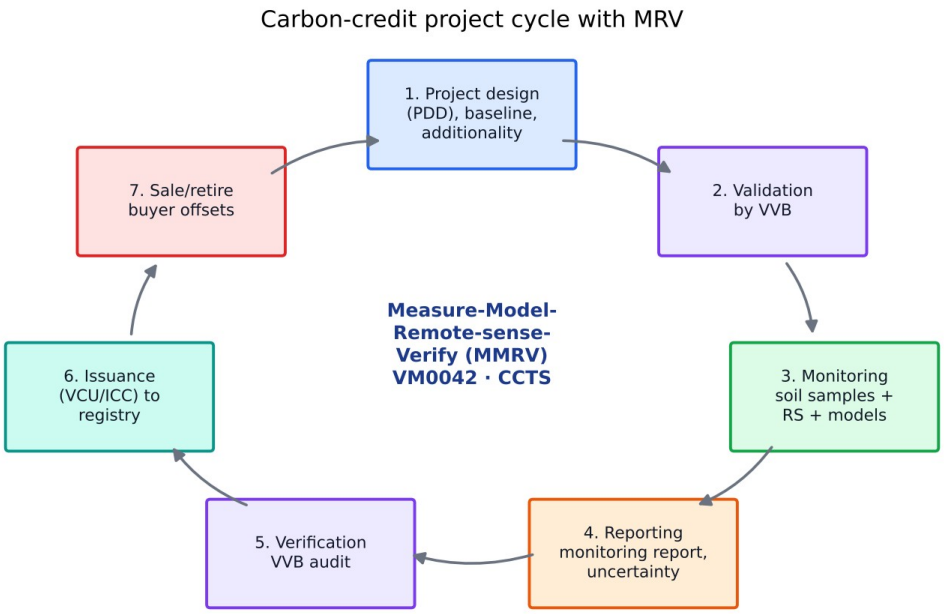


Figure 40. Carbon-credit project cycle with MRV at its core.

1. **Choose a methodology:** e.g. **Verra VM0042 (Improved Agricultural Land Management, v2.0/2.1)**, Gold Standard Soil Organic Carbon Framework, Verra VMD0053 (SOC model calibration), Puro.earth or Isometric Biochar methodologies, or VM0044 (biochar).
2. **Project Design Document (PDD):** project boundary (fields, by GIS polygons), baseline scenario (what would have happened), **additionality** (the practice would not occur without carbon finance), leakage, permanence and risk.
3. **Validation** by an accredited Validation/Verification Body (VVB), then **registration** on the registry.
4. **Implementation and monitoring:** farmer enrolment, a practice-change record (sowing method, residue, N use), soil sampling and **modelling plus remote sensing**.
5. **Calculation:** net removals = (project − baseline) SOC change + emission reductions − leakage − uncertainty deduction − **buffer contribution** (10–20 % non-permanence buffer).
6. **Verification** by the VVB (desk review, field visits, re-sampling), then **issuance** of credits (VCUs on Verra, GS-VERs, or Indian Carbon Credit Certificates).
7. **Sale and retirement:** buyers retire credits against their claims. Revenue is shared with farmers (typically 50–70 % to farmers in good programmes).

Key quality criteria (ICVCM Core Carbon Principles)

- **Additionality:** the activity would not happen anyway (regulatory surplus and financial barriers).
- **Baseline:** a conservative, realistic counterfactual.
- **Permanence:** storage lasts at least 40 years (VM0042 requires monitoring and a buffer). Reversals are compensated.

- **Leakage:** emissions displaced elsewhere are deducted, e.g. yield loss forcing land conversion elsewhere.
- **No double counting:** each tonne is claimed once only. **Corresponding adjustments** under Paris Agreement **Article 6** apply to internationally transferred credits.
- **Robust MRV:** quantification is transparent, conservative and independently verified.
- **Co-benefits and safeguards:** smallholder benefit-sharing, free prior and informed consent, and SDG impacts.

12.3 MRV methodology for soil-carbon projects

Soil carbon changes slowly (about 0.1–1 t C/ha/yr) against a large background stock (30–60 t C/ha in the top 30 cm) with high spatial variability, so measurement alone is expensive. Modern MRV is **MMRV: measure + model + remote sense + verify**.

Table 17. Three quantification approaches in VM0042

Approach	Description	When used
Measure-and-remeasure	Direct stratified soil sampling (SOC % × bulk density × depth, equivalent soil mass basis) at baseline and every 5 years	Costly; gold standard; for calibration
Measure-and-model	Calibrated biogeochemical models (DayCent, RothC, DNDC, APSIM) driven by practice data; soil samples validate the models	Most common; needs model validation for the region (VMD0053)
Default factors	IPCC Tier 1/2 emission factors (e.g. N_2O EF 1 %, rice CH_4 scaling factors)	For N_2O and CH_4 components

Role of remote sensing in MRV

- **Practice verification:** tillage (NDTI, SAR coherence), residue burning (VIIRS fires, dNBR), cover crops (winter NDVI), rice flooding regime for AWD (Sentinel-1 water detection), sowing dates and crop rotation (crop-type maps).
- **Model drivers:** NPP/GPP (LUE, SIF), biomass and residue amount (from yield maps), and soil moisture.
- **Stratification of soil sampling:** use bare-soil composites, DEM and NDVI history to design efficient sampling (cLHS), which reduces the number of cores needed.
- **Direct SOC mapping:** spectral bare-soil SOC prediction (Sentinel-2, hyperspectral PRISMA/EnMAP), which is emerging.
- **Agroforestry biomass:** GEDI LiDAR, NISAR L-band, high-resolution imagery for tree counts, and allometry.
- **Digital MRV (dMRV):** farmer app records, geo-tagged photos and blockchain registries, which cut verification cost. Transaction costs often exceed credit value for smallholders.

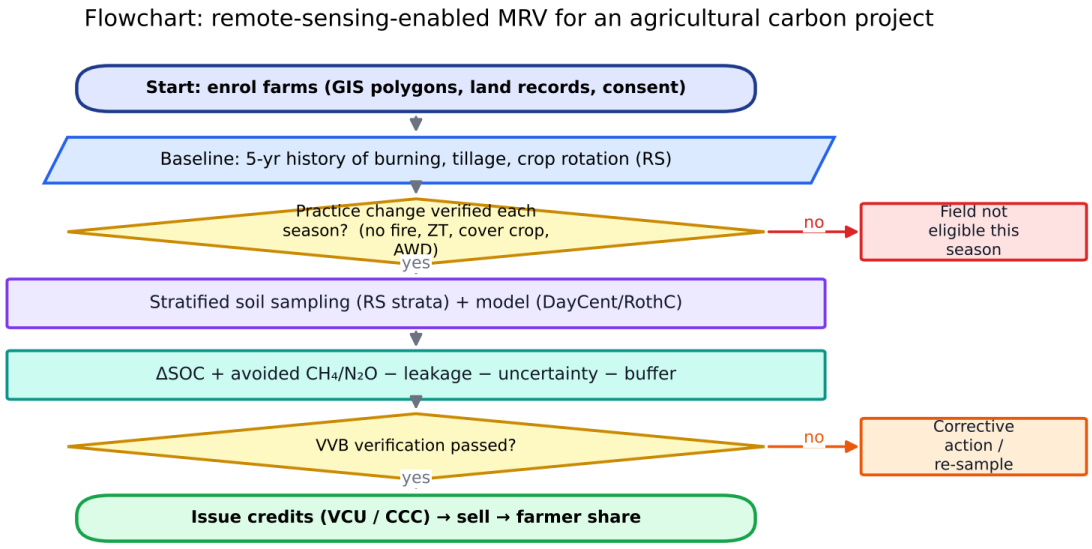


Figure 41. Flowchart of remote-sensing-enabled MRV for an agricultural carbon project.

12.4 Biochar and carbon credits

- **What:** charcoal produced by **pyrolysis** of biomass (300–700 °C, low oxygen). About 50 % of the feedstock C ends up in a highly aromatic, recalcitrant form.
- **Permanence:** H/Corg < 0.7 (EBC/IBI criteria) indicates stability. 100-year persistence is about 70–90 % depending on the H/Corg ratio and temperature.
- **Calculation:** credits = biochar mass × C content × permanence factor − project emissions (feedstock transport, pyrolysis energy, CH_4 from kilns) − leakage.
- **Benefits:** better CEC, water holding, liming effect and microbial habitat; lower N_2O; possible yield gains in acidic and degraded soils.
- **Markets:** biochar is one of the largest categories of durable carbon removal (CDR) purchases, with buyers such as Microsoft and Frontier at about US$120–200/t. India has projects on paddy straw, cotton stalk and Prosopis juliflora, with credits on Puro.earth, Isometric and Verra VM0044.

12.5 Who buys carbon credits?

- **Voluntary carbon market (VCM):** corporations with net-zero or ESG commitments (tech, aviation, oil and gas, FMCG, banks). Examples: Microsoft, Google, Shell, Delta, and Indian firms such as Infosys, Wipro and Tata Steel. Brokers and exchanges include Xpansiv CBL and AirCarbon.
- **Compliance markets:** EU ETS, California cap-and-trade and the **CORSIA** aviation scheme. Article 6.2 and 6.4 buyers include Japan JCM, Singapore and Switzerland.
- **Agriculture-specific buyers:** food and beverage companies reducing Scope 3 emissions through **insetting** within their own supply chains (Nestlé, PepsiCo, Unilever).

12.6 The situation in India

Element	Status
Legal basis	Energy Conservation (Amendment) Act 2022 enables a national carbon market
Carbon Credit Trading Scheme (CCTS) 2023	Notified by the Ministry of Power (June 2023). The Bureau of Energy Efficiency (BEE) is the administrator and Grid Controller of India the registry. Two mechanisms: a compliance mechanism (GHG emission-intensity targets for obligated sectors such as aluminium, cement, chlor-alkali, pulp and paper, and refineries; intensity targets notified in 2025) and an offset mechanism (voluntary projects earn Indian Carbon Credit Certificates, CCC)
Offset methodologies	BEE approved initial methodologies including renewable energy, green hydrogen, mangroves and agroforestry. Agriculture-related methods (rice cultivation, SOC) are under development or consultation
Green Credit Programme (2023, MoEFCC)	Separate non-carbon green credits for tree plantation, water conservation and similar activities, administered by ICFRE
Article 6	India listed activities eligible for international transfer (e.g. green hydrogen, storage, CCUS), with corresponding adjustments only for these
Agri projects in the VCM	Developers such as Varaha, Boomitra, Grow Indigo (Mahyco) and the NABARD-backed pilots: Verra VM0042 projects for DSR, AWD, ZT, no-burn and agroforestry with thousands of smallholders across Punjab, Haryana, MP and Telangana
Challenges	Smallholder fragmentation, high MRV cost per hectare, baseline and additionality on smallholdings, land records, trust and benefit-sharing, credit price volatility
Research (ICAR/IARI)	Cariappa et al. (*Scientific Reports* 2024) used farmer surveys and remote sensing to assess CA eligibility under VM0042 in the IGP. IARI's CREAMS ZT mapping and burning monitoring provide remote-sensing evidence of practice change, the core of MRV

12.7 IARI and Division of Agricultural Physics involvement in MRV

- **Practice monitoring at scale:** fire-event and burnt-area maps (no-burn verification), the Zero-Tillage Mapper (CA adoption), and crop-type and sowing-date maps. These are the activity data of MRV.
- **Flux and soil measurements:** soil-physics and agromet labs measure SOC, bulk density, soil respiration and GHG fluxes (chamber and EC), which provide model calibration data.
- **Modelling:** InfoCrop (IARI), DSSAT and APSIM run for SOC and GHG scenarios. Remote-sensing-derived NPP feeds into RothC.
- **Hyperspectral soil SOC:** spectral libraries and ML models for low-cost SOC estimation.
- **Policy inputs:** technical support to MoA&FW and NITI Aayog on agricultural carbon markets and on standard protocols (as for fire monitoring).
- **The paper by "Dr. Vinay":** most likely by **Dr. Vinay Kumar Sehgal** (Head, Division of Agricultural Physics) and colleagues, on remote-sensing-based monitoring of residue management and conservation practices for climate and MRV applications. **Check the exact title in his Google Scholar list before the interview.**

12.8 Interview questions

Q. What is additionality? Give an agriculture example.

A. A credit is additional only if the emission reduction or removal would not have happened without carbon-credit revenue. Example: a Punjab farmer adopts zero tillage plus residue retention only because the carbon payment covers the Super Seeder rental. If ZT were already common practice or legally mandated, it would not be additional.

Q. How can remote sensing reduce MRV cost?

A. Remote sensing verifies practices (tillage, burning, cover crops, flooding) over every enrolled field without visits. It stratifies soil sampling to reduce the number of cores, provides model drivers (NPP, biomass, soil moisture), and detects reversals. This turns MRV from field-by-field audits into wall-to-wall digital monitoring.

Q. Why is soil carbon controversial as an offset?

A. It is hard to measure accurately (high variability, slow change, depth issues). It is reversible, saturates over time, and has additionality and leakage problems. Model-based estimates can be over-optimistic. Strong MRV, conservative uncertainty deductions and buffer pools are therefore essential.

Q. What is the difference between CCTS compliance and offset mechanisms?

A. Under compliance, obligated industries get emission-intensity targets. Those that beat their targets earn credits, and those that miss must buy them. Under the offset mechanism, non-obligated entities (including agriculture and forestry projects) voluntarily register projects under approved methodologies and earn carbon credit certificates.

13. Precision agriculture and variable-rate technology

One-line definition: precision agriculture (PA) is a management strategy that gathers, processes and analyses temporal, spatial and individual data and combines it with other information. It supports management decisions according to estimated variability, for better resource-use efficiency, productivity, quality, profitability and sustainability (International Society of Precision Agriculture, 2019).

In five words: "right input, right place, right time, right amount" (and right source, the 4R nutrient stewardship framework).

13.1 Principles

1. **Variability exists:** spatial variability within and between fields (soil, topography, pests) and temporal variability between seasons and within a season.
2. **Variability can be measured:** GNSS, sensors and remote sensing.
3. **Variability is manageable:** management zones and variable-rate application.
4. **Managing it pays:** economic and environmental benefits outweigh the cost of the technology.
5. **Closed-loop cycle:** data, analysis, decision, application, evaluation, then back to data.

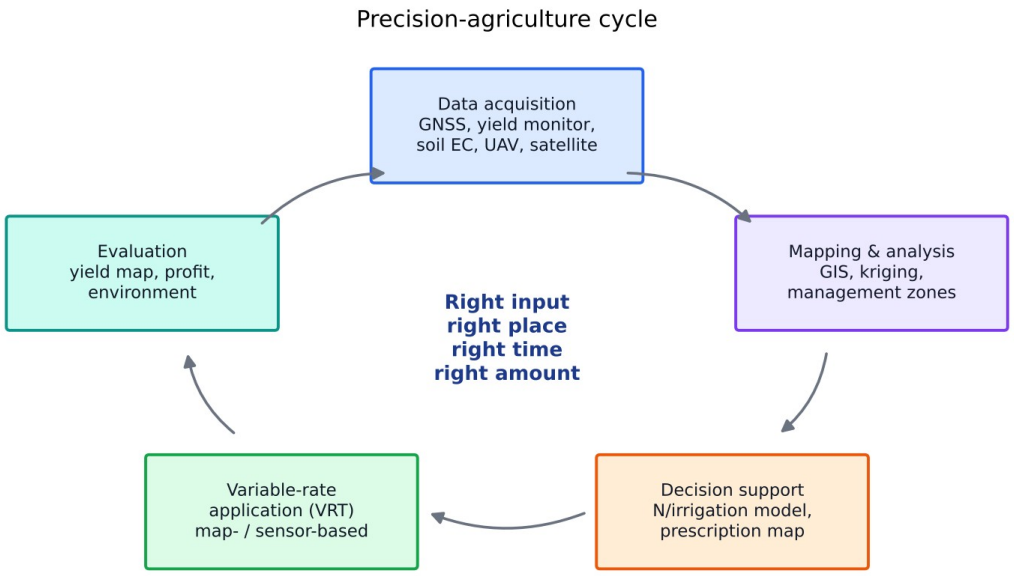


Figure 42. The precision-agriculture management cycle.

13.2 Tools and technologies

Table 18. Precision-agriculture toolkit and applicability in India

Tool	Function	Applicability in India
GNSS (GPS, NavIC/IRNSS, GAGAN)	Positioning; auto-steer (RTK ±2 cm); geo-referencing samples	High (phones, NavIC); RTK limited to large farms and CHCs

Tool	Function	Applicability in India
Remote sensing (satellite)	Crop vigour, stress, N status, ET, yield prediction; Sentinel-2 free 10 m	High: scalable, low cost; cloud cover limits kharif use; holdings are small
UAVs / drones	High-resolution multispectral and thermal imaging; spraying (Kisan drones; Namo Drone Didi scheme)	Growing fast; DGCA rules; spraying viable through custom hiring
Proximal crop sensors	GreenSeeker (NDVI), Crop Circle, SPAD, LCC	High (LCC, GreenSeeker): low cost, proven for N in rice and wheat (PAU, IARI)
Soil sensors	EC mapping (EM-38, Veris), pH, moisture probes (capacitance), NPK sensors	Medium; IoT moisture probes for irrigation spreading
GIS and geostatistics	Kriging, management zones (fuzzy k-means), prescription maps	High (QGIS, Bhuvan, GEE)
Yield monitors	Mass-flow sensor + GNSS on combine; yield maps	Low (custom combines lack them); potential with CHC combines
Variable-rate technology (VRT)	Applies inputs at variable rate from a map or sensor	Low–medium; research prototypes at IARI, PAU and CIAE
Laser land leveller	Precision levelling; ±2 cm	High: widely adopted in Punjab and Haryana; saves 20–25 % water
Decision-support systems	Crop models, Nutrient Expert (IPNI/CIMMYT), irrigation schedulers, AI apps	High via mobile phones (Kisan e-Mitra, Krishi DSS)
IoT and AI	Sensor networks, automated drip and fertigation, pest traps with image AI	Medium; horticulture, polyhouses

13.3 Variable-rate technology (VRT)

- **Map-based VRT:** a prescription map is prepared in advance from soil tests, yield maps or imagery. The applicator reads the map and its GNSS position. Best for P, K, lime and seed rate (slowly varying properties).
- **Sensor-based (on-the-go) VRT:** a real-time sensor (e.g. GreenSeeker NDVI) and an algorithm compute the rate instantly. Best for in-season N (fast-changing properties).
- **Hybrid:** a map defines zones and sensors fine-tune the rate within each zone.
- **Components:** controller, GNSS receiver, actuators (variable-speed hydraulic motors, pulse-width-modulated nozzles), flow sensors and the prescription software.
- **Inputs that can vary:** seed rate (variable-rate seeding), fertiliser (N, P, K), lime, pesticides (spot spraying with machine vision, “see and spray”), irrigation (VRI pivots) and plant-growth regulators.
- **N algorithm example (Raun et al.):** YP_0 = f(INSEY = NDVI/days from sowing). The response index RI = NDVI(N-rich strip)/NDVI(farmer field). N rate = (YP_N − YP_0) × grain N %/NUE.

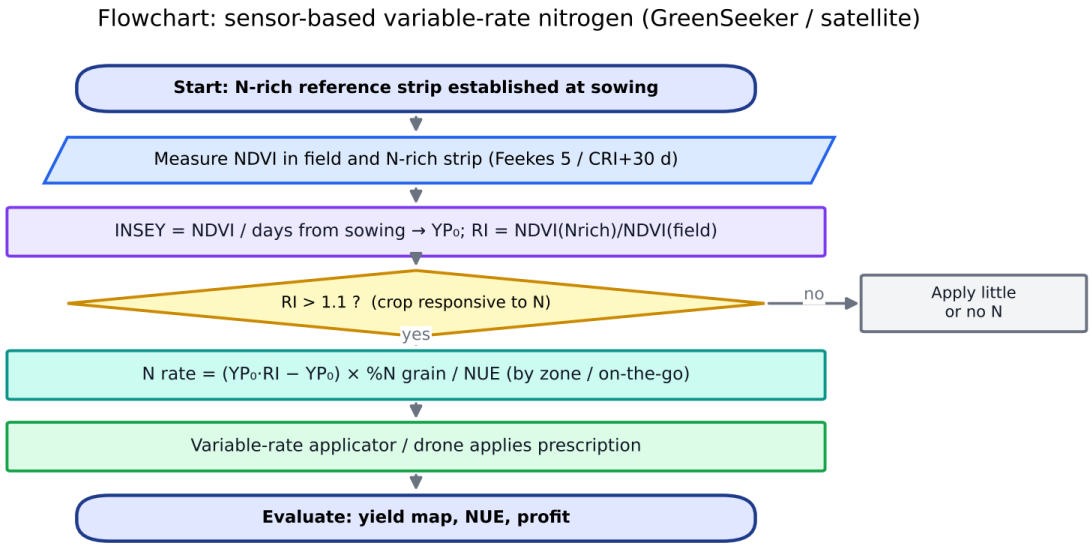


Figure 43. Flowchart of sensor-based variable-rate nitrogen management.

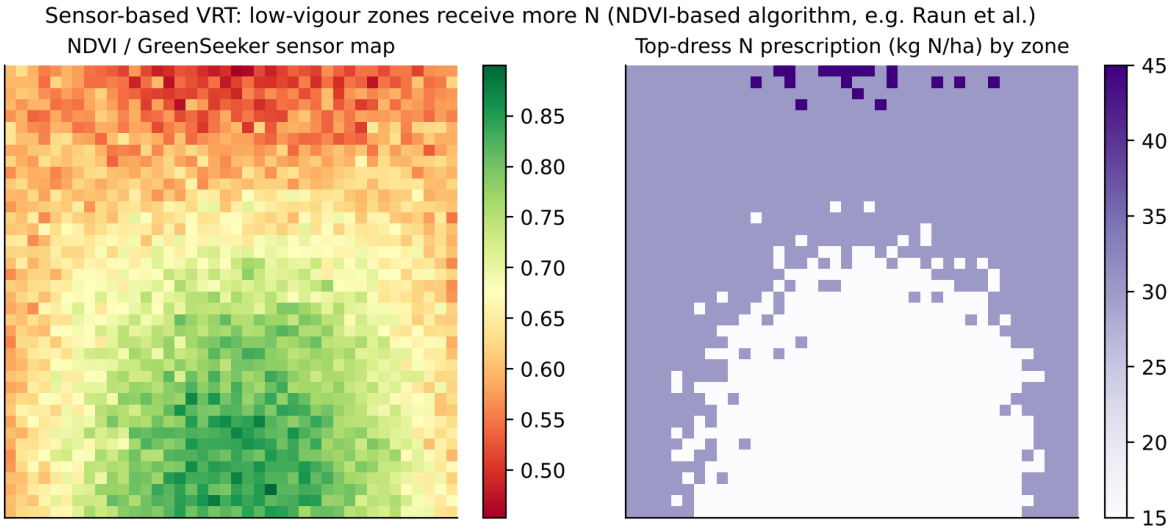


Figure 44. Sensor-based VRT: an NDVI vigour map converted into a zone-wise N top-dressing prescription.

13.4 Applications

- **Nutrient management:** site-specific nutrient management (SSNM), the LCC and GreenSeeker for N in rice and wheat (10–15 % N saving at the same yield), and Soil Health Card based zone recommendations.
- **Water:** soil-moisture-sensor-based drip and sprinkler irrigation, thermal CWSI scheduling, and ET-based scheduling (FAO-56 + Kc from NDVI).
- **Crop protection:** drone spraying, early pest and disease detection (hyperspectral, AI image apps), and weather-based forecast models.
- **Harvest and post-harvest:** yield mapping, quality sensing (protein NIR) and logistics.
- **Horticulture and protected cultivation:** fertigation, climate control, precision orchards (canopy-volume-based spraying).

13.5 Applicability and adoption in India

Constraints	Enablers and the way forward
Small, fragmented holdings (86 % of holdings < 2 ha; average 1.08 ha)	Treat a group of farms or a village as the 'precision unit'; FPO and CHC custom hiring of VRT and drones
High equipment cost; low farmer literacy in digital tools	Low-cost sensors (LCC, phone-based NDVI); subsidies (SMAM, Kisan drone)
Lack of yield monitors and data infrastructure	AgriStack, Digital Crop Survey, Soil Health Card digital records, Bhuvan, NavIC
Heterogeneity within small plots is often smaller than between plots	Satellite-based (Sentinel-2) zone mapping at village scale; between-farm precision
Weak extension linkage	KVKs, apps in local languages (Bhashini), the ICAR-NePPA network

13.6 ICAR Network Programme on Precision Agriculture (NePPA) and IARI's role

- **Launched:** 8 September 2021 by Dr. Trilochan Mohapatra (then Secretary DARE and DG ICAR).
- **Lead institute: ICAR-IARI, New Delhi** (coordination; the then Director, Dr. A. K. Singh). **16 partner institutes** across 7 subject-matter divisions: crop science, NRM, engineering, horticulture, animal science, fisheries and others.
- **Aim:** develop precision ag-techs for better input-use efficiency, a safe production environment and profitable, quality produce. This covers precision crop production as well as **precision livestock farming and precision aquaculture**.
- **Components:** sensor development (soil, crop and nano-sensors), drone and AI applications, IoT-based irrigation and fertigation, VRT machinery, decision-support systems, and data analytics. The **ICAR Data Centre** handles data storage and analytics.
- **IARI's role:** overall coordination, plus Agricultural Physics work on remote sensing, hyperspectral and crop sensors, Agricultural Engineering work on VRT machines and drones, and Water Technology Centre work on precision irrigation. IARI also hosts the steering and monitoring framework (monitoring presented by Dr. Anil Rai, ADG ICT).

13.7 Space-based precision agriculture and the ISRO–SAC context

- **SAC (Space Applications Centre, Ahmedabad)** pioneered crop production forecasting (the CAPE project) and develops methods and pilots for space-based precision farming. These include field-level crop condition, N-stress mapping from red-edge and hyperspectral data, soil moisture, and site-specific advisory.
- **Satellite-based PA in India:** village-scale management zones from multi-year Sentinel-2/Resourcesat NDVI, field-level advisories in local languages (pilot projects with state agriculture departments), and integration with NavIC for geotagging.

- **Future:** NISAR soil moisture (100 m), TRISHNA thermal ET (57 m), hyperspectral missions for N and protein, and FLEX SIF for stress. Together these allow **within-season, field-scale prescriptions**.

13.8 Interview questions

Q. Is precision agriculture relevant for Indian smallholders?

A. Yes, in adapted forms. Low-cost tools (LCC, GreenSeeker, phone apps), satellite-based advisories at field or village level, CHC-based drone and VRT services, and laser levelling deliver precision benefits without each farmer owning equipment. Precision across farms in a village can matter more than variability within a single farm.

Q. Map-based vs sensor-based VRT?

A. Map-based VRT uses a prescription prepared earlier from soil, yield or imagery data. It suits slowly varying properties such as P, K and lime. Sensor-based VRT measures crop status on the go and computes the rate in real time. It suits in-season N.

Q. How does GreenSeeker work?

A. It is an active optical sensor that emits red (≈660 nm) and NIR (≈770 nm) light and measures the reflected fraction to compute NDVI, independent of sunlight. NDVI is related to yield potential through the INSEY algorithm, and comparison with an N-rich reference strip gives the top-dress N rate.

14. Crop insurance and yield estimation (the likely main focus)

One-line definition: crop yield estimation predicts or measures the production per unit area of a crop at a chosen spatial unit (field, GP, district) and time (in season or at harvest). In India these estimates drive **food-security planning (FASAL)** and **crop-insurance claim settlement (PMFBY/YES-TECH)**.

14.1 The Indian crop-statistics and insurance ecosystem

Table 19. Institutions and programmes

Body / programme	Role
MNCFC (Mahalanobis National Crop Forecast Centre), DA&FW	Set up in 2012 in New Delhi. Runs FASAL, CHAMAN, NADAMS and technical support for PMFBY (YES-TECH, smart sampling)
ISRO (SAC, NRSC)	Developed the methods (CAPE → FASAL), provide satellite data (Resourcesat, EOS-04, NISAR), and act as mentors for ML/DL yield models
IMD / IASRI / SAUs	Agro-meteorological yield models; statistical methodology for CCEs (ICAR-IASRI); state-level testing
NSO / DES / state agriculture departments	General Crop Estimation Surveys (GCES): CCEs for official yield statistics
PMFBY (since Kharif 2016)	Area-yield-index insurance for food crops, oilseeds and annual commercial/horticultural crops
RWBCIS	Restructured Weather-Based Crop Insurance Scheme: weather-index cover, popular for horticulture
Digital tools	YES-TECH, WINDS (weather station network), CROPIC (crop photos), DigiClaim, NCIP portal, AIDE app, Krishi Rakshak helpline 14447

14.2 FASAL: Forecasting Agricultural output using Space, Agro-meteorology and Land-based observations

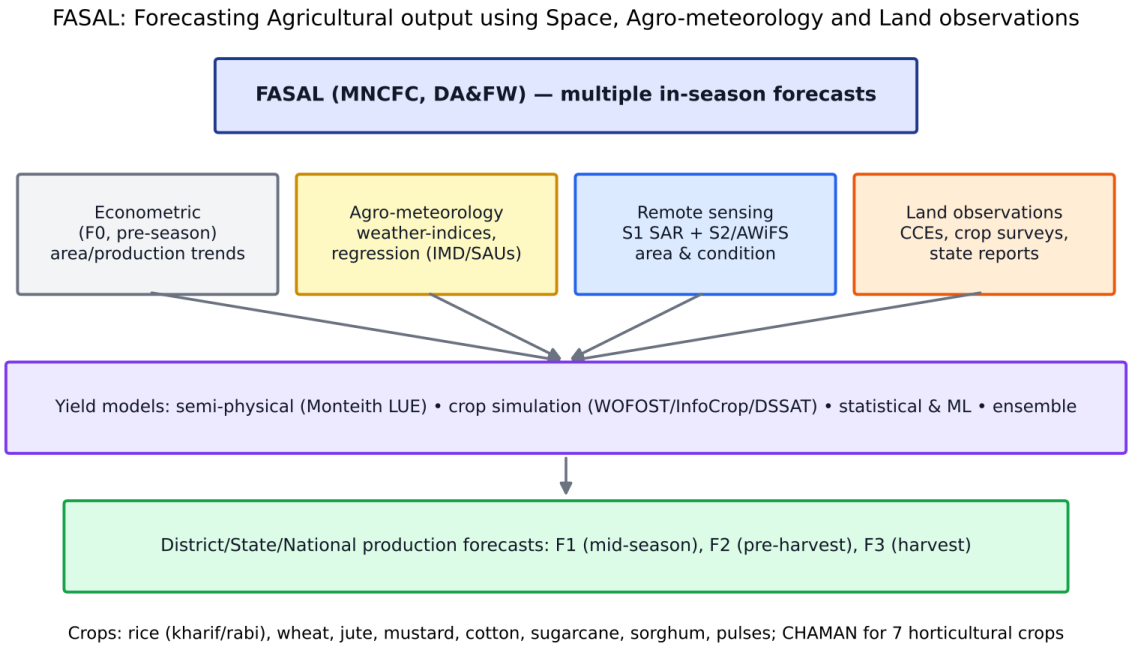


Figure 45. FASAL's multiple-forecast framework combining econometric, agromet, remote-sensing and field data.

- **History:** the CAPE (Crop Acreage and Production Estimation) project at SAC (late 1980s) led to the **FASAL** concept in 2006 and to operational forecasting at MNCFC since 2012.
- **Crops:** kharif and rabi rice, wheat, jute, rapeseed–mustard, cotton, sugarcane, rabi sorghum and rabi pulses, at national, state and district levels.
- **CHAMAN** (Coordinated Horticulture Assessment and Management using geoiNformatics, 2014): area and production of 7 horticultural crops (banana, mango, citrus, potato, onion, tomato, chilli) in major districts. It also covers site suitability and post-harvest infrastructure.
- **Area estimation:** a stratified random sampling design (5 km × 5 km segments). Rabi crops are classified from multi-date Resourcesat AWiFS/LISS-III and Sentinel-2 data, and kharif crops from **SAR (Sentinel-1, EOS-04)** because of clouds. Classification accuracy is above 90 %.
- **Yield estimation:** a multi-model approach (Section 14.3). Production = area × yield.
- **Forecast stages:** a pre-season econometric forecast, then mid-season (RS + weather) and pre-harvest forecasts, and a final estimate, given before the DES advance estimates.

14.3 Yield models used in FASAL and YES-TECH

(a) Semi-physical (light-use-efficiency) model

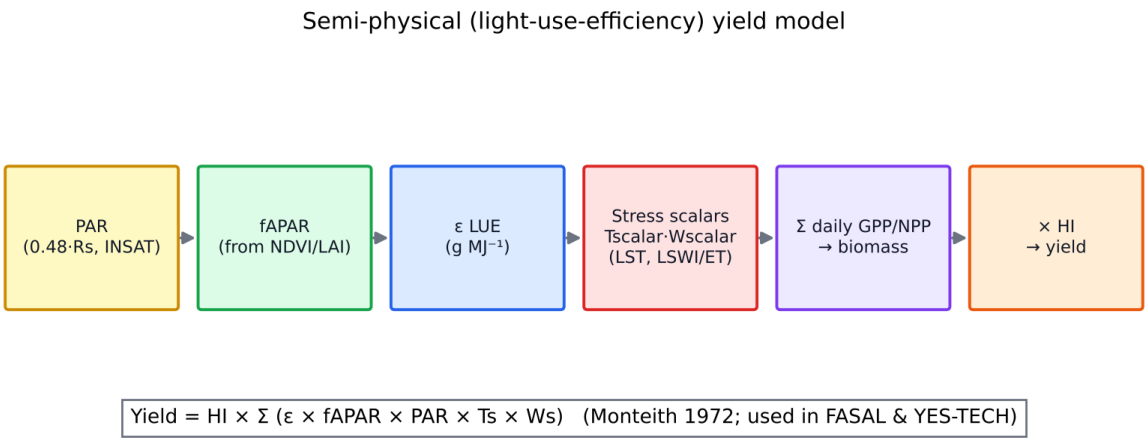


Figure 46. Semi-physical (Monteith LUE) yield model chain (after Monteith 1972).

$$\text{GPP/NPP} = \Sigma\, \varepsilon \cdot \text{fAPAR} \cdot \text{PAR} \cdot \text{Tscalar} \cdot \text{Wscalar}\ ; \quad \text{Yield} = \text{HI} \times \text{Biomass}$$

- **PAR** = about 0.48 × incoming shortwave (INSAT-3D insolation). **fAPAR** from NDVI (fAPAR ≈ 1.24·NDVI − 0.168) or LAI (Beer's law).
- **ε (RUE):** about 2.5–3.0 g MJ^{-1} for wheat and rice (C3) and 3.5–4.0 for maize (C4), reduced by stress scalars: temperature (a CASA/VPM function) and water (LSWI-based or ET/PET-based Ws = 0.5 + 0.5·ETa/ETp).
- **HI:** a crop-specific harvest index (wheat about 0.40–0.45, rice 0.45–0.50), possibly modulated by stress during grain filling.
- **Pros:** physically interpretable and spatially explicit (pixel-wise). It works without CCE calibration and needs only a few parameters.
- **Cons:** the ε and HI assumptions, mixed pixels and cloud gaps. Calibration against CCEs is still recommended.

(b) Crop simulation models with RS assimilation

- **WOFOST, DSSAT and InfoCrop**, run on a 5 km grid or at GP level with gridded weather (IMD), soil (NBSS&LUP) and RS-derived sowing dates.
- **Assimilation** of RS LAI and soil moisture (Chapter 9) corrects the model trajectory.
- **Pros:** handles weather extremes mechanistically and extends the forecast with weather scenarios. **Cons:** needs many inputs (cultivar, management) and calibration.

(c) Statistical and agro-meteorological models

- **Spectral–yield regression:** Yield = a + b·(peak NDVI or Σ NDVI) + c·trend.
- **Weather-indices (Agrawal/IASRI) model:** weekly weather variables (Tmax, Tmin, rain, RH) weighted by their correlation with yield over the crop season, with a time trend. IMD and AMFUs use it for district forecasts.

- **Pros:** simple. **Cons:** stationarity assumption, limited spatial detail, and overfitting with short time series.

(d) Machine-learning models

- RF, XGBoost, SVR and ANN, using features from Sentinel-1/2 time series (NDVI, NDRE, VH, VH/VV), weather (GDD, rain, heat days), soil (texture, SOC), SIF, ET and soil moisture. CCE yields are the labels.
- Deep learning (LSTM, CNN-LSTM, Transformers) on pixel or GP time series.
- **Pros:** captures non-linearity and makes use of many predictors. **Cons:** needs many good-quality CCE labels, transfers poorly between years, and is less interpretable (mitigated by SHAP).

(e) Ensemble

- Combines the models above using simple averaging, inverse-RMSE weighting ($w_i = (1/RMSE_i^2)/\Sigma(1/RMSE_j^2)$), Bayesian model averaging, or **stacking**, where a meta-learner such as ridge or RF is trained on base-model predictions.
- **Why it helps:** different models have different errors, so averaging reduces variance and the risk that one model fails in an unusual year. Ensembles consistently beat their individual members in insurance pilots.
- **Model spread** across ensemble members is a useful **uncertainty indicator**: GPs with high disagreement can be flagged for extra CCEs.

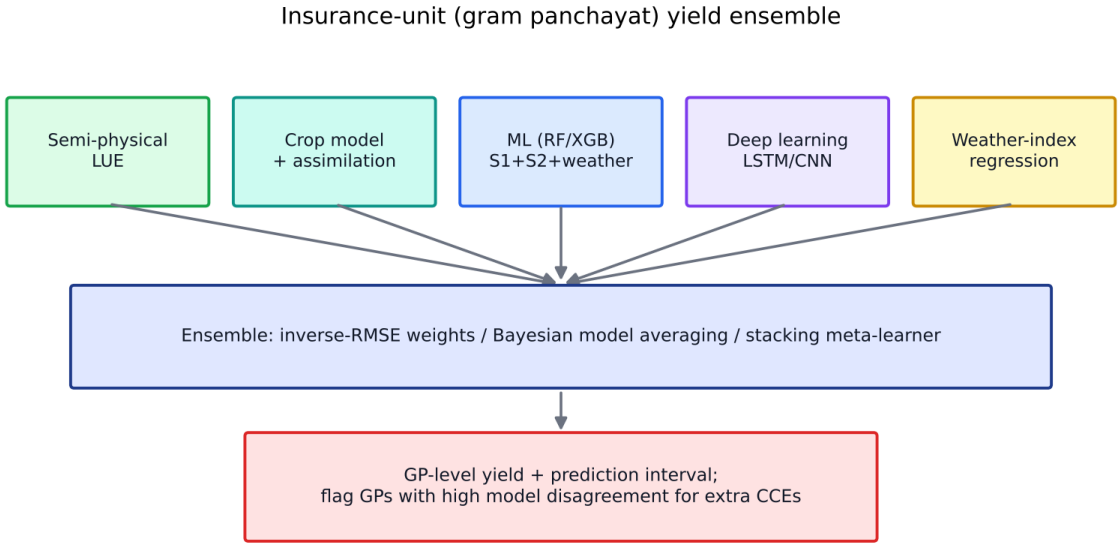


Figure 47. GP-level ensemble yield modelling with uncertainty-driven CCE allocation.

14.4 PMFBY in brief

Feature	Provision
Launch	Kharif 2016; restructured 2020 (voluntary for all farmers); continued to 2025-26 with a technology fund (FIAT, the Fund for Innovation and Technology)

Feature	Provision
Farmer premium	2 % kharif, 1.5 % rabi (food and oilseed crops), 5 % annual commercial and horticultural crops; the rest is shared between Centre and states (90:10 in NE states)
Approach	Area approach: the insurance unit (IU) is the village or gram panchayat for major crops and a larger unit for minor crops
Risks covered	Prevented sowing (claim up to 25 % of sum insured), mid-season adversity (on-account payment of 25 %), standing crop yield loss, post-harvest losses (14 days, individual farm), localised calamities (hail, landslide, inundation, cloudburst, natural fire: individual farm)
Threshold yield (TY)	Average yield of the best 5 of the last 7 years × indemnity level (70/80/90 %)
Claim	Claim = (TY − AY)/TY × sum insured, where AY is the actual yield from CCEs and/or YES-TECH
CCEs	At least 4 CCEs per GP/IU per major crop; recorded on the CCE-Agri app with geotagged photos and timestamps

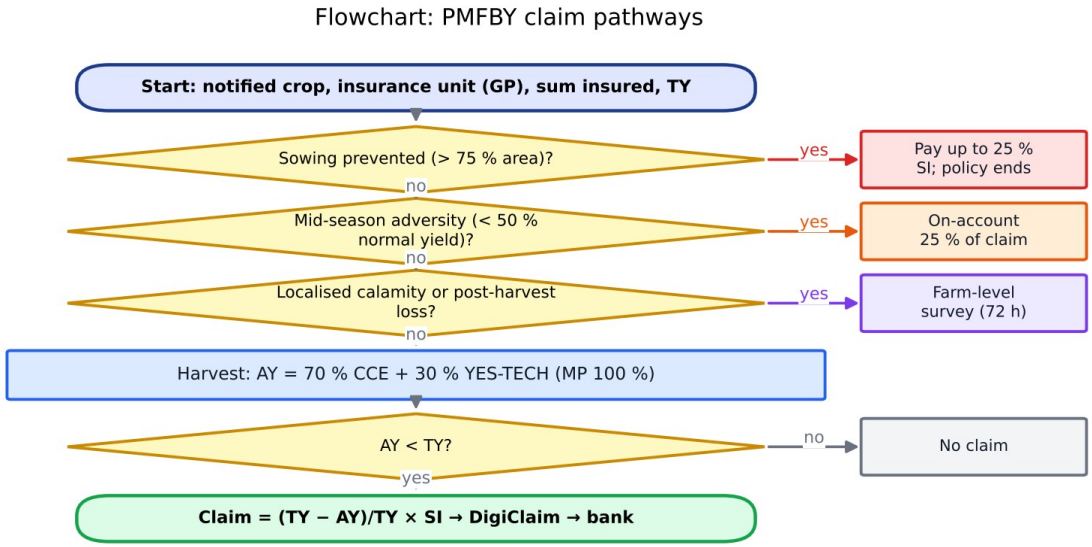


Figure 48. Flowchart of PMFBY claim pathways (prevented sowing, mid-season adversity, localised loss, yield shortfall).

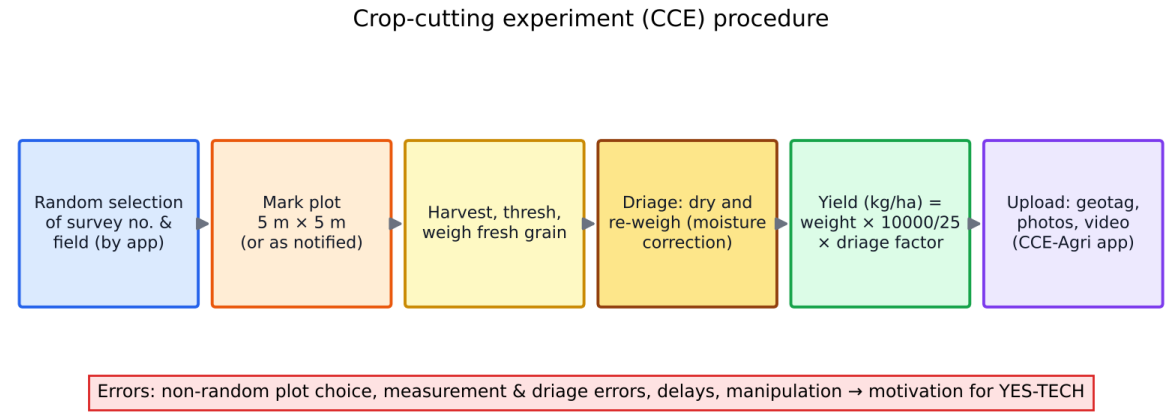


Figure 49. Crop-cutting experiment (CCE) procedure and its sources of error.

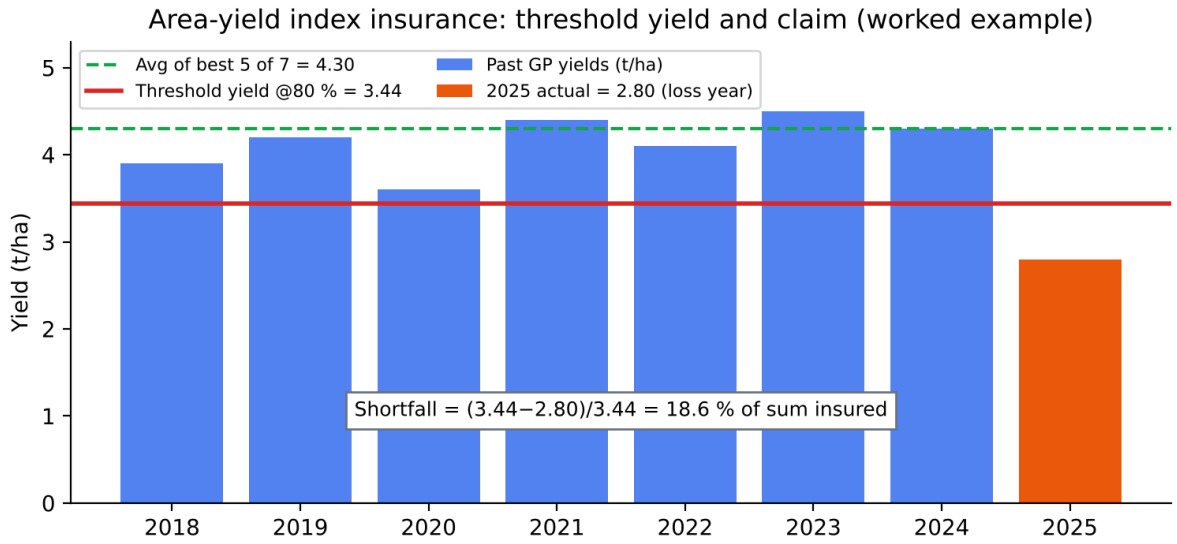


Figure 50. Worked example of threshold yield and claim calculation under the area-yield-index approach.

Practice this: Worked example: past yields over 7 years = 3.9, 4.2, 3.6, 4.4, 4.1, 4.5, 4.3 t/ha. Best 5 average = (4.5 + 4.4 + 4.3 + 4.2 + 4.1)/5 = **4.30**. At 80 % indemnity, TY = **3.44 t/ha**. If AY = 2.80 t/ha, shortfall = (3.44 − 2.80)/3.44 = **18.6 %**. With a sum insured of ₹50,000/ha, the claim is **₹9,300/ha**.

14.5 Yield-index approach for cereals vs weather-based approach for horticulture

Table 20. Area-yield index vs weather index insurance

Aspect	Yield-index (PMFBY, cereals and pulses)	Weather-index (RWBCIS, horticulture)
Index	Actual area yield of the IU (GP) measured by CCEs/technology	Weather parameters (rain deficit or excess, temperature, RH, wind, hail) from reference AWS/WINDS
Trigger	AY < threshold yield (TY = best-5-of-7 average × indemnity level)	Parameter crosses a trigger (e.g. cumulative rain < X mm in a phase), with payout scaled from strike to exit

Aspect	Yield-index (PMFBY, cereals and pulses)	Weather-index (RWBCIS, horticulture)
Why suited	Cereals have long yield records and many CCEs; yield integrates all perils	Perennial and horticultural crops lack yield history; yield measurement is difficult (multiple pickings); weather is the dominant peril
Basis risk	Lower (it measures the outcome) but suffers CCE errors and delays	High: weather at the station may not match farm loss
Settlement speed	Slow (harvest plus CCE compilation)	Fast (data are available immediately)
RS role	Smart sampling, direct yield estimation (YES-TECH)	Satellite rainfall, soil moisture, LST and ET indices to reduce basis risk; gap-filling where AWS are sparse

14.6 YES-TECH: Yield Estimation System based on Technology

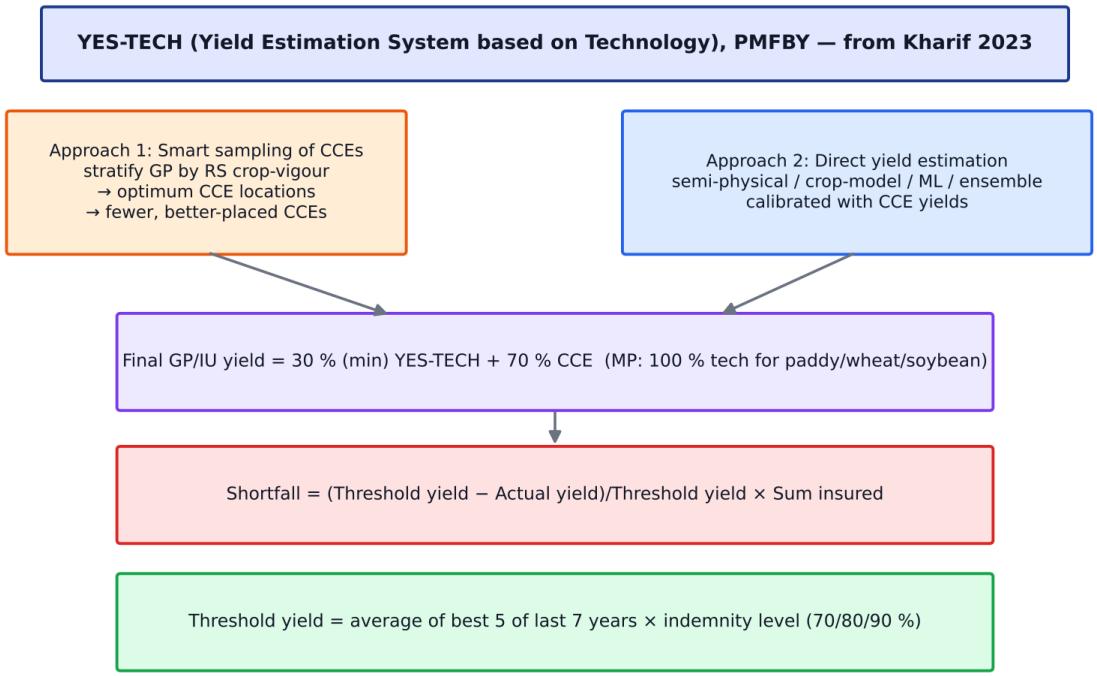


Figure 51. YES-TECH's two pathways (smart sampling and direct estimation) and how they feed claim settlement.

- **Start: Kharif 2023** for paddy and wheat, after pilots run by MNCFC (2019–2022, building on the KISAN project of 2015–16). **Soybean was added from Kharif 2024.** About 10 states take part, including MP, Maharashtra, UP, Odisha, Haryana, Rajasthan and Tamil Nadu.
- **Weightage: at least 30 % of the final yield** must come from YES-TECH, with the rest from CCEs. **Madhya Pradesh uses 100 %** YES-TECH weightage for its notified crops.
- **Governing document:** the **YES-TECH Manual (2023)**, which sets out data, models, accuracy criteria and the empanelment of **Technology Providing Agencies (TPAs)**. MNCFC and NRSC act as technical mentors.

- **Model families allowed:** (1) **index-based / semi-physical** (RS yield proxies, LUE models), (2) **crop simulation models** with RS inputs, (3) **statistical and ML/DL** models, and (4) **hybrid and ensemble** models.
- **Process:** acreage and crop map, then feature extraction (VI time series, SAR, weather, soil), then model training on historical and current CCEs, then GP-level yield, accuracy checks against holdout CCEs, and finally submission to the NCIP portal.
- **Benefits:** faster claim settlement, fewer disputes, less CCE workload, lower manipulation risk, and objective, spatially consistent data.
- **Criticism and challenges:** model accuracy at GP level for heterogeneous smallholdings, cloud cover in kharif, quality of the CCEs used for training, transparency of proprietary models, and the need for third-party audits and standard validation protocols (CSE 2025 commentary).

Two ways remote sensing is used in crop insurance

	1. Smart sampling of CCEs	2. Direct yield estimation
Goal	Improve CCE efficiency: fewer, better-placed CCEs	Estimate GP yield directly with models
Method	Stratify the GP or district by RS crop-vigour classes (NDVI, VCI, LAI, SIF). Allocate CCEs in proportion to stratum area and variance (Neyman allocation). Select plots within strata at random	Semi-physical, crop model, ML and ensemble models calibrated with CCEs
Output	Optimised CCE locations; lower sampling error; optimum number of CCEs	GP-level yield blended at ≥ 30 % with CCE yield
Advantage	Keeps the CCE (ground truth) but reduces bias and cost	Wall-to-wall, timely, objective
Risk	Still depends on CCE execution quality	Model error, especially in anomalous years

$$\text{Neyman allocation:}\quad n_h = n \cdot (N_h S_h) \,/\, \Sigma(N_h S_h) \qquad (N_h = \text{stratum size},\ S_h = \text{stratum SD})$$

14.7 Yield modelling at insurance-unit (GP) level: a design you can present

1. **Crop mask and field boundaries:** a crop map from Sentinel-1/NISAR + Sentinel-2 (RF/LSTM), with field boundaries from a U-Net or Bhu-Naksha cadastral maps, and **Digital Crop Survey** (AgriStack) plot-level crop data for training.
2. **Features per field:** NDVI, NDRE, LSWI and VH time series; phenology (SOS, peak, EOS); **SIF** (TROPOMI now, FLEX from 2027); ET and ESI; NISAR soil moisture; weather (IMD gridded, WINDS GP rain gauges) such as heat days and dry spells; soil (Soil Health Card, DSM).
3. **Base models:** (i) semi-physical LUE, (ii) WOFOST/DSSAT with LAI and SM assimilation, (iii) XGBoost/RF, (iv) LSTM, (v) agromet weather-index regression.
4. **Training:** with multi-year CCE data (geotagged), using **leave-one-year-out CV** and **spatial blocks**.

5. **Ensemble:** stacking with uncertainty (quantile regression or ensemble spread). Aggregate field-level yields to GP with area weighting.
6. **Validation:** against an independent holdout of CCEs. Report RMSE and MAPE (target < 10–15 % at GP level), bias, and the share of GPs within ±20 %.
7. **Decision layer:** blend with CCEs (≥ 30 %), flag high-uncertainty GPs for additional CCEs, and give early-warning loss estimates at the mid-season stage.

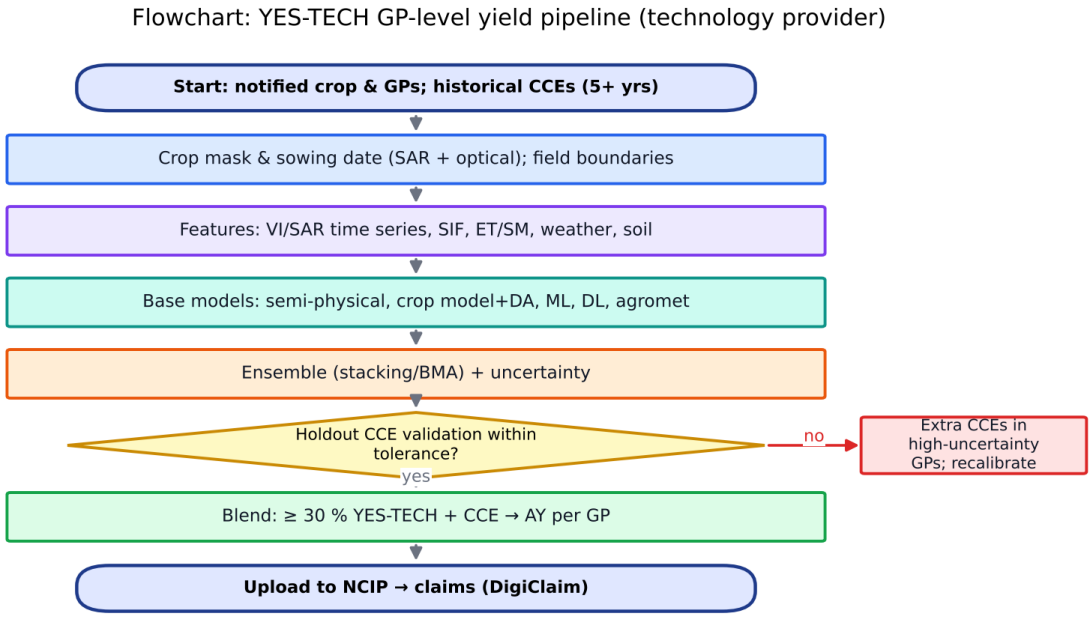


Figure 52. Flowchart of a YES-TECH GP-level yield pipeline run by a technology provider.

14.8 What we (ISRO and research) can contribute to crop insurance

- **Better yield models:** a GP-level ensemble with uncertainty, and integrating new sensors (NISAR, FLEX SIF, TRISHNA ET).
- **Designing insurance for the areas remote sensing can cover:**
- Extend cover to **non-notified and minor crops** and to **tenant and sharecropper farmers** using RS-based area and yield indices.
- **Hybrid index products:** a weather index plus a satellite index (soil moisture, ET deficit or ESI, NDVI/VCI or SIF anomaly) to reduce basis risk for horticulture and rainfed crops.
- **Parametric triggers** for **localised calamities**: flood inundation from SAR (NISAR, EOS-04), hail damage from sudden NDVI or backscatter change, and fire from VIIRS.
- **Prevented-sowing assessment** from SAR-based sowing detection by the cut-off date.
- **Pasture and fodder insurance for livestock** using NDVI anomalies (as in Kenya's IBLI index-based livestock insurance).
- **Premium rating and risk zonation:** long-term RS yield variability maps for actuarial pricing at GP level.
- **Fraud and dispute reduction:** check claimed sown area against the RS crop map, use CROPIC photos with AI verification, and audit CCE plot locations.

- **Faster settlement:** DigiClaim combined with early RS estimates.
- **Capacity building:** standard validation protocols, open benchmark datasets and audits of TPA models.

14.9 Key concepts in insurance economics

- **Basis risk:** the mismatch between the index payout and the farmer's actual loss. It is the main weakness of index insurance. Reduce it with finer units (GP), better indices (RS) and hybrid triggers.
- **Moral hazard:** insured farmers neglect management. Area-yield insurance limits this because an individual's actions barely affect the GP average.
- **Adverse selection:** high-risk farmers enrol more. Mandatory or bundled enrolment and risk-based premiums reduce it.
- **Loss ratio** = claims paid / gross premium. Under PMFBY it has typically been about 60–90 %, varying by state and season.
- **Actuarial premium rate (APR):** set by insurers through bidding. The farmer pays a capped share and the government pays the rest.

14.10 Interview questions on insurance and yield

Q. What is YES-TECH and how does it work?

A. YES-TECH is the technology-based yield estimation system under PMFBY, started in Kharif 2023 for paddy and wheat and extended to soybean in 2024. Empanelled agencies use remote sensing, weather and models (semi-physical, crop simulation, ML or ensemble), calibrated with CCEs, to estimate GP-level yield. That yield gets at least 30 % weight in the final yield used for claims (100 % in MP).

Q. What is smart sampling?

A. Remote sensing stratifies an insurance unit or district by crop condition. CCEs are then allocated across strata to reduce sampling error (Neyman allocation) and placed at random within strata. This gives a more representative yield with fewer CCEs.

Q. Why combine models into an ensemble?

A. Individual models have different structural errors. Semi-physical models miss management effects, ML models fail outside their training range, and crop models depend on their inputs. Weighted or stacked ensembles reduce variance and the risk of one model failing in an unusual year, and the spread between models gives an uncertainty estimate.

Q. How do you compute threshold yield?

A. The average of the best 5 of the last 7 years' yields of the insurance unit, multiplied by the indemnity level (70, 80 or 90 %). The claim is the shortfall fraction (TY − AY)/TY times the sum insured.

Q. Why is weather-based insurance used for horticulture?

A. Horticultural crops have multiple pickings, long durations, poor yield records and difficult CCEs, and their main perils (frost, high temperature, excess rain at flowering, wind) are

captured well by weather parameters. RWBCIS therefore uses weather triggers for quick payouts, at the cost of higher basis risk.

Q. What accuracy is acceptable for GP-level yield?

A. Operational targets are usually RMSE or MAPE of 10–15 % of mean yield and low bias. The model should also rank GPs correctly (high Spearman correlation), because claims depend on shortfall relative to threshold yield.

Q. How can FLEX or NISAR improve insurance?

A. NISAR gives cloud-free sowing, crop-area and soil-moisture data every 12 days, which helps prevented-sowing and mid-season adversity claims and rice yield models. FLEX SIF at 300 m detects photosynthetic stress early and directly improves yield prediction, especially under heat or drought.

15. Worked numerical examples

Boards often ask candidates to "show the calculation". Each example below uses realistic Indian values. Practise them on paper until you can do them without notes.

15.1 FAO-56 reference ET (ET_o)

Given (Delhi, March): mean T = 25 °C, Rn = 15 MJ m^{-2} day^{-1}, G ≈ 0, u_2 = 2 m s^{-1}, VPD (es − ea) = 1.2 kPa, P = 101.3 kPa.

1. es(25) = 0.6108·exp(17.27×25/(25 + 237.3)) = 0.6108 × 5.187 = **3.168 kPa**
2. Δ = 4098·es/(T + 237.3)^2 = 4098 × 3.168/68,801 = **0.1887 kPa °C^{-1}**
3. γ = 0.665×10^{-3} × P = **0.0674 kPa °C^{-1}**
4. Numerator = 0.408×0.1887×15 + 0.0674×(900/298)×2×1.2 = 1.155 + 0.489 = **1.643**
5. Denominator = 0.1887 + 0.0674×(1 + 0.34×2) = 0.1887 + 0.1132 = **0.3019**
6. **ET_o = 1.643/0.3019 ≈ 5.4 mm day^{-1}**. For wheat at mid-season (Kc = 1.15), ETc = 6.3 mm day^{-1}.

15.2 SEBAL instantaneous and daily ET for one pixel

Given (Landsat overpass, 10:30): Rs↓ = 850 W m^{-2}, albedo α = 0.18, surface emissivity ε_o = 0.97, Ts = 305 K (31.85 °C), RL↓ = 380 W m^{-2}, NDVI = 0.70.

Step	Calculation	Result
Outgoing longwave	RL↑ = 0.97 × 5.67×10^{-8} × 305^4	476 W m^{-2}
Net radiation	Rn = (1 − 0.18)×850 + 380 − 476 − 0.03×380	590 W m^{-2}
G/Rn (Bastiaanssen)	(31.85/0.18)(0.0038×0.18 + 0.0074×0.18^2)(1 − 0.98×0.7^4)	0.125
Soil heat flux	G = 0.125 × 590	74 W m^{-2}
Hot pixel	Ts = 318 K, H = Rn − G = 480 W m^{-2}, rah = 30 s m^{-1} → dT = H·rah/ρcp = 480×30/1155	12.5 K
Cold pixel	Ts = 300 K, dT = 0 (H = 0)	0 K
Linear dT–Ts	b = 12.5/(318 − 300) = 0.693; a = −0.693×300	dT = −207.9 + 0.693·Ts
Our pixel dT	0.693 × (305 − 300)	3.46 K
Sensible heat	H = 1155 × 3.46/35 (rah = 35 s m^{-1} after stability iteration)	114 W m^{-2}
Latent heat	λET = 590 − 74 − 114	402 W m^{-2}
Evaporative fraction	EF = 402/(590 − 74)	0.78
Daily ET (EF method)	ET24 = 86400 × 0.78 × Rn24 (180 W m^{-2})/2.45×10^6	4.9 mm day^{-1}

METRIC alternative: ETinst = 402/2.45×10^6 × 3600 = 0.59 mm h^{-1}. With ETr,inst = 0.75 mm h^{-1} the reference fraction is ETrF = 0.79. With daily ETr,24 = 7.0 mm, **ET24 = 0.79 × 7.0 = 5.5 mm day^{-1}**. The

ETrF method gives the higher value because it accounts for the afternoon increase in evaporative demand (advection), which the EF method misses.

15.3 Spectral indices

Case	Reflectances	Index
NDVI, healthy wheat	NIR 0.40, Red 0.06	(0.40 − 0.06)/(0.46) = 0.74
NBR, green crop before burning	NIR 0.38, SWIR2 0.12	0.26/0.50 = 0.52
NBR, burnt field	NIR 0.13, SWIR2 0.21	−0.08/0.34 = −0.24
dNBR, green → burnt	0.52 − (−0.24)	0.76 → high severity
NBR, harvested stubble (pre-burn)	NIR 0.22, SWIR2 0.20	0.05
dNBR, stubble → burnt	0.05 − (−0.24)	0.29 → moderate-low

> **Lesson:** For residue burning the right pre-fire image is harvested stubble, not the green crop. Using the green crop overstates severity and confuses harvest with burning. This is a common examiner trap.

15.4 Fire radiative power, biomass and emissions

1. **VIIRS pixel area:** 375 m × 375 m = 140,625 m^2.
2. **MIR excess radiance** over the background: ΔL_4 = 2.0 W m^{-2} sr^{-1} μm^{-1}. Take a = 3.0×10^{-9} W m^{-2} sr^{-1} μm^{-1} K^{-4}.
3. **FRP** = $A\cdot\sigma/a\cdot\Delta L_4$ = 140,625 × ($5.67\times10^{-8}/3.0\times10^{-9}$) × 2.0 = **5.3 MW**.
4. If the fire burns at about this rate for 30 minutes: **FRE** = 5.3 MW × 1800 s = **9,540 MJ**.
5. **Biomass burnt** = 0.368 kg MJ^{-1} × 9,540 = **3.5 t**, roughly 0.6 ha of paddy straw at 6 t ha^{-1}.
6. **Emissions** (rice-straw EFs: CO_2 about 1,460, CO about 93, PM2.5 about 8.3 g kg^{-1}): CO_2 = 5.1 t, CO = 326 kg, **PM2.5 = 29 kg**.

Bottom-up regional emission (IPCC equation)

$$E = A \times B \times Cf \times EF \qquad (A = \text{burnt area, } B = \text{residue load, } Cf = \text{combustion factor})$$

Example (Punjab): A = 1.0 Mha burnt, B = 7 t ha^{-1} and Cf = 0.8 give **5.6 Mt dry matter burnt**. That produces about **8.2 Mt CO_2** and **about 46,000 t PM2.5** (EF 8.3 g/kg). This is why accurate burnt-area mapping (A) matters as much as fire counts.

15.5 Semi-physical (LUE) wheat yield

Given: 120-day season, mean Rs = 18 MJ m^{-2} day^{-1}, PAR = 0.48·Rs = 8.64, mean fAPAR = 0.50, ε = 2.8 g MJ^{-1} APAR, mean stress scalar Ts × Ws = 0.85, HI = 0.40.

$$\text{Biomass} = 8.64 \times 0.50 \times 2.8 \times 0.85 \times 120 = 1{,}234 \text{ g m}^{-2} = 12.3 \text{ t ha}^{-1}\ ; \quad \text{Yield} = 0.40 \times 12.3 = 4.9 \text{ t ha}^{-1}$$

Sensitivity: a heat wave in March lowers the mean stress scalar to 0.75 and the yield to 4.3 t/ha, a 12 % loss. Under the insurance example in Chapter 14 this could be near or below the threshold yield. The model therefore links weather shocks directly to insurance claims.

15.6 Insurance claim and smart sampling

(a) Claim

Past 7-year GP yields: 3.9, 4.2, 3.6, 4.4, 4.1, 4.5, 4.3 t/ha. Best 5 = 4.5, 4.4, 4.3, 4.2, 4.1, with a mean of **4.30**. Indemnity is 80 %, so **TY = 3.44**. Actual yield blended: 30 % × YES-TECH 2.70 + 70 % × CCE 2.85 = **2.805**. Shortfall = (3.44 − 2.805)/3.44 = **18.5 %**. With a sum insured of ₹50,000/ha, the **claim is ₹9,230/ha**.

(b) Number of CCEs needed

$$n = (z \cdot CV/E)^2 = (1.96 \times 30/10)^2 \approx 35 \text{ CCEs}$$ (district-level, CV 30 %, ±10 % error at 95 % confidence)

(c) Neyman vs proportional allocation of 12 CCEs in a GP

Vigour stratum	Area Nh (ha)	Yield SD Sh (t/ha)	Proportional nh	Neyman nh
Low	200	0.8	2.4 → 2	12×160/540 = 3.6 → 4
Medium	500	0.4	6.0 → 6	12×200/540 = 4.4 → 4
High	300	0.6	3.6 → 4	12×180/540 = 4.0 → 4

Neyman allocation moves CCEs toward the **more variable low-vigour stratum**, where stressed and lodged patches make yield uncertain. This reduces the variance of the GP mean for the same total number of CCEs.

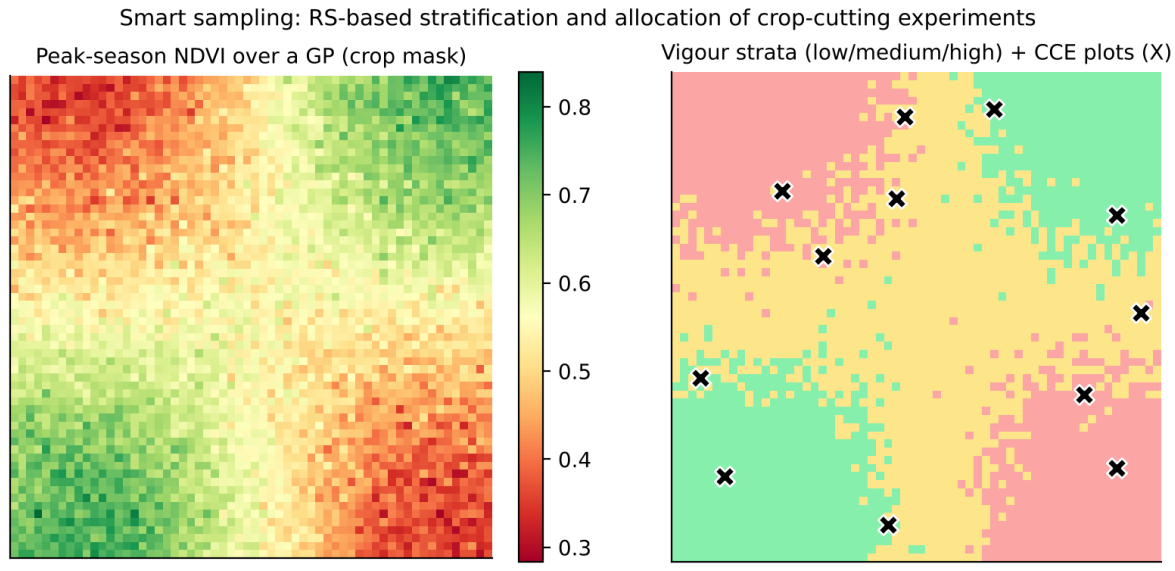


Figure 53. Smart sampling: NDVI-based stratification of an insurance unit and allocation of CCE plots across strata.

15.7 Carbon credits from a zero-till + no-burn wheat farm

Component	Calculation	tCO_2e/ha/yr
SOC sequestration (removal)	0.3 t C/ha/yr × 44/12	1.10
Avoided CH_4 from burning	4.8 t burnt × 2.7 kg CH_4/t × GWP 27	0.35
Avoided N_2O from burning	4.8 t × 0.07 kg N_2O/t × GWP 273	0.09
Diesel saved (no ploughing)	40 L × 2.68 kg CO_2/L	0.11
Gross		1.65
Uncertainty deduction (15 %)		−0.25
Buffer pool (15 % of removals)	0.15 × 1.10	−0.17
Net credits issued		≈ 1.24

At US$15/credit this is about US$18.6/ha (≈ ₹1,550/ha). The farmer's 60 % share is about **₹930/ha/yr**. This shows why **aggregation of thousands of farmers** and **low-cost remote-sensing MRV** are essential for smallholder carbon projects. Note that CO_2 from burning straw is **biogenic** and is not counted, while CH_4 and N_2O are.

Biochar credit

$$1\text{ t biochar} \times 0.70\text{ C} \times 0.80\text{ permanence} \times 3.667 = 2.05\ tCO_2e - 0.15\text{ project emissions} \approx 1.9\text{ credits}$$

15.8 Accuracy statistics from a confusion matrix

Using the confusion matrix in Chapter 2: N = 1400, diagonal sum = 1280, so **OA = 91.4 %**. Expected chance agreement pe = Σ(row × column totals)/N^2 = 512,250/1,960,000 = 0.261. **κ = (0.914 − 0.261)/(1 − 0.261) = 0.88**. Rice producer's accuracy = 412/441 = 93.4 % and user's accuracy = 412/440 = 93.6 %.

15.9 GDD and phenology

Wheat sown on 10 November with Tmax 24 °C, Tmin 8 °C and Tbase 5 °C gives a daily GDD of (24 + 8)/2 − 5 = 11 °C·day. If anthesis needs about 1,100 GDD, it occurs after about 100 days (mid-February) at that average. Warmer temperatures bring anthesis forward and shorten grain filling, so terminal heat risk rises with late sowing. This is the agronomic reason **zero-till early sowing** helps.

15.10 Water-cloud model inversion

Given: VV σ^0 = −12 dB (0.0631 linear), θ = 35° (cos θ = 0.819), wheat LAI = 2.5 ($V_1 = V_2$ = LAI), A = 0.0012, B = 0.091.

1. γ^2 = exp(−2×0.091×2.5/0.819) = exp(−0.556) = **0.574**
2. σ^0veg = 0.0012 × 2.5 × 0.819 × (1 − 0.574) = **0.00105**
3. σ^0soil = (0.0631 − 0.00105)/0.574 = **0.108** (−9.7 dB)

4. Use an empirical soil relation calibrated locally, e.g. σ^0soil(dB) = −20 + 35·θv, giving θv = (−9.7 + 20)/35 ≈ **0.29 m^3 m^{-3}** (moist soil after irrigation).

16. Research-paper summaries (concepts and evidence)

Each summary gives the **objective**, the **data and method**, the **key finding** and **how to use it in the interview**. Mentioning one or two of these by author and year shows depth.

16.1 Evapotranspiration

SEBAL: the original energy-balance algorithm

Bastiaanssen, Menenti, Feddes & Holtslag (1998), J. Hydrology 212–213

- **Objective:** estimate regional ET from satellite data with minimal ground data.
- **Method:** residual SEB; dT linearly related to Ts and calibrated with hot and cold pixels; Monin–Obukhov iteration for rah.
- **Finding:** ET within about 10–20 % (instantaneous) and 5 % (seasonal) of field measurements in Egypt, Spain and Niger.
- **Use:** the 'internal calibration' idea removes most of the absolute Ts bias, which is why SEBAL spread worldwide.

METRIC

Allen, Tasumi & Trezza (2007), J. Irrig. Drain. Eng. 133(4)

- **Objective:** an operational, high-resolution ET model for water-rights management.
- **Method:** SEBAL-based, but the cold pixel is calibrated to 1.05 ETr (alfalfa reference) from weather stations, and the ETrF method is used for daily and seasonal scaling.
- **Finding:** seasonal ET within about 4–7 % of lysimeters in Idaho.
- **Use:** explains why ETrF (AET/ET_0) scaling beats EF scaling under advection.

ALEXI two-source time-integrated model

Anderson, Norman, Diak, Kustas & Mecikalski (1997), Remote Sens. Environ. 60

- **Objective:** remove the need for air temperature and accurate absolute Ts.
- **Method:** TSEB coupled with an ABL growth model using the morning rise in GOES Ts.
- **Finding:** robust continental-scale fluxes; the basis of the ESI drought product.
- **Use:** cite it when asked how to estimate ET without ground meteorology, for example with INSAT data.

Evaporative fraction self-preservation

Crago (1996), J. Hydrology 180

- **Finding:** EF is approximately constant during daytime under fair weather, with small concave-up deviations (lower at mid-day).
- **Use:** the theoretical basis for instantaneous-to-daily scaling and its known 5–10 % underestimation.

16.2 SIF and FLEX

Global crop photosynthesis from SIF

Guanter et al. (2014), PNAS 111(14)

- **Objective:** test whether satellite SIF captures cropland photosynthesis better than models.
- **Data:** GOME-2 SIF compared with flux-tower GPP in the US Corn Belt.
- **Finding:** SIF-based GPP peaks were 40–60 % higher than those of carbon-cycle models, which underestimate agricultural productivity.
- **Use:** SIF captures the high intensity of irrigated, high-input crops (relevant to Punjab and Haryana).

SIF improves crop productivity monitoring

Guan et al. (2016), Global Change Biology 22

- **Finding:** GOME-2 SIF explained US county maize and soybean yield and GPP better than EVI and NDVI. C4 vs C3 LUE differences must be accounted for.
- **Use:** supports SIF as a yield predictor in YES-TECH-type models once FLEX data are available.

Drought onset detected by SIF

Sun et al. (2015), J. Geophys. Res. Biogeosci. 120

- **Finding:** during the 2012 US flash drought, GOME-2 SIF declined earlier and more strongly than NDVI or EVI, tracking the GPP drop.
- **Use:** early-warning and mid-season adversity triggers for PMFBY.

50 years of SIF remote sensing (review)

Mohammed et al. (2019), Remote Sens. Environ. 231

- **Content:** physiology, the leaf-to-canopy scaling (fesc), retrieval methods (FLD, SFM, SVD), instruments and missions, and open challenges.
- **Use:** the go-to reference for explaining why SIF ≠ GPP everywhere (APAR dominance, structural effects).

FLEX mission concept

Drusch et al. (2017), IEEE TGRS 55(3)

- **Content:** FLORIS design (500–780 nm, 0.3 nm in the O_2 bands), tandem with Sentinel-3, 300 m resolution and the products (full SIF spectrum, PRI and NPQ, photosynthesis).
- **Use:** cite it for mission specifications; FLEX launched on 15 Sep 2026.

16.3 SAR and soil moisture

Water Cloud Model

Attema & Ulaby (1978), Radio Science 13

- **Idea:** the canopy is modelled as a cloud of identical water droplets held in place by the vegetation. Backscatter = volume term + attenuated soil term.
- **Use:** the standard vegetation correction for SAR soil moisture (Sentinel-1, NISAR).

Change detection soil moisture

Wagner, Lemoine & Rott (1999), Remote Sens. Environ. 70

- **Idea:** normalise backscatter between the historical dry and wet references to get relative soil moisture; roughness is assumed static.
- **Use:** the basis of the ASCAT and Copernicus SSM products and one of the NISAR algorithms.

Topp's equation

Topp, Davis & Annan (1980), Water Resour. Res. 16

- **Idea:** an empirical polynomial between the dielectric constant and volumetric water content, valid across mineral soils.
- **Use:** the basis of TDR probes and dielectric-to-moisture conversion in radar retrievals.

16.4 Fire and burnt area

MODIS Collection 6 active fire algorithm

Giglio, Schroeder & Justice (2016), Remote Sens. Environ. 178

- **Method:** a contextual algorithm using the 4 µm and 11 µm brightness temperatures, a background characterisation window and false-alarm rejection. FRP is computed from MIR radiance.
- **Use:** describes the algorithm behind FIRMS data used in the IARI bulletin.

FRP to biomass combustion

Wooster, Roberts, Perry & Kaufman (2005), J. Geophys. Res. 110

- **Finding:** laboratory burns gave a combustion factor of **0.368 ± 0.015 kg MJ^{-1}**, independent of fuel type.
- **Use:** converts FRE to biomass burnt and emissions (see the Chapter 15 example).

Normalised Burn Ratio and CBI

Key & Benson (2006), USDA FIREMON RMRS-GTR-164

- **Method:** NBR from Landsat NIR and SWIR2; dNBR calibrated against the field Composite Burn Index.
- **Use:** gives the dNBR severity thresholds; adapt them for stubble (low pre-fire biomass) using RdNBR or RBR.

MODIS fire characteristics in north India

Vadrevu, Ellicott, Badarinath & Vermote (2011), Environmental Pollution 159

- **Finding:** strong seasonal peaks of paddy-residue fires in Punjab in Oct–Nov, linked to rises in aerosol optical depth over the IGP.
- **Use:** early evidence of the regional air-quality link to burning.

Missing emissions from post-monsoon fires

Liu, Marlier et al. (2020), Environ. Res. Commun. 2

- **Finding:** MODIS burned-area and active-fire products miss a large share of small agricultural fires in NW India, so emissions are underestimated. Finer sensors (VIIRS, Sentinel-2) detect much more.
- **Use:** supports your argument for combining VIIRS counts with Sentinel-2 burnt area, and with GEO data for late-afternoon fires.

16.5 Crop yield and parameter retrieval

Monteith light-use efficiency

Monteith (1972), J. Appl. Ecol. 9

- **Idea:** crop dry matter is proportional to intercepted (absorbed) solar radiation; ε is fairly conservative for a given crop type.
- **Use:** the foundation of the semi-physical yield model in FASAL and YES-TECH.

Scalable Crop Yield Mapper (SCYM)

Lobell et al. (2015), Remote Sens. Environ. 164

- **Method:** run a crop model (APSIM) many times with varied weather, soil and management; fit a regression of simulated yield on simulated VIs and weather; apply the regression to satellite VIs.
- **Finding:** field-level maize and soybean yield without any ground calibration.
- **Use:** a model for calibration-free GP-level yield in data-poor areas, a hybrid of crop model and ML.

Smallholder wheat yields and sowing dates in India

Jain et al. (2016), Remote Sensing 8(10), 860

- **Data:** SkySat and Landsat micro-satellite data with crop-cut data in Bihar.
- **Finding:** field-scale yield and sowing-date mapping is feasible for smallholders. Late sowing reduces yield, consistent with terminal heat.
- **Use:** evidence that field-level remote sensing works in Indian smallholder systems.

Assimilating remote sensing into crop models (review)

Huang et al. (2019), Agric. For. Meteorol. 276–277

- **Content:** forcing, calibration and updating (EnKF, PF, 4D-Var) strategies; variables (LAI, SM, ET); gains in yield accuracy.
- **Use:** cite it when explaining the crop-model-to-digital-twin link.

Retrieval of vegetation properties (review)

Verrelst et al. (2015), ISPRS J. Photogramm. 108

- **Content:** parametric, non-parametric, physical and hybrid retrieval methods compared, with their strengths and weaknesses.
- **Use:** the framework in Chapter 5.1 comes from this review.

PROSPECT + SAIL review

Jacquemoud et al. (2009), Remote Sens. Environ. 113

- **Content:** 20 years of PROSAIL use, inversion approaches and the ill-posed problem.
- **Use:** cite it for RTM basics.

Optical-sensor-based N management

Raun et al. (2002), Agron. J. 94

- **Finding:** GreenSeeker NDVI-based in-season N (the INSEY algorithm) increased NUE by more than 15 % in winter wheat compared with uniform rates.
- **Use:** the standard example of sensor-based VRT.

16.6 Soil, carbon and conservation agriculture

On digital soil mapping

McBratney, Mendonça Santos & Minasny (2003), Geoderma 117

- **Idea:** the scorpan framework S = f(s, c, o, r, p, a, n) + spatially autocorrelated residuals.
- **Use:** the conceptual basis of DSM (Dr. Bappa Das's line of work).

The contentious nature of soil organic matter

Lehmann & Kleber (2015), Nature 528

- **Idea:** SOM is a continuum of progressively decomposed compounds. Persistence depends on accessibility and mineral protection rather than intrinsic 'humic' recalcitrance.
- **Use:** explains why MAOM and aggregation (helped by zero tillage) matter for permanence in carbon credits.

Tillage mapping with Sentinel-2 in the IGP

Zhou et al. (2021), Remote Sensing 13(24), 5108

- **Method:** Sentinel-2 time series, tillage indices (NDTI and related), Random Forest on Google Earth Engine across the Indo-Gangetic Plains.
- **Finding:** field-level ZT and CT discrimination is feasible at regional scale in smallholder systems.
- **Use:** the same principle as the IARI CREAMS Zero-Tillage Mapper (Dhakar & Sehgal).

CA eligibility for carbon credits

Cariappa et al. (2024), Scientific Reports

- **Method:** farmer surveys and remote sensing to assess whether CA practices in the IGP meet Verra VM0042 requirements (additionality, baseline, practice change).
- **Use:** shows that remote-sensing-based practice verification is central to agricultural MRV in India.

Area estimation and accuracy assessment good practice

Olofsson et al. (2014), Remote Sens. Environ. 148

- **Idea:** pixel counting is biased. Use a stratified sample of reference data to estimate class area with confidence intervals (error-adjusted area).
- **Use:** the right way to report crop area (FASAL) and burnt area.

17. Mock interview: long-form model answers and programme timeline

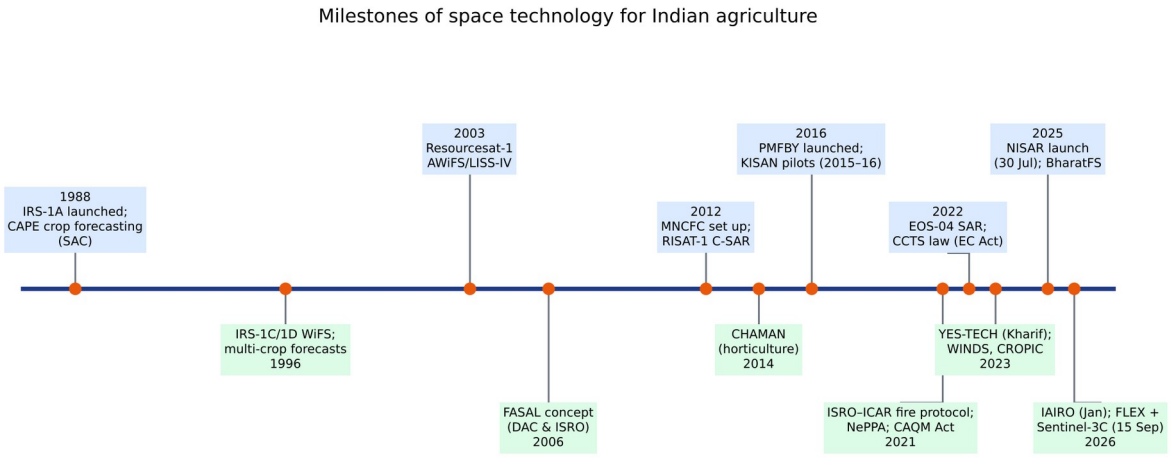


Figure 54. Milestones of space technology applications for Indian agriculture (1988–2026).

Use the **PREP structure** for every long answer: **P**oint (one-line definition), **R**eason (the physics or principle), **E**xample (Indian programme or paper), **P**ath forward (limitation and your contribution). Aim for 90–120 seconds per answer.

17.1 Insurance and yield (expected main focus)

Q. You have to estimate paddy yield for 500 gram panchayats in Odisha for Kharif 2026 under YES-TECH. How will you do it?

A. Point: I would build a GP-level ensemble yield system calibrated with CCEs. **Steps:** (1) A rice mask from Sentinel-1 and NISAR 12-day VH/HH time series, which work through monsoon clouds, with transplanting dates taken from the flooding minimum. (2) Features: SAR backscatter trajectories, cloud-free Sentinel-2 NDVI/NDRE where available, TROPOMI SIF (FLEX when operational), IMD gridded rainfall, heat-stress days, NISAR soil moisture and soil properties. (3) Five base models: an LUE semi-physical model, WOFOST or ORYZA with LAI assimilation, XGBoost, LSTM and a weather-index regression. (4) Train on 5 years of geotagged CCEs with leave-one-year-out validation. (5) Stack the models, and use the spread between members as uncertainty. (6) Validate on holdout CCEs, targeting MAPE below 12 %. (7) Deliver GP yields with the 30 % blending, and flag high-uncertainty GPs for additional CCEs. **Caveats:** cloud cover, cyclone and flood years (which need SAR inundation layers), and CCE quality.

Q. What are the weaknesses of CCE-based yield estimation, and how does technology help?

A. CCEs are expensive and labour-intensive (lakhs per season). They can be biased by non-random plot selection and manipulation. Measurement and driage errors occur, results arrive late (which delays claims), and a few points per GP may not represent heterogeneous fields. Technology helps in two ways. **Smart sampling** stratifies by RS vigour so CCEs are fewer and

more representative. **Direct estimation** produces wall-to-wall model yields. CROPIC photos and the CCE-Agri app add transparency through geotagging and video.

Q. Design an insurance product for a horticulture belt where CCEs are not feasible.

A. A **hybrid weather and satellite index**. Weather triggers (from WINDS or AWS) cover frost, high temperature at flowering and excess rain. A satellite layer (Sentinel-2 or FLEX canopy stress, LST from Sentinel-3 or TRISHNA, NISAR soil moisture) reduces basis risk by confirming crop-level impact. Payouts are tiered from strike to exit. The product is priced from 20+ years of reanalysis weather and RS anomaly history, and parametric settlement could happen within 15 days. A localised-loss window covers hail, verified with CROPIC photos and SAR change detection.

Q. Why is basis risk a problem and how would you reduce it?

A. Basis risk is the mismatch between the index payout and a farmer's actual loss. A farmer who loses the crop but gets no payout loses trust, and adoption falls. It can be reduced with **smaller units** (GP rather than block), **better indices** (field-level RS yield instead of station weather), **multi-index triggers** and **gap-filling** of weather data with satellite rainfall and soil moisture. Audits and grievance channels that use satellite evidence also help.

17.2 Residue burning

Q. The Ministry says fire counts in Punjab fell 90 % but Delhi's AQI is still poor in November. Explain.

A. (1) **Detection gaps:** polar orbiters pass at about 10:30 and 13:30, and farmers have shifted burning to the late afternoon, so many fires are missed. Small, short fires and fires under smoke or cloud are also missed. (2) **Burnt area** from Sentinel-2 dNBR shows a smaller decline than counts, so actual burning fell less than the counts suggest. (3) **Meteorology:** low November boundary-layer height, calm winds and inversions trap pollutants regardless of the source. (4) **Other sources:** vehicles, construction dust, industry, biomass for heating and firecrackers. (5) **Long-range transport** depends on the wind direction (north-westerlies carry smoke to Delhi). **Solution:** combine VIIRS counts, GEO fire detection, Sentinel-2 burnt area and FRP-based emission inventories, and attribute sources with chemical transport models.

Q. Explain how you would map burnt area for the 2026 paddy season and validate it.

A. Start with a paddy mask (Sentinel-1). Build 5-day Sentinel-2 composites from 1 October to 30 November and compute NBR, NBR2, MIRBI and BAIS2. Detect the transition from harvested stubble to burnt as a dNBR increase, plus NBR2 and MIRBI thresholds, refined with a Random Forest trained on VIIRS-guided and field samples. Validate with stratified random points interpreted on PlanetScope imagery and field photos, and report the error-adjusted area (Olofsson 2014) by district.

17.3 ET, SIF and NISAR

Q. You have only a Landsat image and one weather station. How do you get daily ET for wheat fields?

A. Run METRIC. Compute albedo, NDVI, LAI and emissivity, and LST from TIRS. Calculate Rn, then G, then select hot and cold pixels, with the cold pixel set to ET = 1.05 ETr from the station's hourly data. Iterate H with Monin–Obukhov stability to get instantaneous λET, then ETrF. Daily ET = ETrF × ETr,24 from the same station. For the season, interpolate ETrF between cloud-free images and multiply by daily ETr.

Q. How does FLEX help Indian agriculture specifically?

A. (1) **Early stress detection** for heat waves in March during wheat grain filling and for dry spells in rainfed kharif crops. (2) **GPP and yield** at 300 m, closer to the GP scale than TROPOMI. (3) **Separating structure from physiology** using both fluorescence peaks, PRI and LST. (4) **Inputs to yield models and insurance.** (5) **Science:** C3/C4 differences and irrigation effects on photosynthesis. India can contribute ground cal/val super-sites (FloX instruments next to EC towers at IARI and NRSC sites) and help develop algorithms for heterogeneous smallholder landscapes.

Q. Compare optical and SAR for kharif crop mapping.

A. Optical sensors give spectral and biochemical detail (chlorophyll, water) but clouds block them in July–September. SAR is all-weather and sensitive to structure, moisture and flooding, which suits rice, but it is affected by speckle and roughness. In practice, SAR (Sentinel-1, EOS-04 and now NISAR L/S) forms the backbone of kharif mapping and optical data are fused in where available.

17.4 Carbon, CA and precision agriculture

Q. Can a Punjab farmer earn carbon credits for not burning straw? What would MRV look like?

A. Yes, under methodologies such as VM0042, through avoided CH_4 and N_2O from burning, SOC gains from residue retention and zero tillage, and reduced diesel. **MRV:** baseline burning history from 5+ years of VIIRS and Sentinel-2 data; project-period monitoring showing no fires or burn scars, a ZT signature (NDTI, early green-up) and wheat sowing; soil sampling stratified by remote sensing; a calibrated model (DayCent or RothC) for SOC; then verification by a VVB. Credits are small per hectare (about 1–1.5 tCO_2e), so **aggregation** and **low-cost digital MRV** are essential. IARI's CREAMS fire and ZT mapping already supplies the practice-verification layer.

Q. What is your view on precision agriculture for small farms?

A. Precision should be **knowledge-intensive, not capital-intensive**. Satellite-based field and village zoning, LCC and GreenSeeker-based N management, laser levelling, sensor-based irrigation and drone spraying through CHCs give most of the benefit. ICAR-NePPA (led by IARI) and AgriStack data integration can scale these approaches.

17.5 Opening and closing questions

Q. Why ISRO and not academia?

A. ISRO turns science into operational national services (FASAL, NADAMS, fire bulletins, NISAR products). I want my retrieval and modelling research to reach millions of farmers through such services, with the scale, data and mission discipline that only ISRO offers.

Q. What is one limitation of your own research?

A. Say it honestly, then give a remedy. For example: my model was validated over two seasons in one agro-climatic zone. To make it operational I would test leave-one-year-out across zones, add SAR for cloud-prone years and quantify uncertainty.

Q. Where do you see agricultural remote sensing in 10 years?

A. Field-scale, daily, multi-sensor (optical, SAR including NISAR, thermal including TRISHNA, SIF including FLEX, hyperspectral) data fused by foundation models and assimilated into digital twins. It will provide GP-level yield, water and carbon accounting that directly drives insurance, advisories and carbon markets, with India building sovereign AI capability through bodies such as IAIRO.

18. Indian programmes, data portals and upcoming missions

18.1 Operational and digital-agriculture programmes

Programme	What to know
NADAMS	National Agricultural Drought Assessment and Monitoring System. Started by NRSC in 1989 and now with MNCFC. Monthly district and sub-district drought assessment from NDVI, NDWI, VCI and SASI (shortwave angle slope index), soil moisture, rainfall and sown area. Feeds drought declaration under the Manual for Drought Management (2016, revised 2020): rainfall and dry-spell triggers, then impact indicators (remote-sensing vegetation indices, sown area, soil moisture, hydrology), then ground truth.
KISAN (2015–16)	C(K)rop Insurance using Space technology And geoiNformatics. Pilots in several states testing smart sampling of CCEs and RS and crop-model yield estimation. A precursor to YES-TECH.
Krishi-DSS (Aug 2024)	Krishi Decision Support System: a geospatial platform of DA&FW and ISRO integrating crop maps, drought, soil, weather and reservoir data for policy decisions.
AgriStack / Digital Agriculture Mission (2024)	Digital public infrastructure: Farmers' Registry (Kisan ID), geo-referenced village maps and Crop Sown Registry (the Digital Crop Survey, plot-level crop data via mobile app, which provides ideal training labels for crop mapping and yield models).
CROPIC	Collection of Real-time Observations and Photos of Crops. Geotagged smartphone photos taken 4–5 times per season, with AI analysis of crop stage, health and damage. Supports PMFBY loss assessment.
WINDS	Weather Information Network and Data Systems: AWS at block level and rain gauges at GP level, with data shared to a national server for insurance and advisories.
DigiClaim	Digital claim calculation and direct bank transfer under PMFBY, integrated with PFMS.
Soil Health Card (2015)	Grid-based soil testing (12 parameters) with fertiliser recommendations; data useful for DSM.
GKMS / Meghdoot	Agromet advisories from IMD and ICAR; the Meghdoot app shows them in local languages.
ISRO–ICAR fire protocol (2021)	Standard protocol for satellite monitoring of residue burning; the basis of the daily CREAMS bulletin.

18.2 Data portals

Portal	Content
Bhuvan (NRSC, 2009)	ISRO geoportal: imagery, thematic maps (LULC, wasteland), Bhuvan-Fasal and crop-related services, NDVI and fire layers, a mobile geotagging app
Bhoonidhi (NRSC)	ISRO's EO data hub: Resourcesat, Cartosat, EOS-04, NISAR L/S products (including 100 m soil moisture) and some open data
MOSDAC (SAC)	Meteorological and oceanographic data: INSAT-3D/3DR/3DS insolation, LST, rainfall, ET, Oceansat-3
VEDAS (SAC)	Visualisation of Earth observation Data and Archival System: analysis-ready products

Portal	Content
	and agricultural apps
NICES (NRSC)	National Information System for Climate and Environment Studies: essential climate variables
FIRMS (NASA)	Near-real-time MODIS and VIIRS active fires and FRP
Copernicus Data Space / GEE	Sentinel-1/2/3/5P data; Google Earth Engine for large-area analysis
ASF DAAC (NASA)	NISAR L-band global products, Sentinel-1

18.3 Recent and upcoming missions relevant to agriculture

Table 21. Missions to mention when asked "what next?"

Mission	Status	Agricultural relevance
NISAR (NASA–ISRO)	Launched 30 Jul 2025	L/S SAR: crop area every 12 days, soil moisture at 100–200 m, biomass, floods
FLEX (ESA)	Launched 15 Sep 2026	SIF at 300 m: photosynthesis, early stress, GPP
Sentinel-3C	Launched 15 Sep 2026 (with FLEX)	OLCI and SLSTR continuity; FLEX tandem partner
Sentinel-2C / 1C	Launched Sep 2024 / Dec 2024	Continuity of 10 m optical and C-band SAR
Biomass (ESA, P-band)	Launched Apr 2025	Forest biomass and carbon (agroforestry MRV)
TRISHNA (ISRO–CNES)	Upcoming	TIR about 57 m, near-daily revisit: field-scale ET, crop water stress, irrigation
CHIME (ESA)	Upcoming (late 2020s)	Hyperspectral 400–2500 nm: N, protein, soil properties
LSTM (ESA)	Upcoming	High-resolution thermal for ET and water productivity
ROSE-L (ESA)	Upcoming	L-band SAR continuity with NISAR
Resourcesat-3 series, Oceansat-3A	Planned (ISRO)	AWiFS and LISS continuity for FASAL; ocean inputs for the monsoon
EOS-08 GNSS-R	Launched Aug 2024	Soil-moisture and flood sensing from reflected GNSS signals

18.4 Topic-wise question bank (practise aloud)

Remote-sensing basics

- Why is the red edge useful for nitrogen? What is the REP?
- What is the difference between radiance, reflectance and brightness temperature?
- Explain BRDF and why Sentinel-2 data need normalisation (NBAR).
- What is atmospheric correction? Name two methods (Sen2Cor, 6S, FLAASH).
- Why do we use surface reflectance rather than DN for time-series analysis?

SIF and FLEX

- Draw the SIF spectrum and mark the O_2 bands.
- Why is the red peak smaller at canopy level?
- How is SIF related to APAR and LUE? What is SIF yield?
- What does Sentinel-3 contribute to FLEX?
- How would you set up a FLEX cal/val site at IARI?

NISAR and SAR

- Explain speckle and how to reduce it.
- Why does rice show a low backscatter at transplanting?
- What is InSAR coherence and how can it detect tillage and harvest?
- Write the water cloud model and explain each term.
- L-band vs C-band for biomass: which saturates first, and why?

ET and eddy covariance

- Write the energy-balance equation and show which terms satellites measure.
- How are hot and cold pixels selected in METRIC? What errors arise from a wrong choice?
- Why use ETrF instead of EF in arid regions?
- What is the flux footprint? How does it affect validation?
- What is energy-balance closure and why is it never 100 %?

Burning, CA and carbon

- Explain the contextual fire-detection algorithm in steps.
- What is FRP? How do you convert it to emissions?
- Which Sentinel-2 bands and indices map burnt area? Why use stubble, not the green crop, as the pre-fire reference?
- Compare residue incorporation and retention for SOC and wheat yield.
- Explain additionality, permanence and leakage with agricultural examples.
- Who buys agricultural carbon credits, and what is India's CCTS?

Insurance and yield

- Calculate threshold yield and the claim for a given yield series.
- Explain smart sampling with Neyman allocation.
- List the model types allowed under YES-TECH and the pros and cons of each.
- What is basis risk? How would you reduce it for horticulture?
- How would you validate GP-level yield estimates?
- Why might ML yield models fail in an El Niño year?

19. Rapid revision: formula sheet, abbreviations and rapid-fire Q&A

19.1 One-page formula sheet

Quantity	Formula
Wien / Stefan–Boltzmann	$\lambda max = 2898/T\ \mu m$; $M = \varepsilon\sigma T^4$
NDVI / EVI	$(NIR - R)/(NIR + R)$; $2.5(NIR - R)/(NIR + 6R - 7.5B + 1)$
NDRE / CIre	$(NIR - RE)/(NIR + RE)$; $NIR/RE - 1$
NBR / dNBR	$(B8 - B12)/(B8 + B12)$; NBRpre − NBRpost
NDTI	$(B11 - B12)/(B11 + B12)$
RVI (SAR)	$8\sigma HV/(\sigma HH + \sigma VV + 2\sigma HV)$
Water cloud model	$\sigma^0 = A \cdot V_1 \cdot \cos\theta(1 - \gamma^2) + \gamma^2\sigma^0 soil$; $\gamma^2 = \exp(-2BV_2/\cos\theta)$
Energy balance	$Rn = G + H + \lambda ET$; $H = \rho cp \cdot dT/rah$
SEBAL dT	$dT = a + b \cdot Ts$ (from hot and cold pixels)
EF / ETrF	$EF = \lambda ET/(Rn - G)$; $ETrF = ETinst/ETr,inst$
Daily ET	$ET24 = ETrF \times ETr,24$ or $86400 \cdot EF \cdot Rn24/\lambda$
FAO-56 Kc	$ETc = Kc \cdot ET_o$; $ETa = Ks \cdot Kc \cdot ET_o$
EC flux	$F = \rho\langle w'c' \rangle$
SIF	$SIF = PAR \cdot fAPAR \cdot \Phi F \cdot fesc$
FLD	$F = (Eout \cdot Lin - Ein \cdot Lout)/(Eout - Ein)$
LUE yield	$Yield = HI \cdot \Sigma(\varepsilon \cdot fAPAR \cdot PAR \cdot Ts \cdot Ws)$
Beer's law	$fAPAR = 1 - \exp(-k \cdot LAI)$
FRP → biomass	Biomass (kg) = 0.368 × FRE (MJ)
Threshold yield	TY = mean(best 5 of 7 years) × indemnity level
Claim	$(TY - AY)/TY \times$ sum insured
Neyman allocation	$nh = n \cdot NhSh/\Sigma NhSh$
RMSE / nRMSE	$\sqrt{(\Sigma(P - O)^2/n)}$; $RMSE/\bar{O} \times 100$
Kappa	$(po - pe)/(1 - pe)$
GDD	$\Sigma[(Tmax + Tmin)/2 - Tbase]$
CWSI	$(Tc - Twet)/(Tdry - Twet)$
Topp equation	$\theta v = -0.053 + 0.0292\varepsilon - 5.5\times10^{-4}\varepsilon^2 + 4.3\times10^{-6}\varepsilon^3$

19.2 Abbreviations

Abbreviation	Expansion
ALEXI	Atmosphere–Land EXchange Inverse

Abbreviation	Expansion
AWD	Alternate Wetting and Drying
BAI	Burned Area Index
CAQM	Commission for Air Quality Management in NCR and adjoining areas
CBG	Compressed Bio-Gas
CCE	Crop Cutting Experiment
CCTS	Carbon Credit Trading Scheme (India, 2023)
CHAMAN	Coordinated Horticulture Assessment and Management using geoiNformatics
CREAMS	Consortium for Research on Agroecosystem Monitoring and Modelling from Space (IARI)
CROPIC	Collection of Real-time Observations and Photos of Crops
DSM	Digital Soil Mapping
EC	Eddy Covariance / Electrical Conductivity
ESI	Evaporative Stress Index
FASAL	Forecasting Agricultural output using Space, Agro-meteorology and Land-based observations
FIRMS	Fire Information for Resource Management System (NASA)
FLEX / FLORIS	FLuorescence EXplorer / FLuORescence Imaging Spectrometer
FRP / FRE	Fire Radiative Power / Energy
GKMS	Gramin Krishi Mausam Sewa
IAIRO	India AI Research Organisation (GIFT City)
ICVCM	Integrity Council for the Voluntary Carbon Market
IOD	Indian Ocean Dipole
LUE / RUE	Light / Radiation Use Efficiency
METRIC	Mapping EvapoTranspiration at high Resolution with Internalized Calibration
MNCFC	Mahalanobis National Crop Forecast Centre
MRV / dMRV	Monitoring (Measurement), Reporting and Verification / digital MRV
NBR	Normalised Burn Ratio
NePPA	Network Programme on Precision Agriculture (ICAR)
NISAR	NASA–ISRO Synthetic Aperture Radar
ONI	Oceanic Niño Index
PMFBY	Pradhan Mantri Fasal Bima Yojana
PROSAIL	PROSPECT + SAIL radiative transfer model
RWBCIS	Restructured Weather Based Crop Insurance Scheme
SATAT	Sustainable Alternative Towards Affordable Transportation
SEBAL	Surface Energy Balance Algorithm for Land
SIF	Sun-Induced (chlorophyll) Fluorescence
TPA	Technology Providing Agency (YES-TECH)

Abbreviation	Expansion
TRISHNA	Thermal infraRed Imaging Satellite for High-resolution Natural resource Assessment (ISRO–CNES)
TSEB	Two-Source Energy Balance
VCU	Verified Carbon Unit (Verra)
VRT	Variable-Rate Technology
VVB	Validation/Verification Body
WINDS	Weather Information Network and Data Systems
YES-TECH	Yield Estimation System based on TECHnology

19.3 Rapid-fire questions (one-line answers)

Q. When did FLEX launch?

A. 15 September 2026, on Vega-C VV30 from Kourou, together with Sentinel-3C.

Q. FLEX spatial resolution and swath?

A. 300 m; 150 km.

Q. FLEX spectral range?

A. 500–780 nm (FLORIS), 0.3 nm resolution in the O_2 bands.

Q. Wavelengths of the SIF peaks?

A. About 685 nm (red) and 740 nm (far-red).

Q. When did NISAR launch?

A. 30 July 2025, GSLV-F16, Sriharikota.

Q. NISAR bands and their builders?

A. L-band 24 cm (NASA) and S-band 9.4 cm (ISRO).

Q. NISAR repeat cycle and swath?

A. 12 days; about 242 km.

Q. NISAR soil-moisture resolution in India?

A. 100 m L4 products (L and S), distributed via Bhoonidhi.

Q. Which polarisation is most sensitive to biomass?

A. Cross-pol (HV/VH), from volume scattering.

Q. Why is HH high for rice?

A. Double-bounce between the water surface and vertical stems.

Q. Hot pixel in SEBAL?

A. Dry bare field where $\lambda ET \approx 0$, so $H = Rn - G$.

Q. Cold pixel in METRIC?

A. Well-watered crop with ET = 1.05 ETr.

Q. Basis of daily scaling?

A. EF or ETrF (AET/ET_0) stays nearly constant through the day.

Q. Which model uses the morning rise in Ts?

A. ALEXI.

Q. What is energy-balance closure?

A. $(H + \lambda E)/(Rn - G)$; typically 0.7–0.9 for EC towers.

Q. Which VIIRS band detects fire?

A. I4 (3.74 µm) at 375 m, with I5 (11.45 µm) for background.

Q. Unit of FRP?

A. Megawatts (MW).

Q. Which Sentinel-2 bands are used for NBR?

A. B8 (NIR) and B12 (SWIR2); NBR2 uses B11 and B12.

Q. dNBR threshold for burnt?

A. > 0.1 (low severity from 0.1 to 0.27).

Q. Who issues the daily residue-burning bulletin?

A. The CREAMS lab, Division of Agricultural Physics, IARI, under the ISRO–ICAR protocol.

Q. States covered by the bulletin?

A. Punjab, Haryana, UP, Delhi, MP and Rajasthan.

Q. What is the PUSA decomposer?

A. A fungal-consortium spray from IARI that decomposes paddy stubble in about 20–25 days.

Q. Where is the 2G ethanol plant that uses paddy straw?

A. IOCL Panipat, Haryana.

Q. Three CA principles?

A. Minimum soil disturbance, permanent soil cover, crop diversification.

Q. Index for residue cover?

A. NDTI (B11, B12) or CAI (hyperspectral).

Q. One carbon credit equals?

A. 1 tCO_2e reduced or removed.

Q. Verra methodology for agricultural soil carbon?

A. VM0042 Improved Agricultural Land Management.

Q. Who administers India's CCTS?

A. The Bureau of Energy Efficiency (BEE), Ministry of Power.

Q. Biochar stability criterion?

A. Molar H/Corg < 0.7.

Q. What is additionality?

A. The reduction would not occur without carbon finance.

Q. Lead institute of NePPA?

A. ICAR-IARI; launched 8 September 2021 with 16 partners.

Q. Map-based vs sensor-based VRT?

A. A prescription prepared in advance vs a real-time on-the-go rate.

Q. Which agency runs FASAL?

A. MNCFC (DA&FW), with ISRO methodology.

Q. CHAMAN crops?

A. Banana, mango, citrus, potato, onion, tomato and chilli.

Q. YES-TECH start and crops?

A. Kharif 2023 for paddy and wheat; soybean from Kharif 2024.

Q. Minimum YES-TECH weightage?

A. 30 % (100 % in MP).

Q. Threshold yield?

A. Mean of best 5 of 7 years × indemnity level.

Q. PMFBY farmer premiums?

A. 2 % kharif, 1.5 % rabi, 5 % commercial and horticultural crops.

Q. Insurance unit for major crops?

A. Village or gram panchayat.

Q. Semi-physical model?

A. Monteith LUE: Σ ε·fAPAR·PAR·stress × HI.

Q. Super El Niño?

A. ONI ≥ +2.0 °C (an informal term).

Q. Niño 3.4 region?

A. 5° N–5° S, 170°–120° W.

Q. What can offset El Niño's effect on the monsoon?

A. A positive Indian Ocean Dipole.

Q. Monsoon LPA?

A. 87 cm (1971–2020).

Q. Digital twin vs crop model?

A. The twin is a live, closed-loop, data-assimilating replica, while a crop model is an open-loop simulation.

Q. EnKF?

A. Ensemble Kalman Filter, a data-assimilation method.

Q. PROSPECT inputs?

A. N, Cab, Car, Cw, Cm, Cbrown.

Q. scorpan?

A. Soil, climate, organisms, relief, parent material, age, position (DSM).

Q. Saline vs sodic soil?

A. Saline: EC > 4 dS/m, pH < 8.5. Sodic: ESP > 15, pH > 8.5.

Q. Who is IAIRO's founding director?

A. Dr. Amit Sheth.

20. References and further reading

Read the original papers below for the topics you mention in your self-introduction. For the IARI-specific papers (Dr. Rajkumar Dhakar on CA mapping, Dr. V. K. Sehgal, Dr. Bappa Das), confirm exact titles in Google Scholar or the IARI repository before quoting them.

1. Allen, R.G., Pereira, L.S., Raes, D., Smith, M. (1998). Crop evapotranspiration — FAO Irrigation and Drainage Paper 56. FAO, Rome.
2. Allen, R.G., Tasumi, M., Trezza, R. (2007). Satellite-based energy balance for mapping evapotranspiration with internalized calibration (METRIC) — Model. J. Irrig. Drain. Eng. 133(4), 380–394.
3. Anderson, M.C., Norman, J.M., Diak, G.R., Kustas, W.P., Mecikalski, J.R. (1997). A two-source time-integrated model for estimating surface fluxes using thermal infrared remote sensing. Remote Sens. Environ. 60, 195–216.
4. Anderson, M.C. et al. (2007). A climatological study of evapotranspiration and moisture stress across the continental United States based on thermal remote sensing (ALEXI). J. Geophys. Res. 112, D10117.
5. Attema, E.P.W., Ulaby, F.T. (1978). Vegetation modeled as a water cloud. Radio Science 13(2), 357–364.
6. Bastiaanssen, W.G.M., Menenti, M., Feddes, R.A., Holtslag, A.A.M. (1998). A remote sensing surface energy balance algorithm for land (SEBAL). J. Hydrol. 212–213, 198–212.
7. Breiman, L. (2001). Random forests. Machine Learning 45, 5–32.
8. Cariappa, A.G.A. et al. (2024). Study of conservation-agriculture eligibility for carbon credits under Verra VM0042 using farmer surveys and remote sensing. Scientific Reports (April 2024). [Title paraphrased — verify.]
9. Chuvieco, E., Martín, M.P., Palacios, A. (2002). Assessment of different spectral indices in the red–near-infrared spectral domain for burned land discrimination. Int. J. Remote Sens. 23, 5103–5110.
10. Crago, R.D. (1996). Conservation and variability of the evaporative fraction during the daytime. J. Hydrol. 180, 173–194.
11. Drusch, M. et al. (2017). The FLuorescence EXplorer mission concept — ESA's Earth Explorer 8. IEEE Trans. Geosci. Remote Sens. 55(3), 1273–1284.
12. ESA (2026). FLEX and Sentinel-3C launched on Vega-C (VV30), 15 September 2026. esa.int.
13. Frankenberg, C. et al. (2011). New global observations of the terrestrial carbon cycle from GOSAT: patterns of plant fluorescence with gross primary productivity. Geophys. Res. Lett. 38, L17706.
14. Giglio, L., Schroeder, W., Justice, C.O. (2016). The collection 6 MODIS active fire detection algorithm and fire products. Remote Sens. Environ. 178, 31–41.
15. Guan, K. et al. (2016). Improving the monitoring of crop productivity using spaceborne solar-induced fluorescence. Global Change Biology 22, 716–726.

16. Guanter, L. et al. (2014). Global and time-resolved monitoring of crop photosynthesis with chlorophyll fluorescence. PNAS 111(14), E1327–E1333.
17. Huang, J. et al. (2019). Assimilation of remote sensing into crop growth models: current status and perspectives. Agric. For. Meteorol. 276–277, 107609.
18. Jacquemoud, S. et al. (2009). PROSPECT + SAIL models: a review of use for vegetation characterization. Remote Sens. Environ. 113, S56–S66.
19. Kalambukattu, J.G., Kumar, S., Das, B., Roy, T. et al. (2025). Machine-learning based mapping of soil organic carbon in the Indian Himalayan region. Discover Soil.
20. Key, C.H., Benson, N.C. (2006). Landscape assessment: ground measure of severity, the Composite Burn Index; and remote sensing of severity, the Normalized Burn Ratio. USDA FIREMON, RMRS-GTR-164-CD.
21. Lehmann, J., Kleber, M. (2015). The contentious nature of soil organic matter. Nature 528, 60–68.
22. McBratney, A.B., Mendonça Santos, M.L., Minasny, B. (2003). On digital soil mapping. Geoderma 117, 3–52.
23. Meroni, M. et al. (2009). Remote sensing of solar-induced chlorophyll fluorescence: review of methods and applications. Remote Sens. Environ. 113, 2037–2051.
24. Mohammed, G.H. et al. (2019). Remote sensing of solar-induced chlorophyll fluorescence (SIF) in vegetation: 50 years of progress. Remote Sens. Environ. 231, 111177.
25. Monteith, J.L. (1972). Solar radiation and productivity in tropical ecosystems. J. Appl. Ecol. 9, 747–766.
26. Norman, J.M., Kustas, W.P., Humes, K.S. (1995). Source approach for estimating soil and vegetation energy fluxes in observations of directional radiometric surface temperature. Agric. For. Meteorol. 77, 263–293.
27. Olofsson, P. et al. (2014). Good practices for estimating area and assessing accuracy of land change. Remote Sens. Environ. 148, 42–57.
28. Porcar-Castell, A. et al. (2014). Linking chlorophyll a fluorescence to photosynthesis for remote sensing applications. J. Exp. Bot. 65, 4065–4095.
29. Raun, W.R. et al. (2002). Improving nitrogen use efficiency in cereal grain production with optical sensing and variable rate application. Agron. J. 94, 815–820.
30. Sehgal, V.K., Dhakar, R. et al. (2025). Scaling geospatial research for operational agricultural applications in India (Zero-Tillage Mapper; residue-burning monitoring). Geospatial World Forum / GSI 2025 presentation, CREAMS Lab, ICAR-IARI.
31. Sun, Y. et al. (2015). Drought onset mechanisms revealed by satellite solar-induced chlorophyll fluorescence. J. Geophys. Res. Biogeosci. 120, 2427–2440.
32. Topp, G.C., Davis, J.L., Annan, A.P. (1980). Electromagnetic determination of soil water content. Water Resour. Res. 16, 574–582.
33. Trigg, S., Flasse, S. (2001). An evaluation of different bi-spectral spaces for discriminating burned shrub-savannah. Int. J. Remote Sens. 22, 2641–2647.

34. Verrelst, J. et al. (2015). Optical remote sensing and the retrieval of terrestrial vegetation bio-geophysical properties — a review. ISPRS J. Photogramm. Remote Sens. 108, 273–290.
35. Wagner, W., Lemoine, G., Rott, H. (1999). A method for estimating soil moisture from ERS scatterometer and soil data. Remote Sens. Environ. 70, 191–207.
36. Wooster, M.J., Roberts, G., Perry, G.L.W., Kaufman, Y.J. (2005). Retrieval of biomass combustion rates and totals from fire radiative power observations. J. Geophys. Res. 110, D24311.
37. Zhou, W. et al. (2021). Using Sentinel-2 to track field-level tillage practices at regional scales in smallholder systems. Remote Sensing 13(24), 5108.
38. Government and programme sources: PMFBY Operational Guidelines (2020) and YES-TECH Manual (2023), DA&FW; MNCFC FASAL and CHAMAN documentation; ISRO NISAR soil-moisture product page (isro.gov.in); ICAR press release on NePPA (8 Sep 2021); BEE Carbon Credit Trading Scheme 2023; Verra VM0042 v2.1; ISRO–ICAR standard protocol for crop-residue burning monitoring (2021); iFOREST report on stubble burning (2025); Business Today/The Week reports on the IAIRO launch (30 Jan 2026); IMD monsoon forecasts 2026.

> **Final advice:** Spend the last day on Chapter 14 (insurance), Chapter 10 (burning) and Chapters 15 and 19 (worked examples and rapid revision). Draw the energy-balance, SIF-fate, NBR and YES-TECH diagrams from memory once each.